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(19) **United States**(12) **Patent Application Publication**
Miccio(10) **Pub. No.: US 2025/0281304 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **INTERVERTEBRAL SPINAL IMPLANT**(71) Applicant: **GLOBUS MEDICAL, INC.**,
AUDUBON, PA (US)(72) Inventor: **Mark Miccio**, West Collingswood
Heights, NJ (US)(21) Appl. No.: **19/217,618**(22) Filed: **May 23, 2025****Related U.S. Application Data**

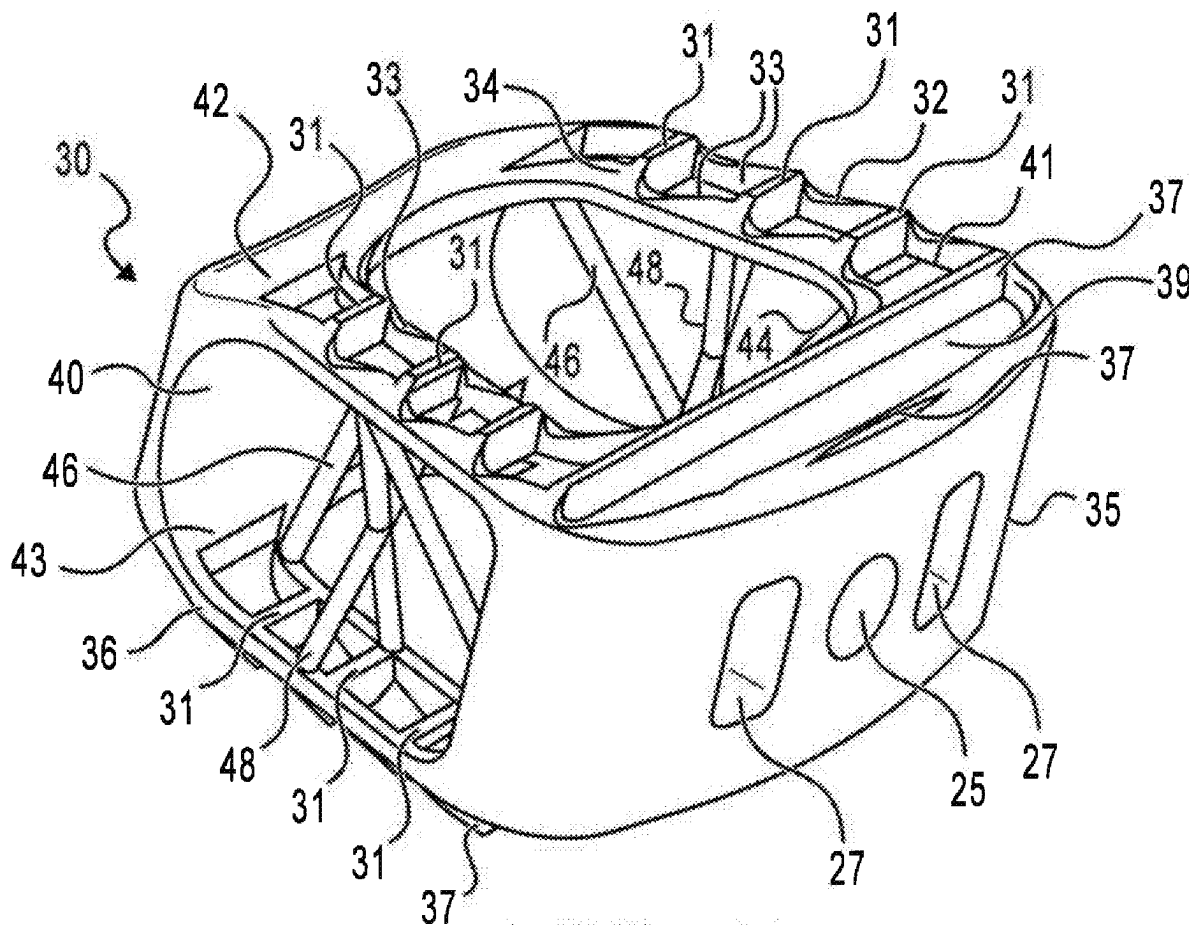
(63) Continuation of application No. 17/933,560, filed on Sep. 20, 2022, now Pat. No. 12,310,859, which is a continuation of application No. 16/600,632, filed on Oct. 14, 2019, now Pat. No. 11,478,361, which is a continuation of application No. 15/973,756, filed on May 8, 2018, now Pat. No. 10,555,819, which is a continuation of application No. 15/973,609, filed on May 8, 2018, now Pat. No. 11,141,286.

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(2013.01); **A61F 2002/30182** (2013.01); **A61F**
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2002/30904 (2013.01); **A61F 2002/3097**
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(57)

ABSTRACT

An intervertebral implant for implantation in an intervertebral space between vertebrae. The implant includes a body extending from an upper surface to a lower surface. The body has a front end, a rear end and a pair of spaced apart first and second side walls extending between the front and rear walls such that an interior chamber is defined within the front and rear ends and the first and second walls. The body defines an outer perimeter and an inner perimeter extending about the internal chamber. At least one of the side walls is defined by a solid support structure and an integral porous structure, the porous structure extending from the outer perimeter to the inner perimeter. The porous structure embeds or encapsulates at least a portion of the solid support structure.



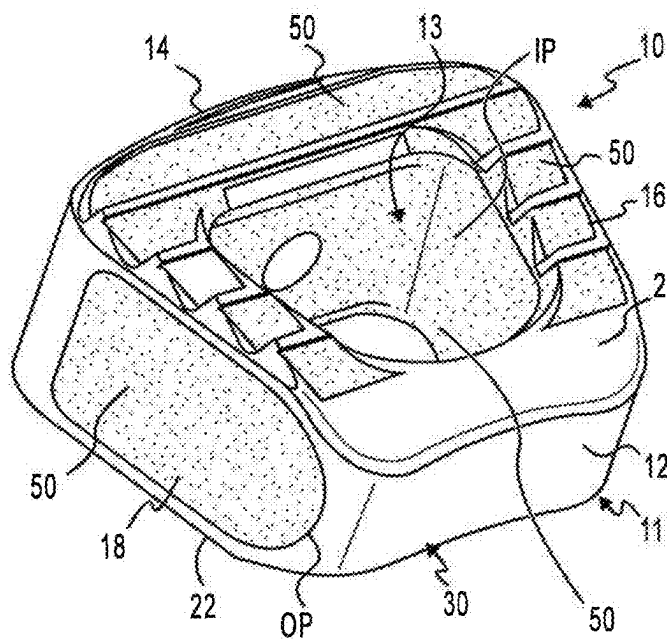


FIG. 1

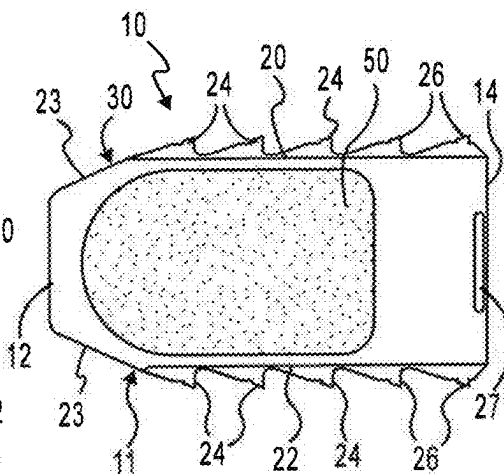


FIG. 2

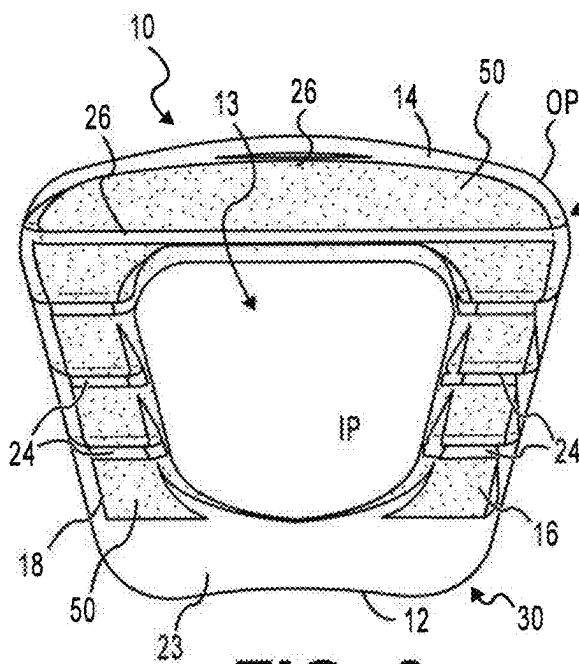


FIG. 3

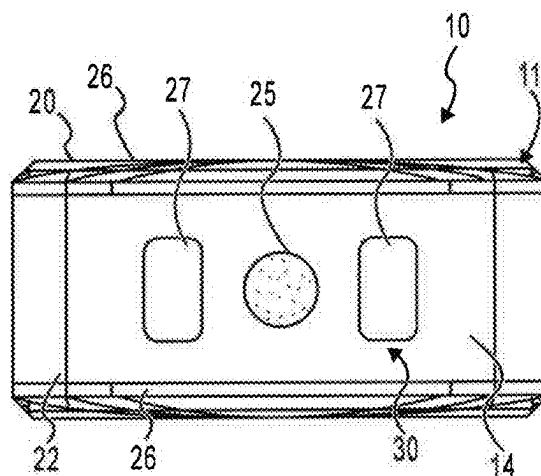


FIG. 4

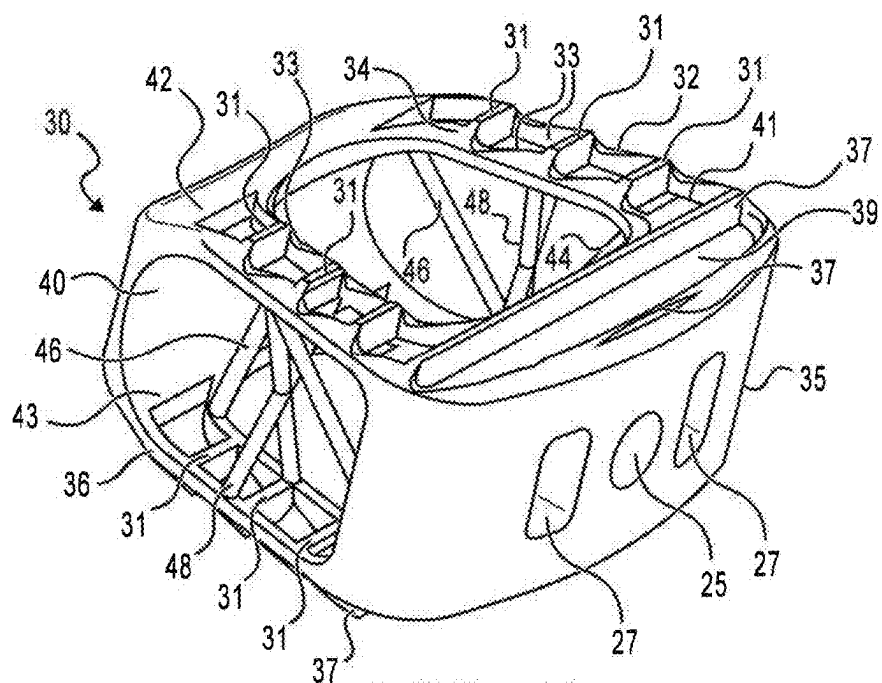


FIG. 5

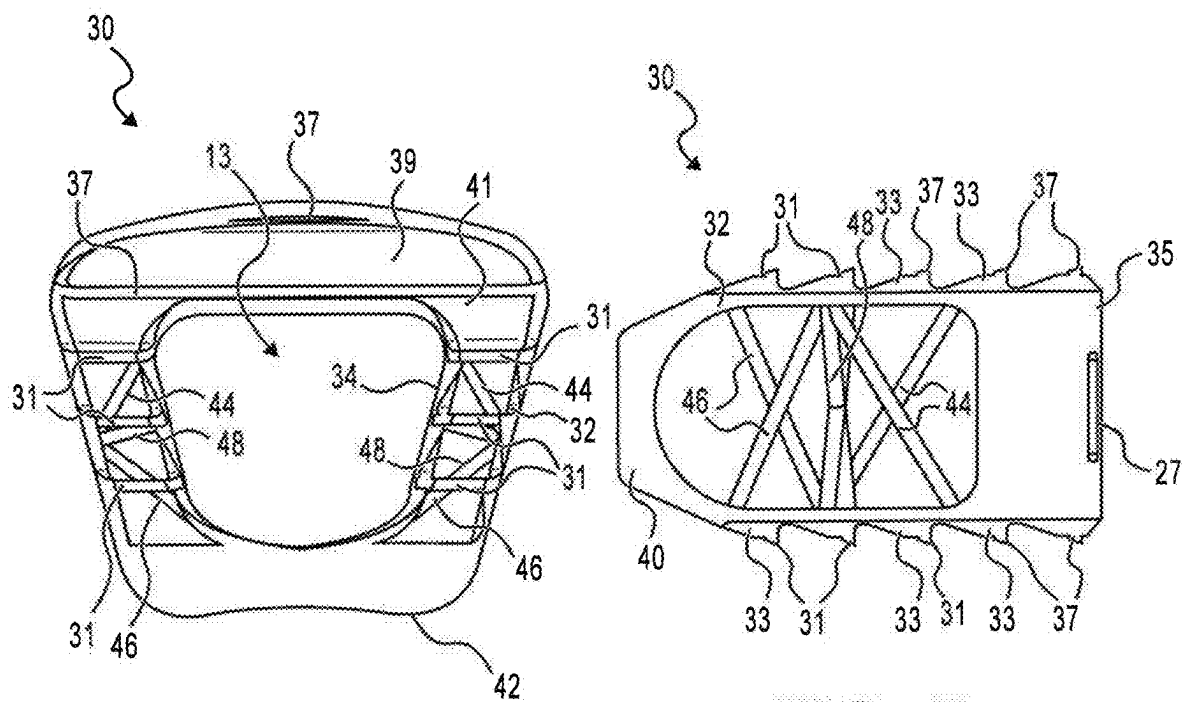


FIG. 6

FIG. 7

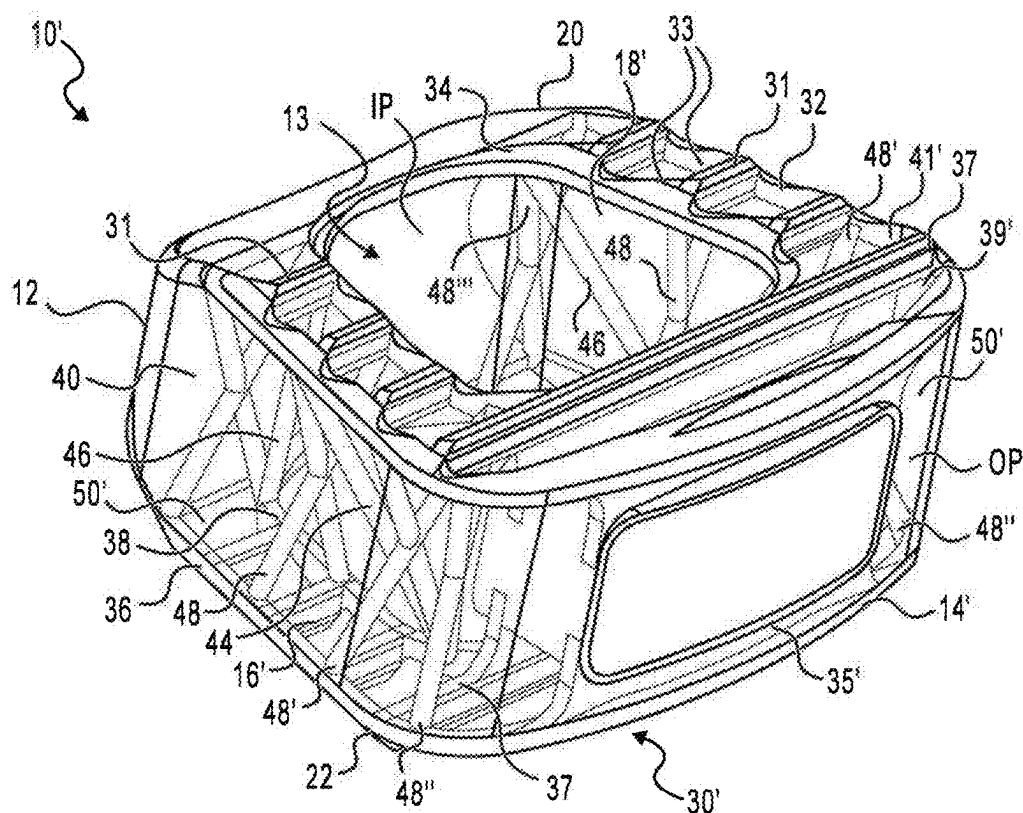


FIG. 8

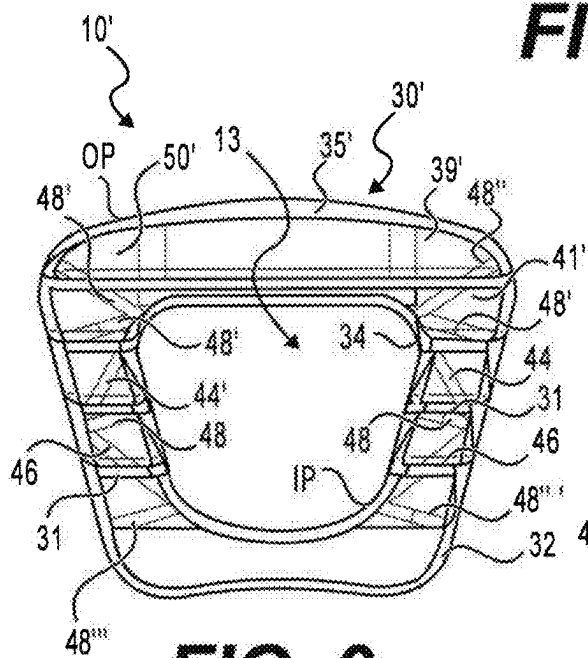


FIG. 9

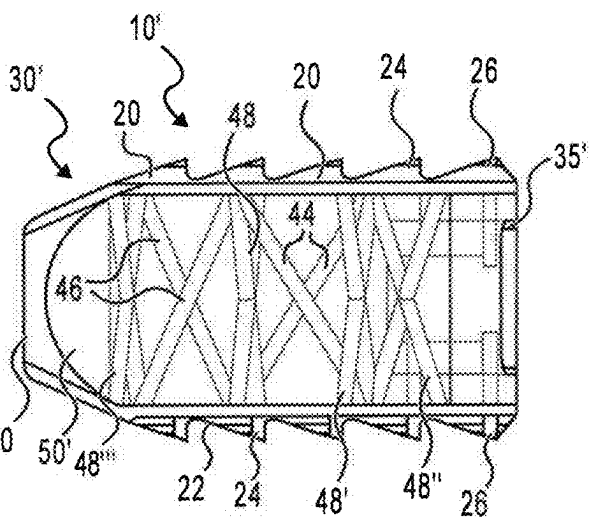


FIG. 10

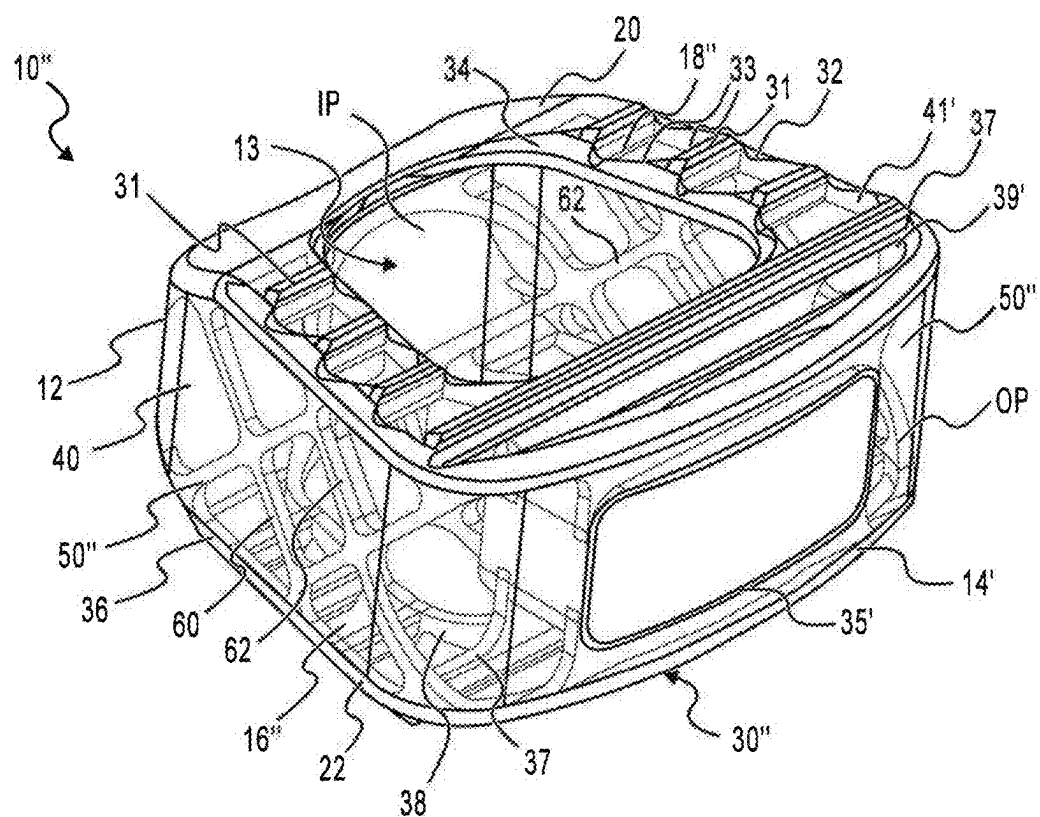


FIG. 11

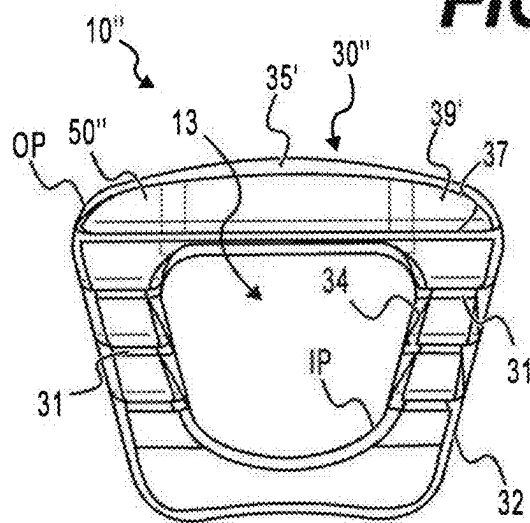


FIG. 12

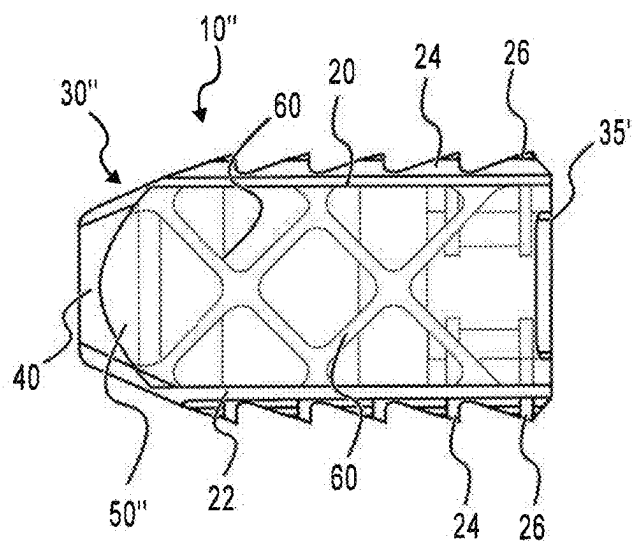


FIG. 13

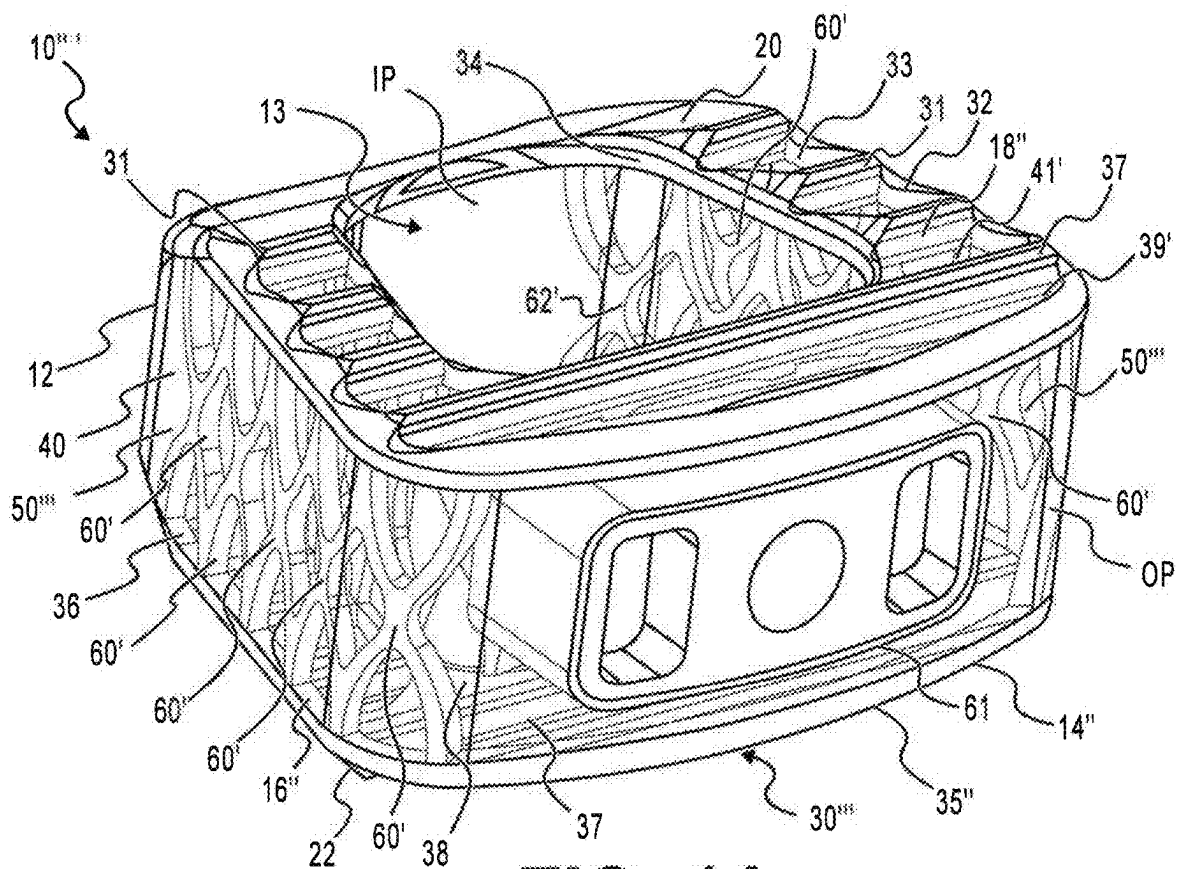


FIG. 14

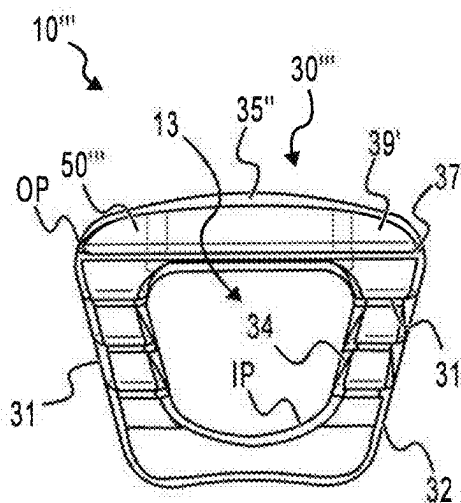


FIG. 15

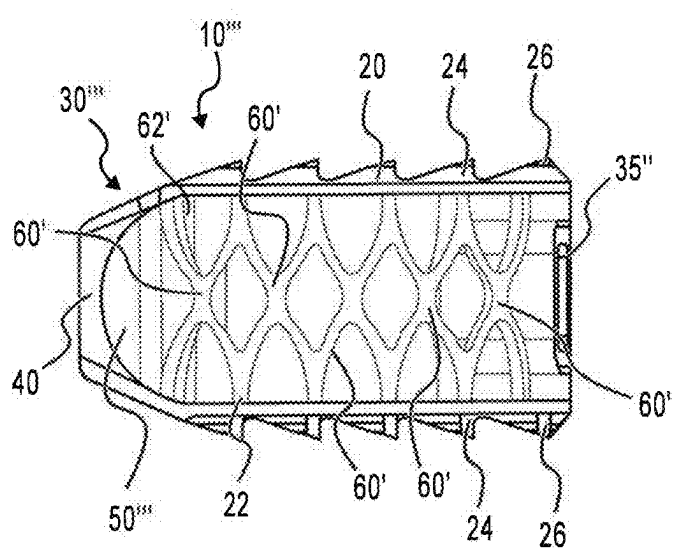


FIG. 16

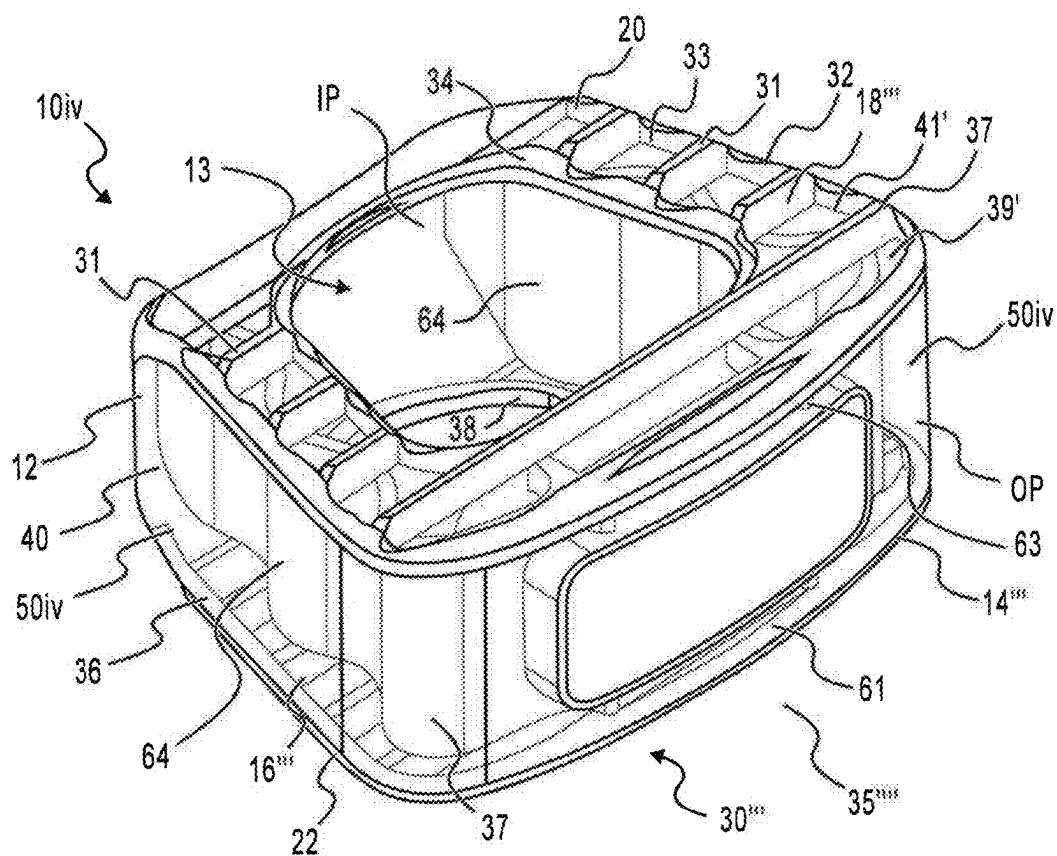


FIG. 17

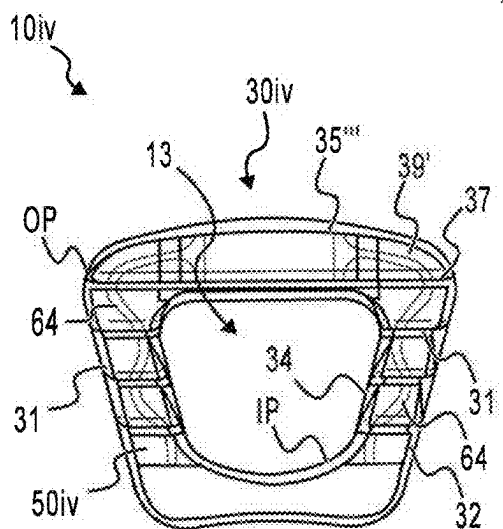


FIG. 18

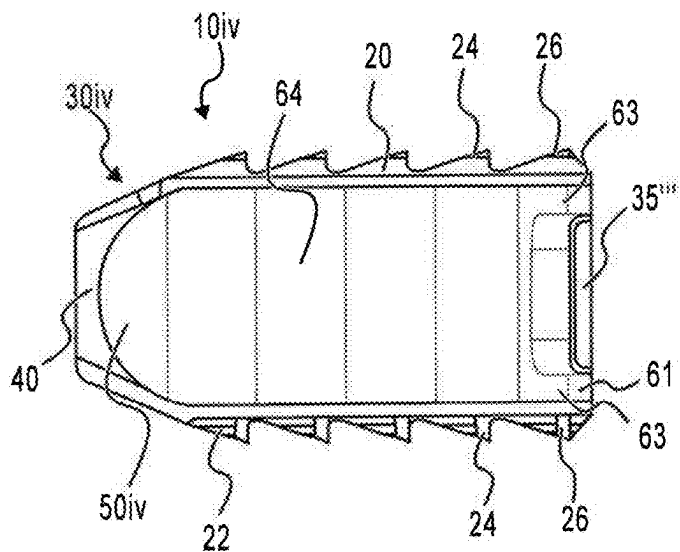


FIG. 19

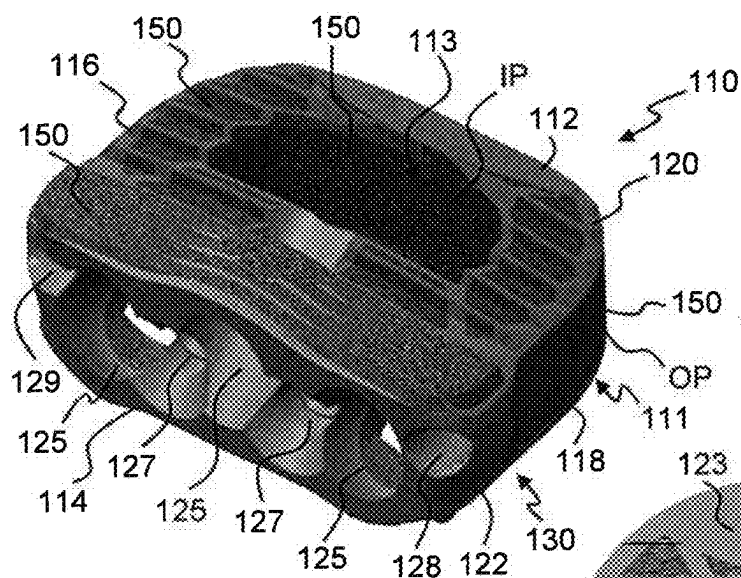


Fig. 20

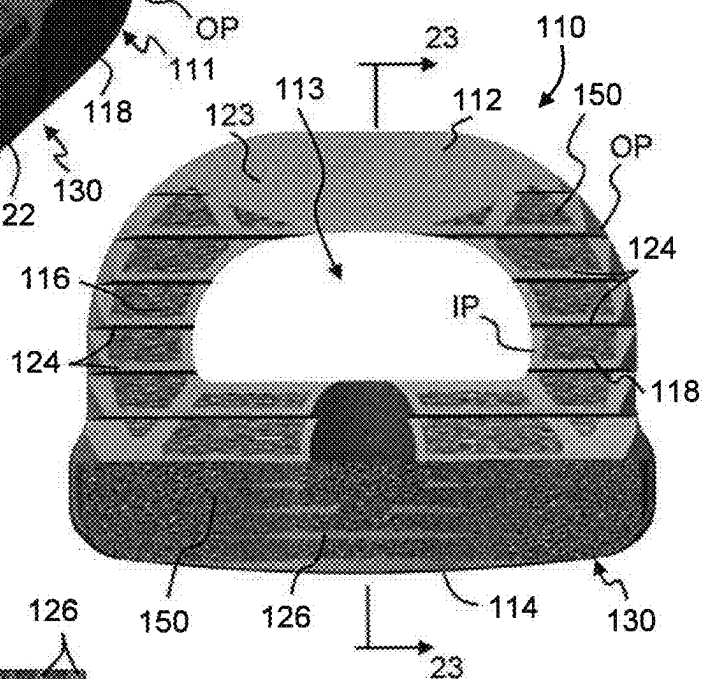


Fig. 21

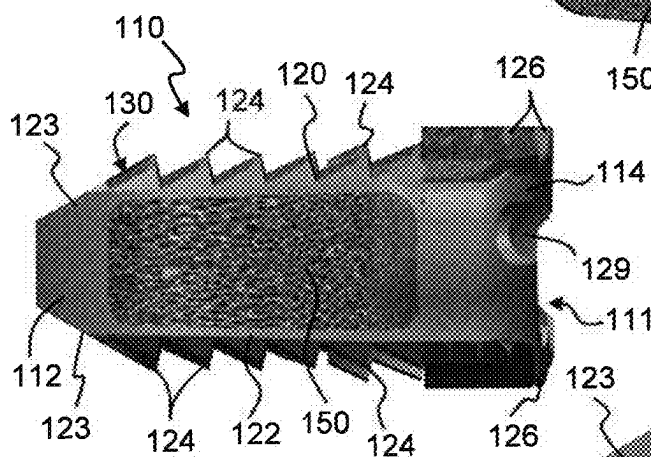


Fig. 22

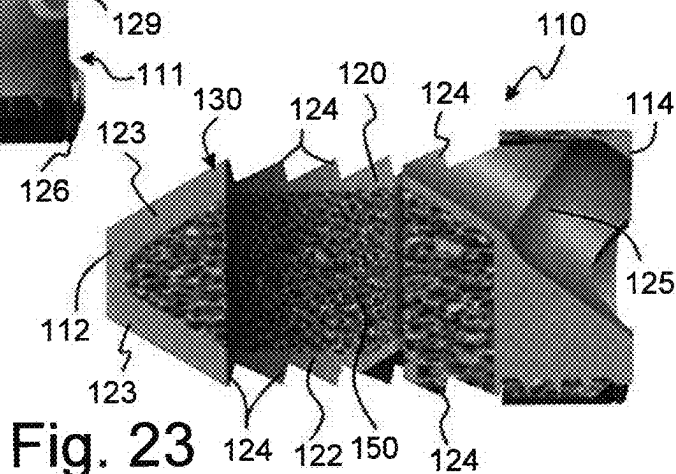


Fig. 23

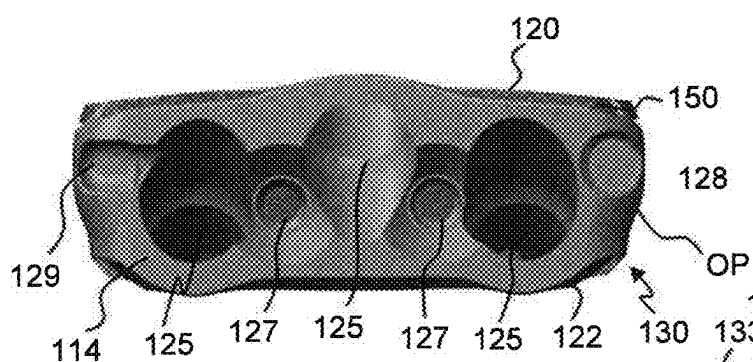


Fig. 24

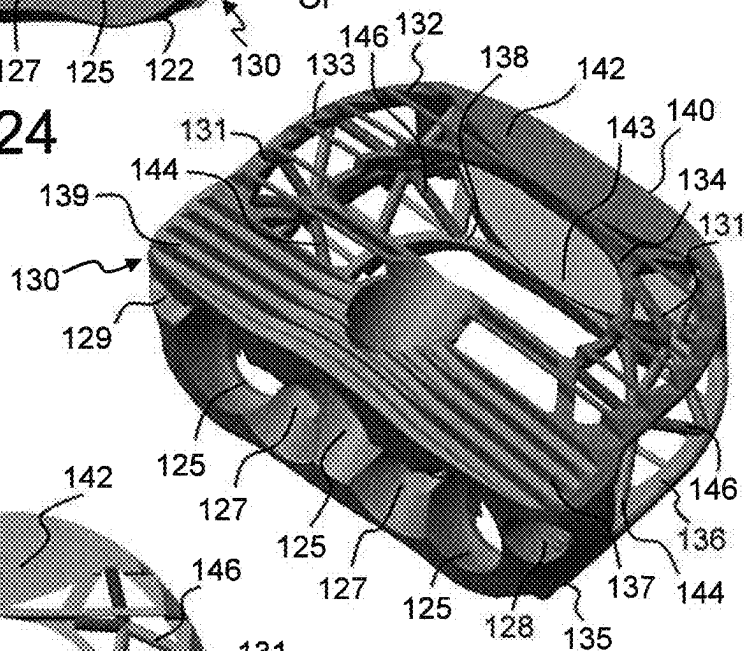


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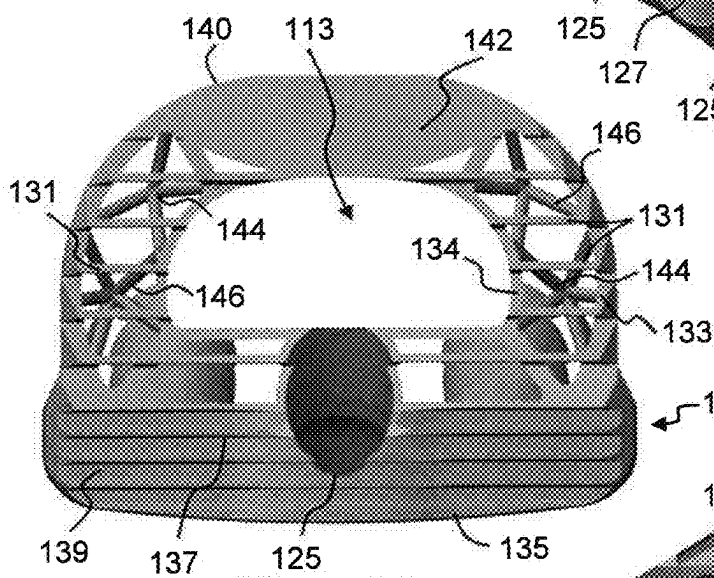


Fig. 26

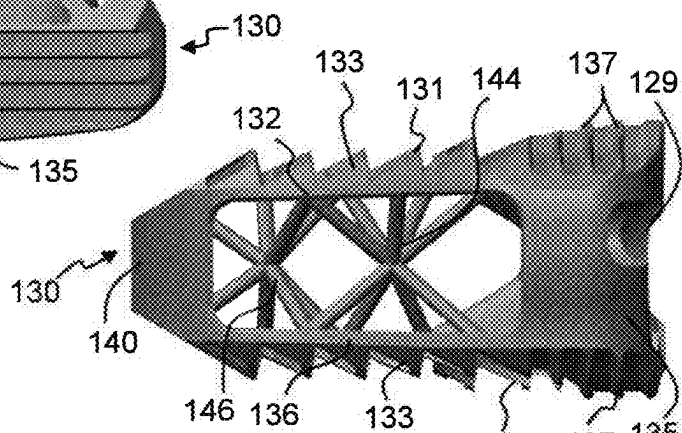
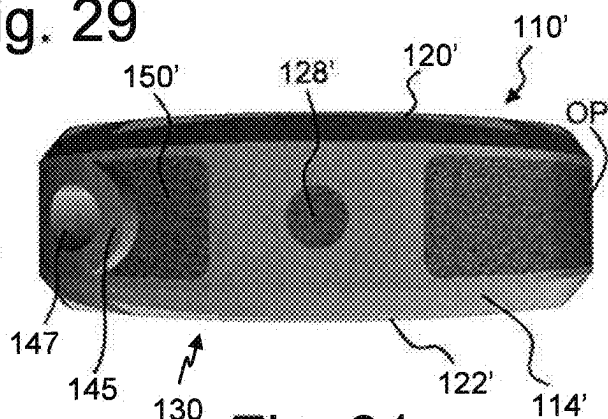
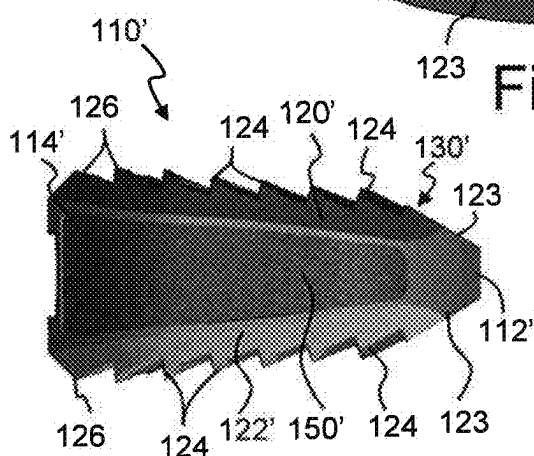
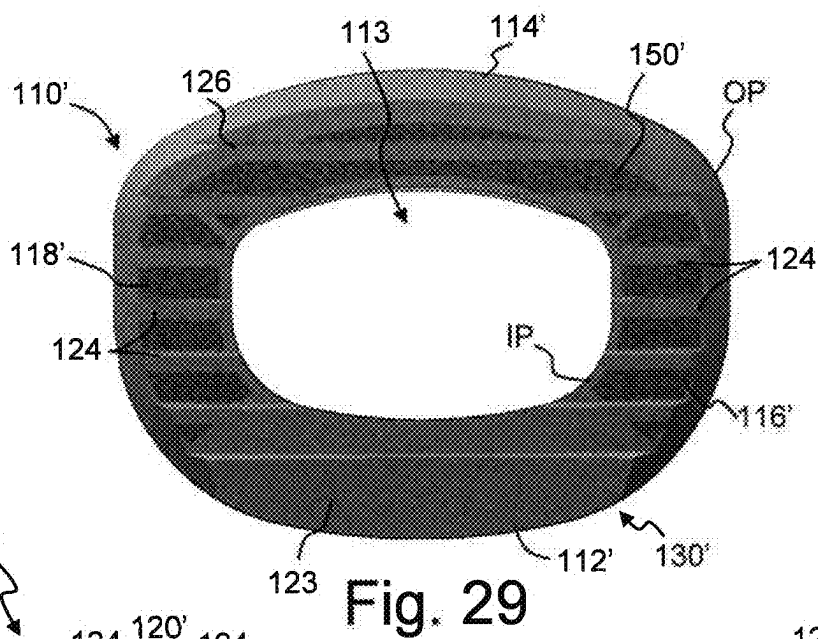
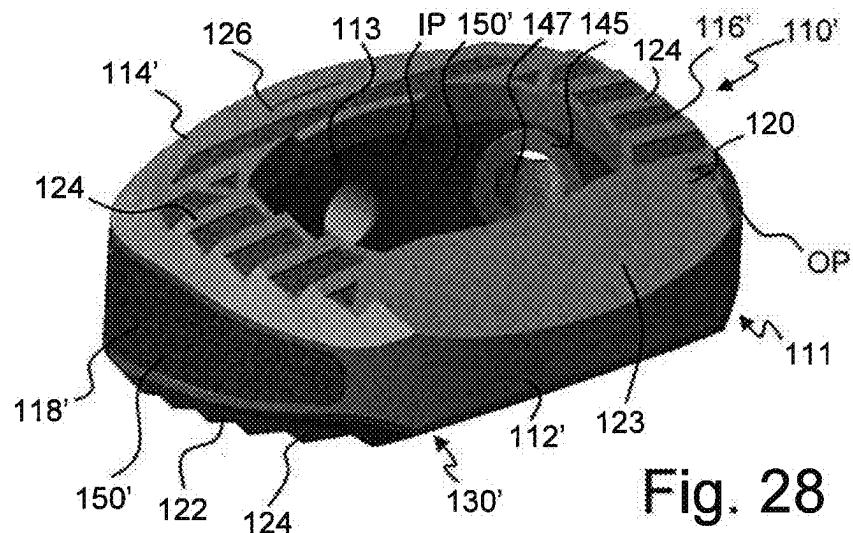
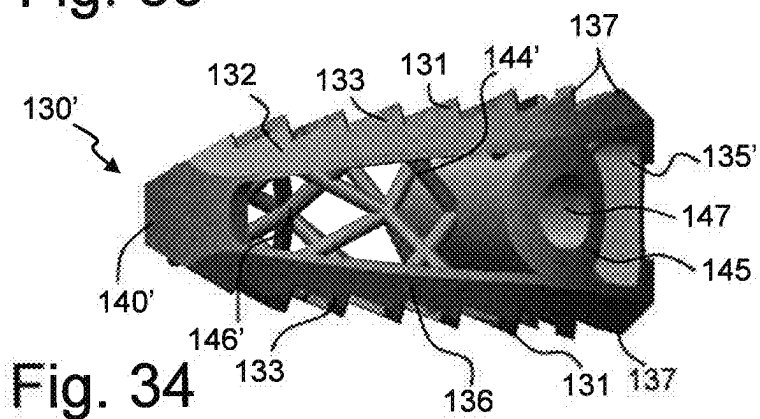
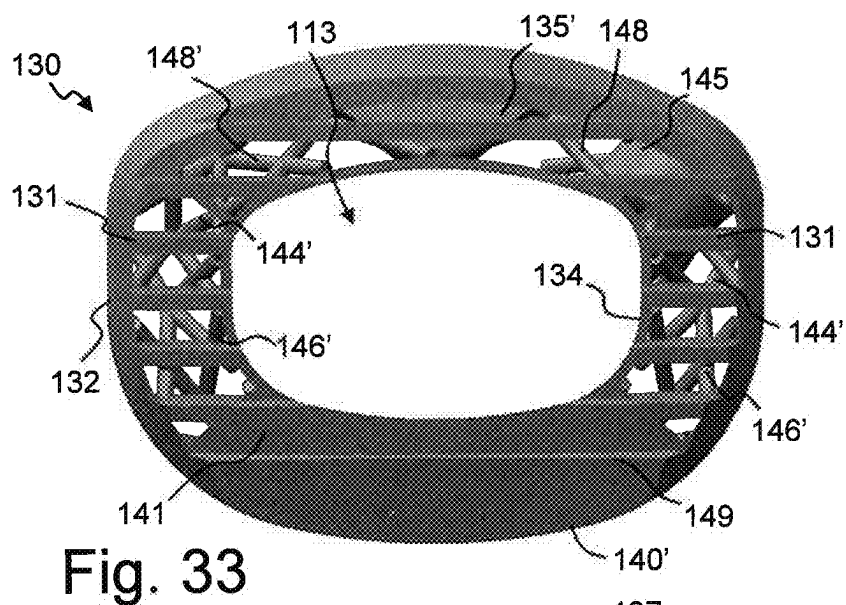
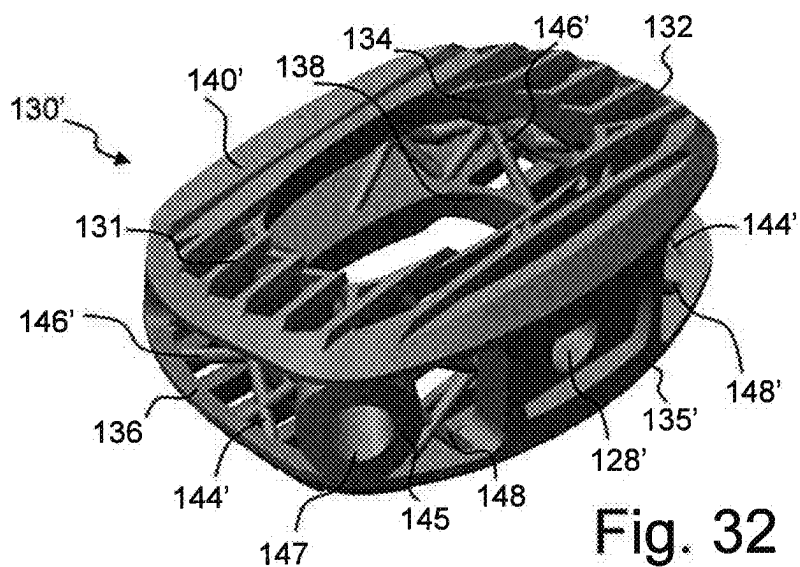


Fig. 27





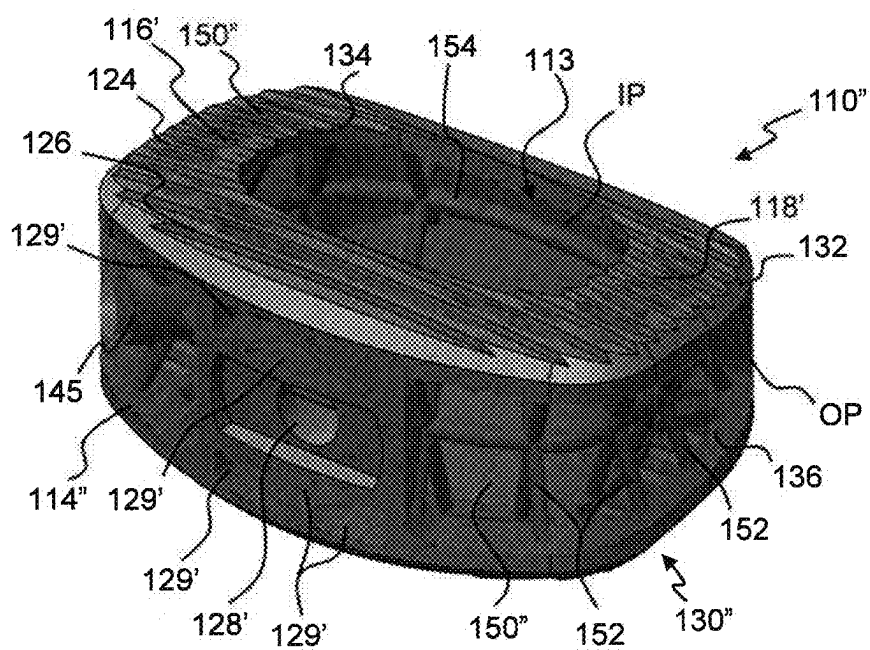


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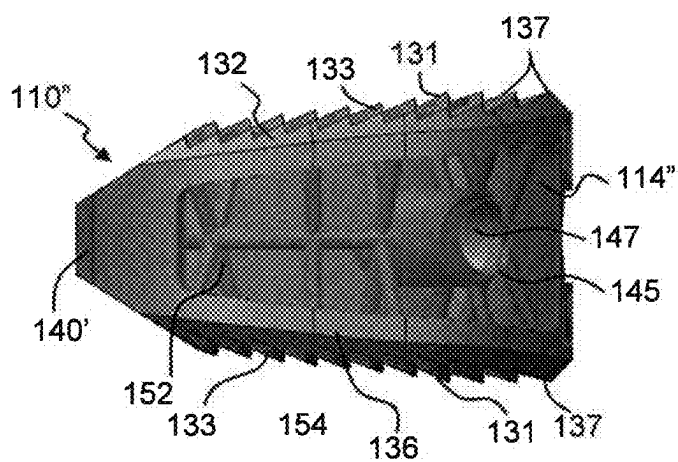
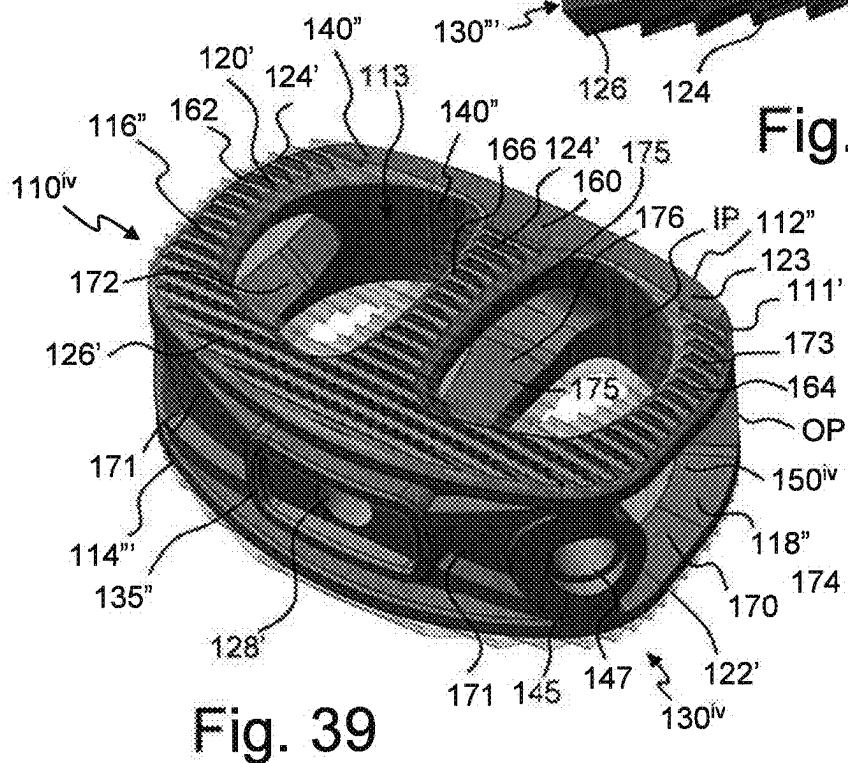
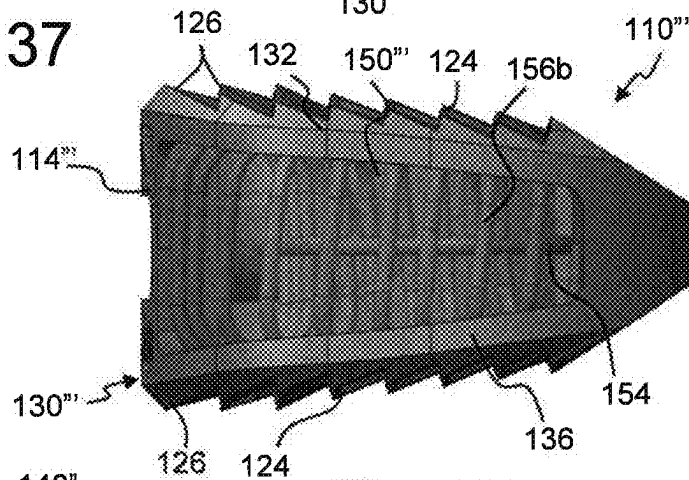
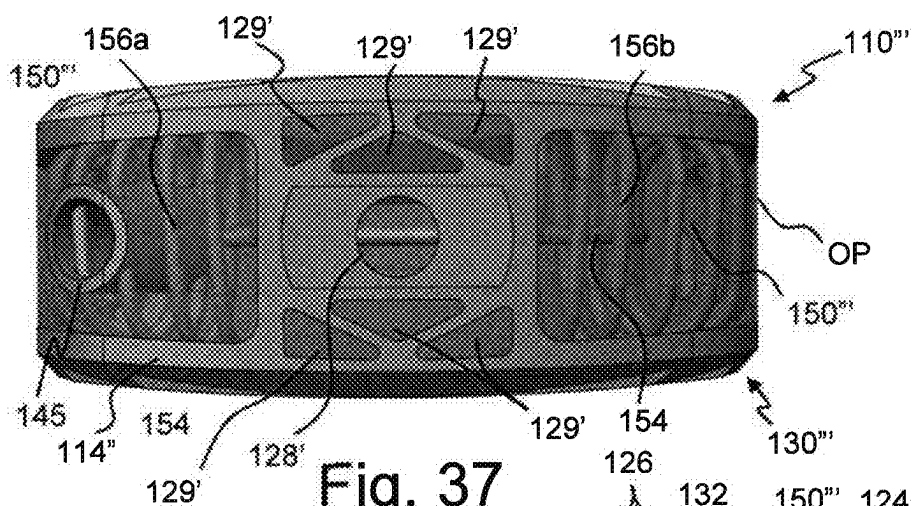


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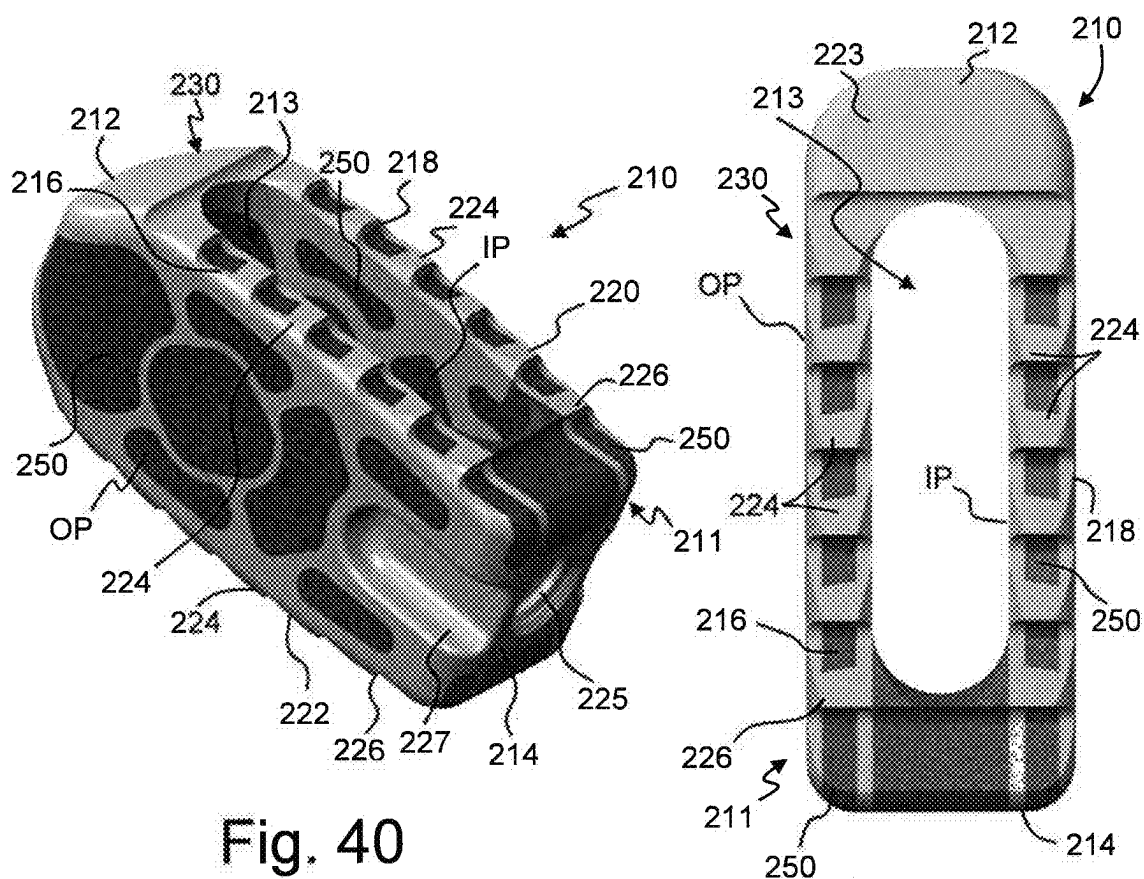


Fig. 40

Fig. 41

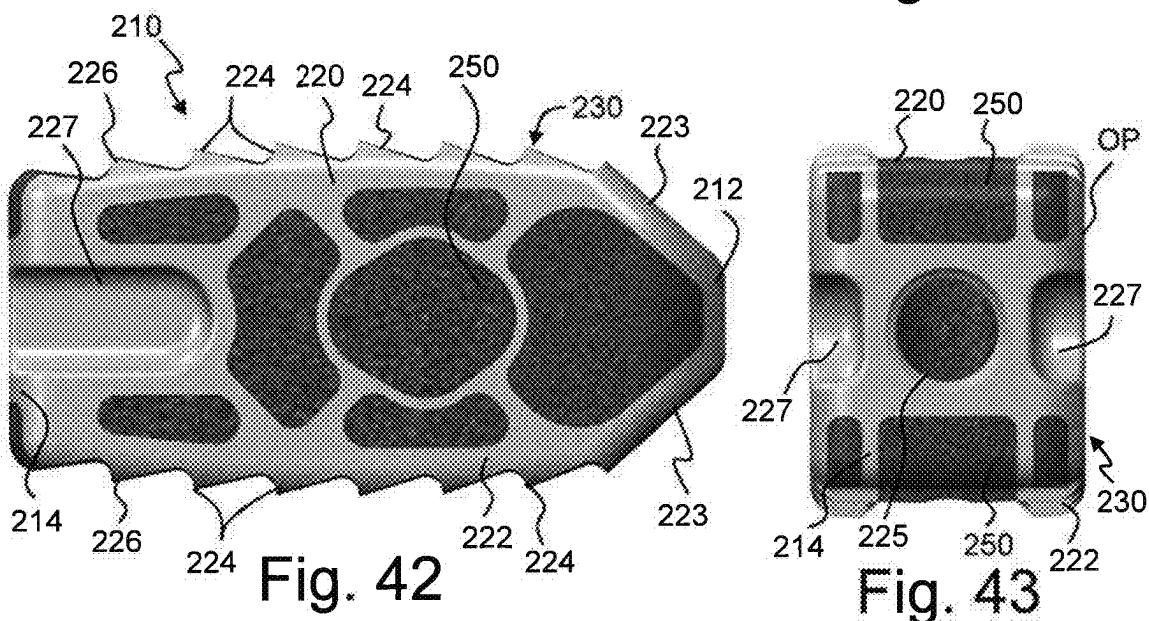


Fig. 42

Fig. 43

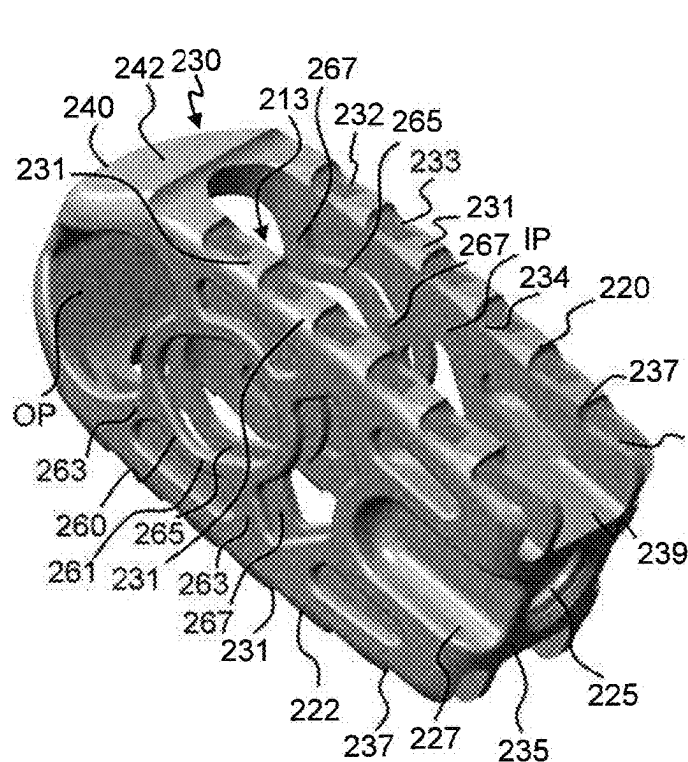


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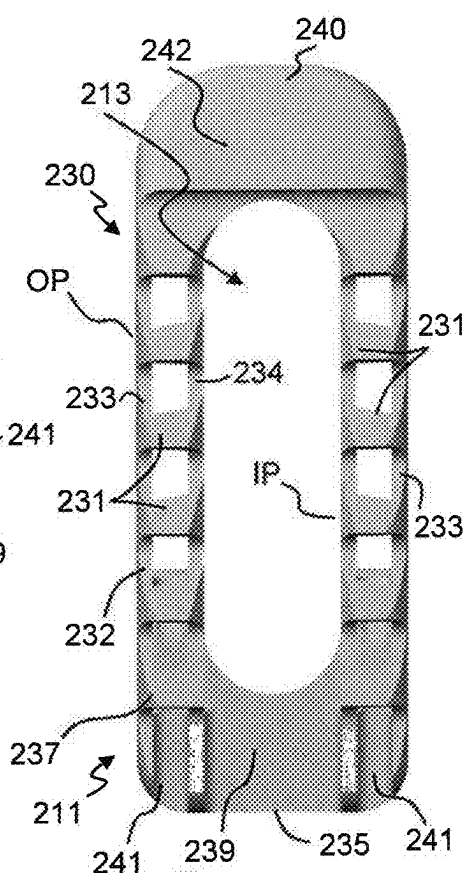


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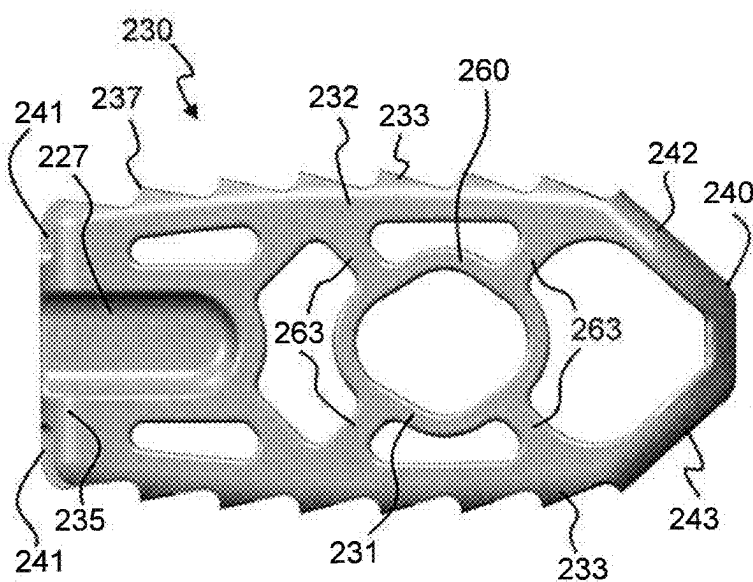


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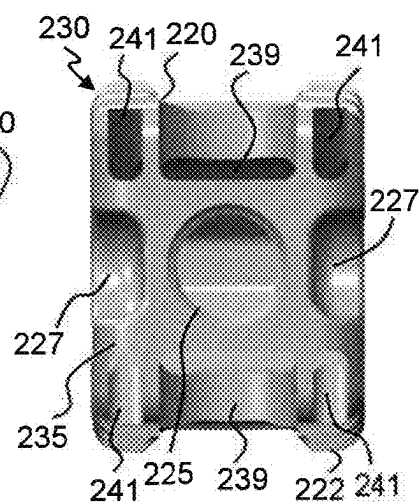


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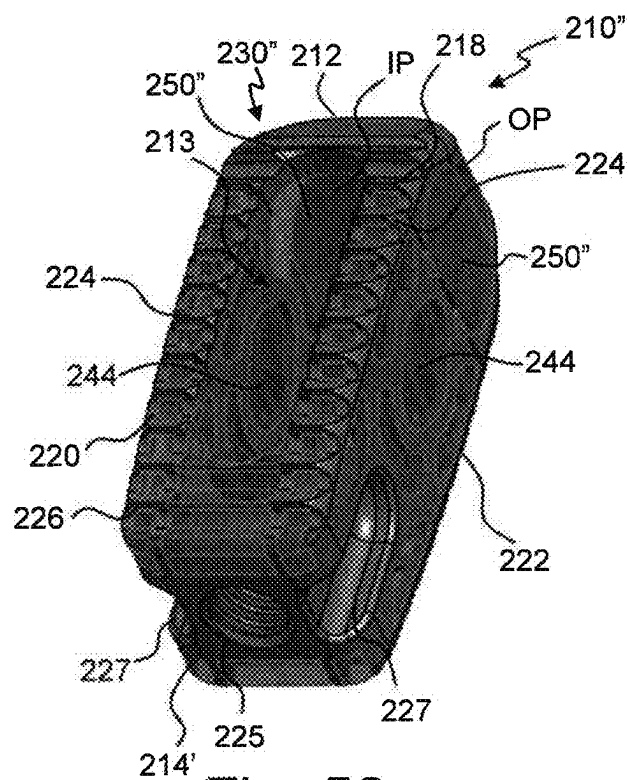


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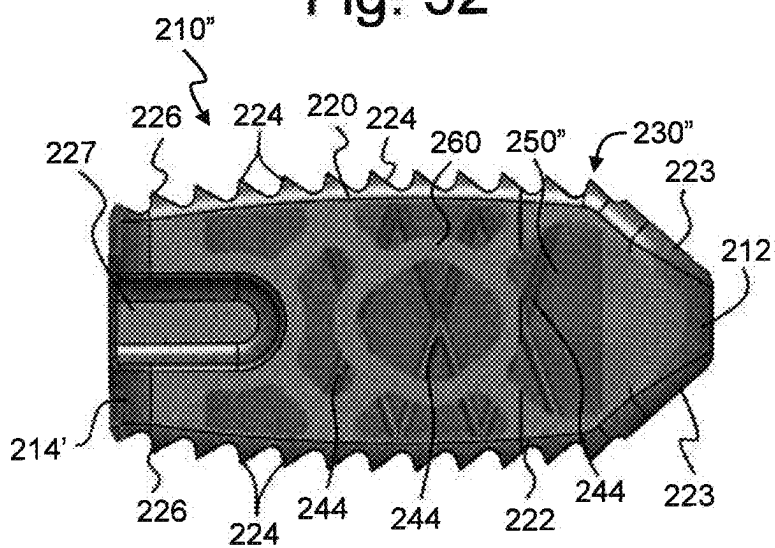


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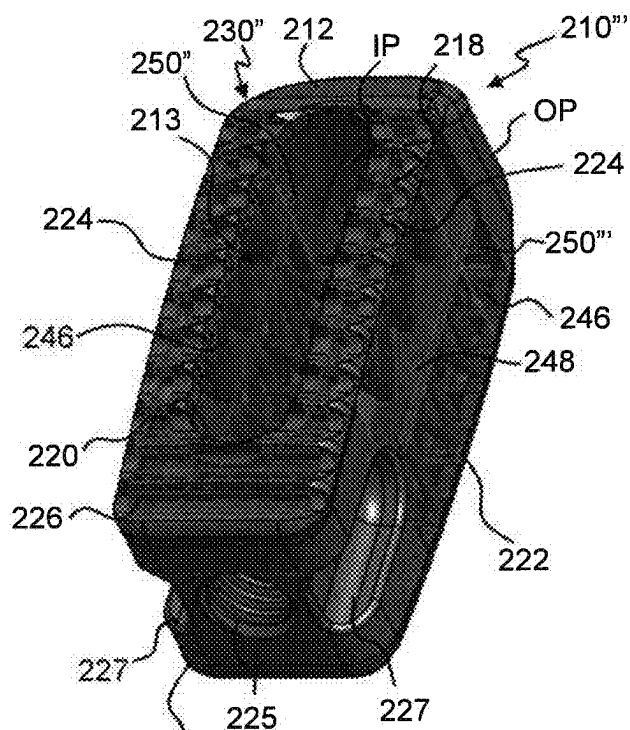


Fig. 54

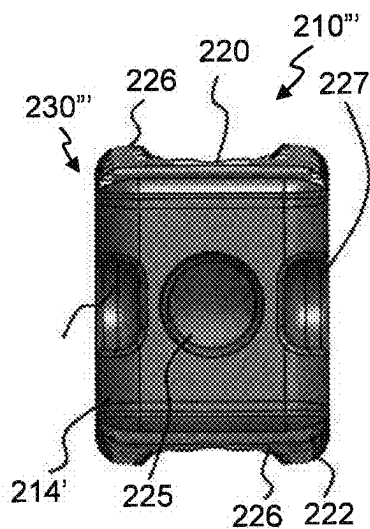


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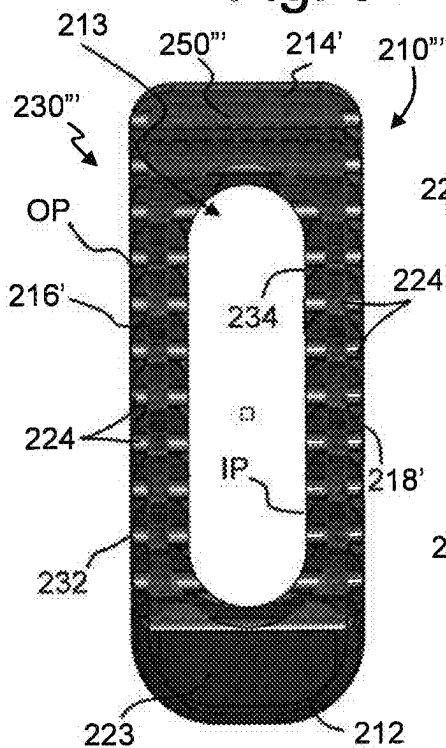


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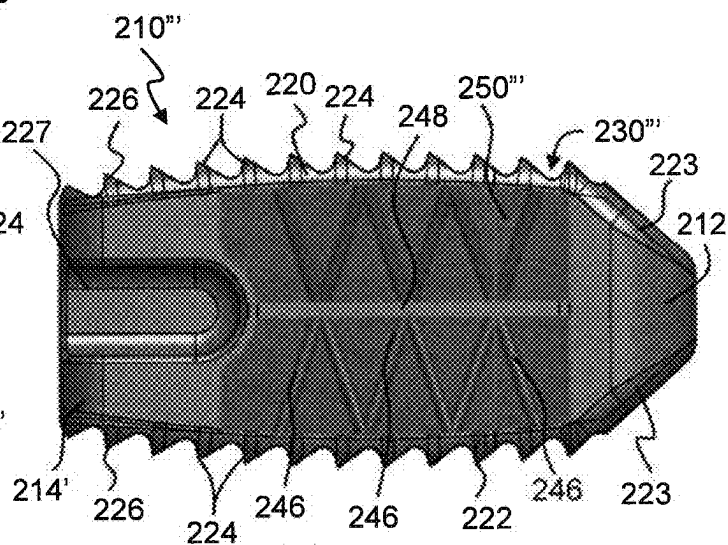
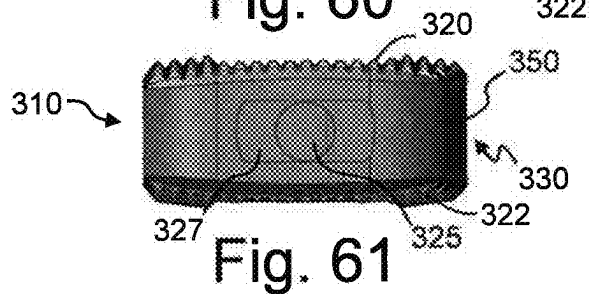
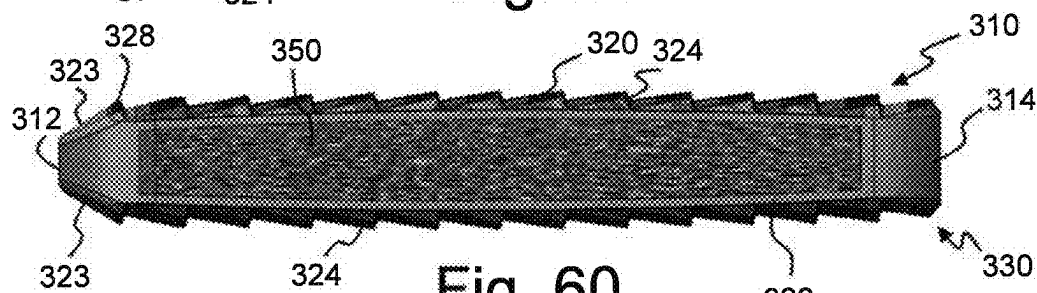
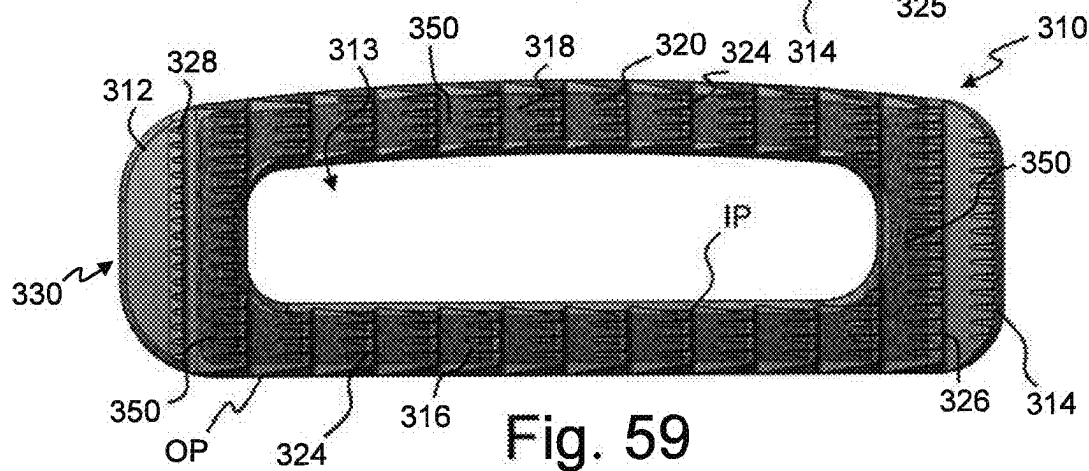
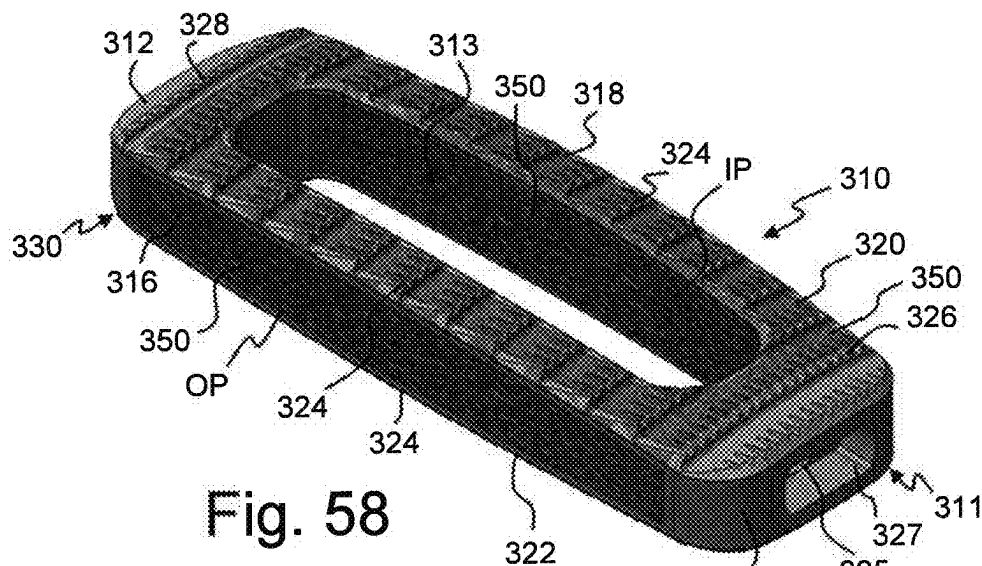
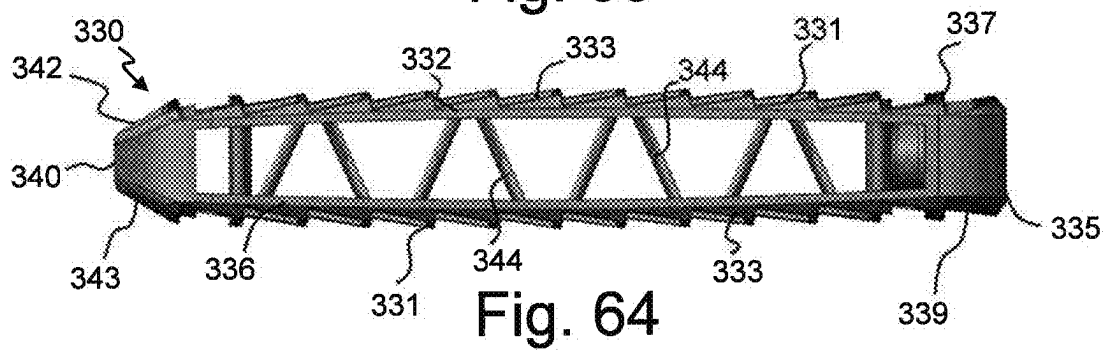
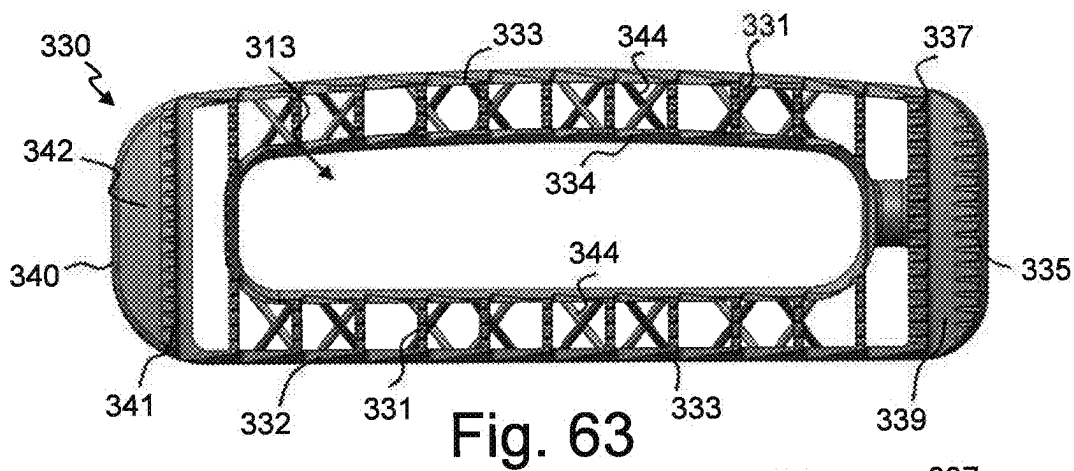
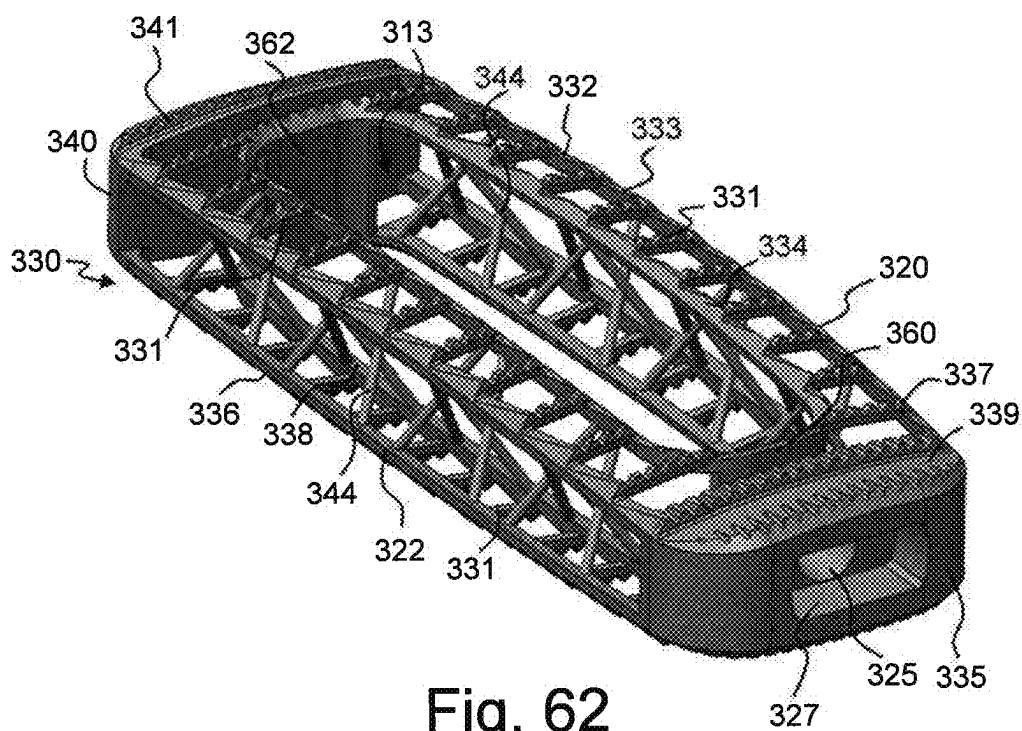


Fig. 57





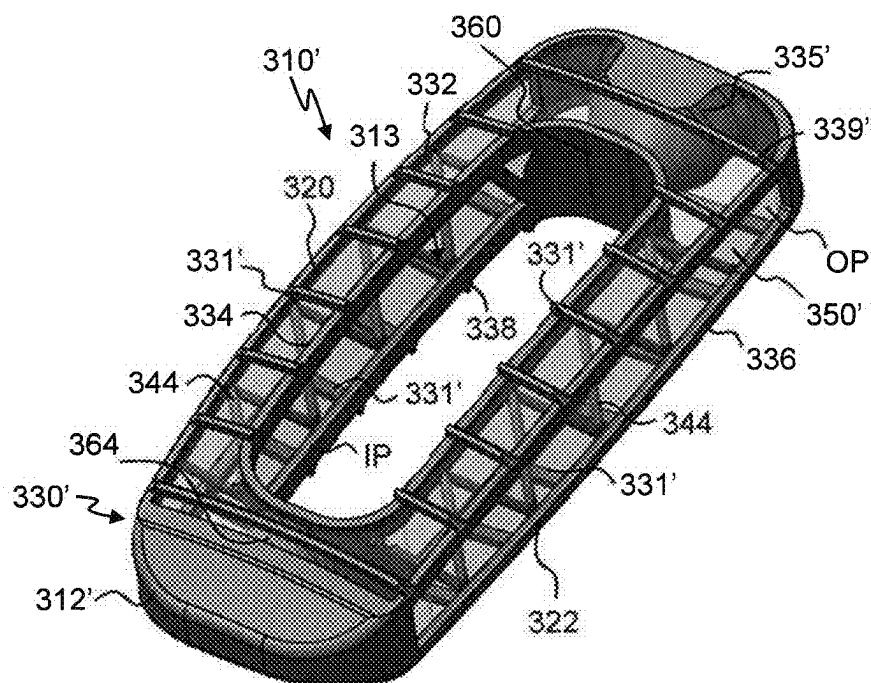


Fig. 65

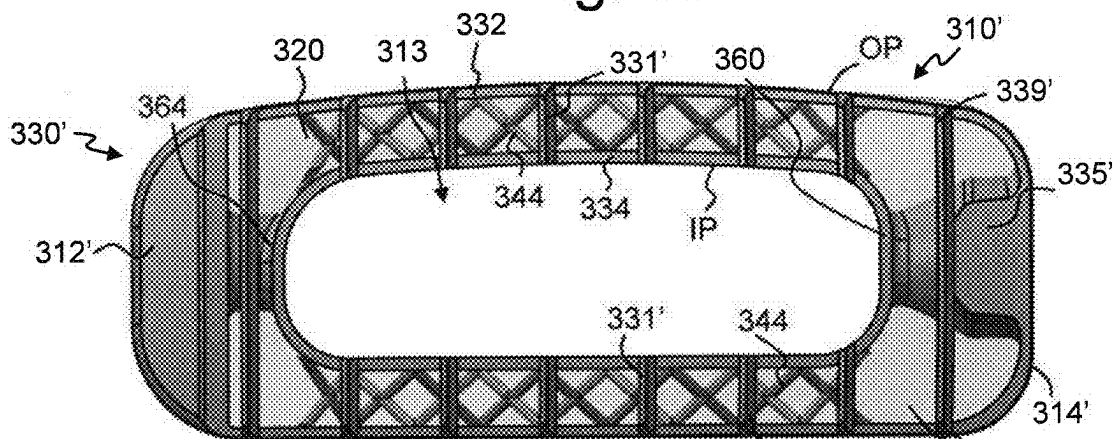


Fig. 66

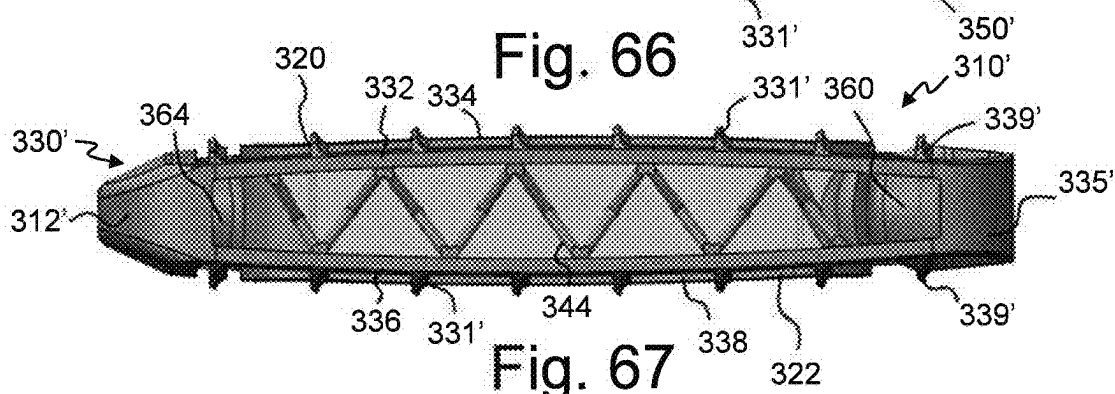


Fig. 67

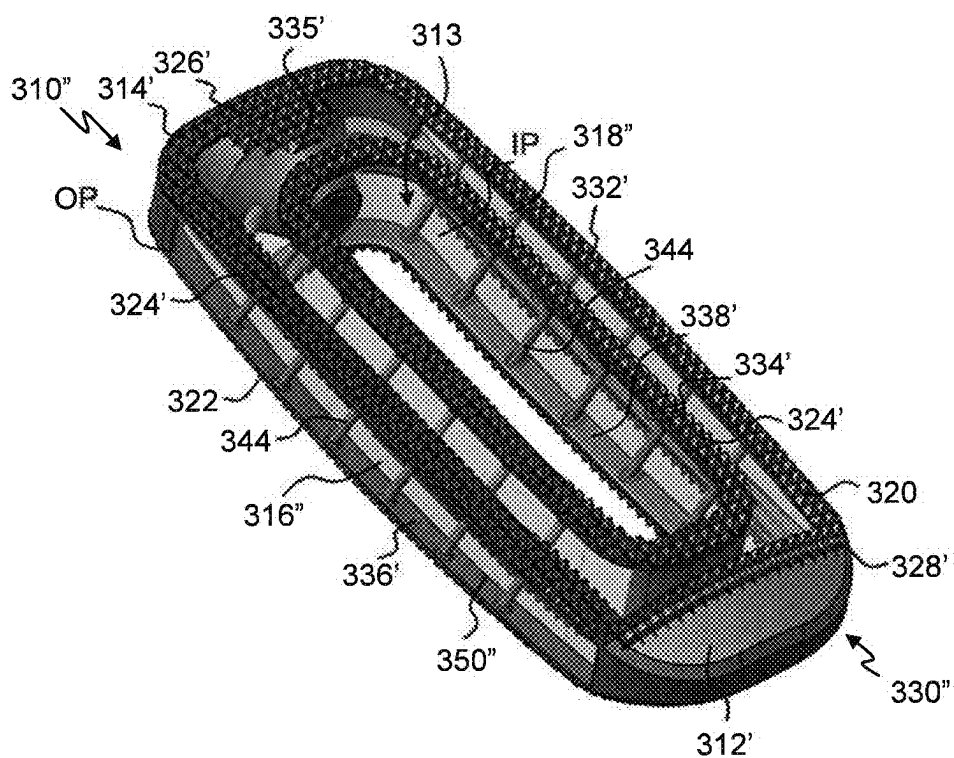


Fig. 68

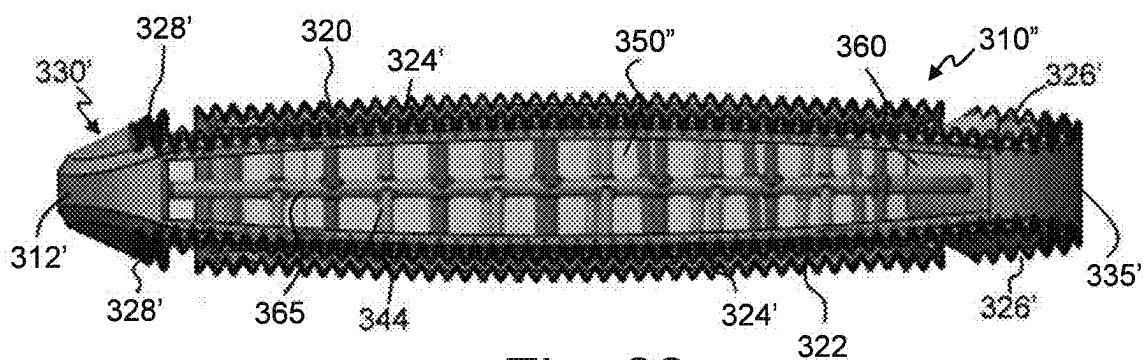


Fig. 69

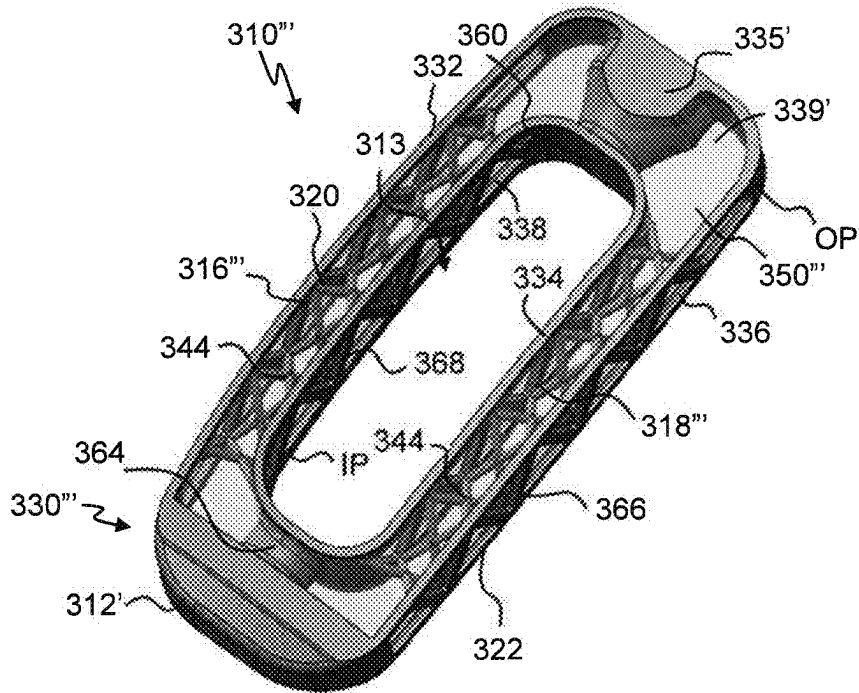


Fig. 70

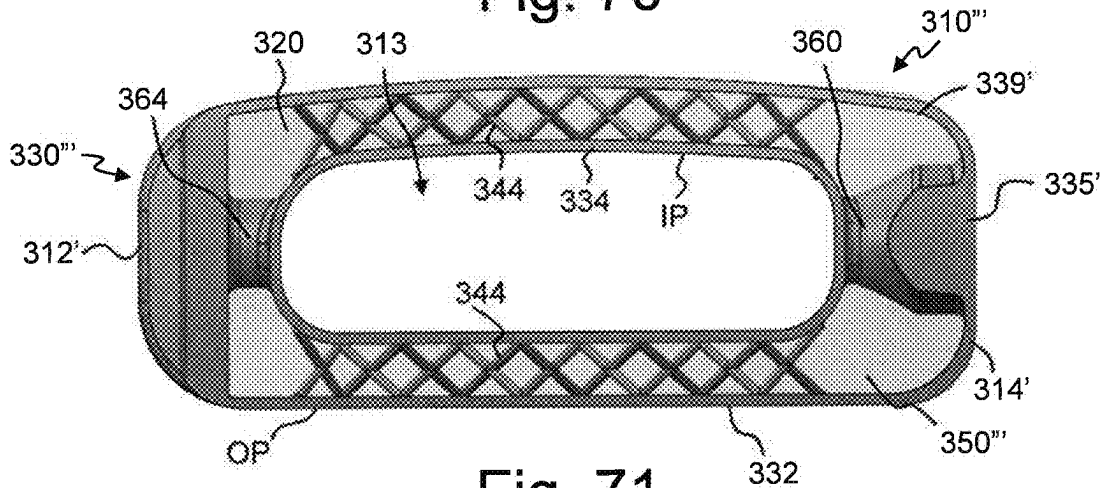


Fig. 71

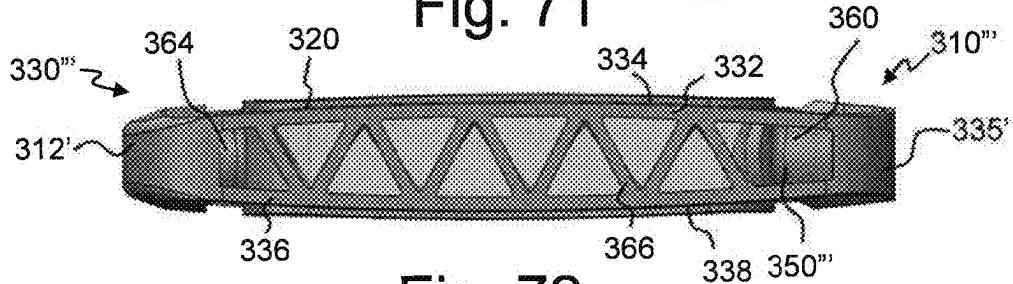


Fig. 72

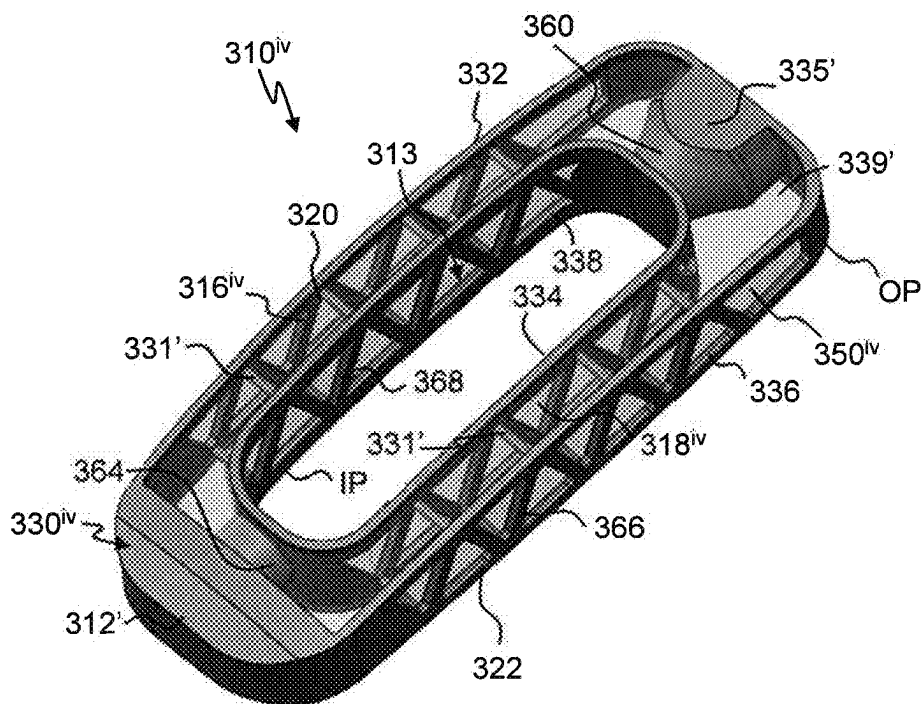


Fig. 73

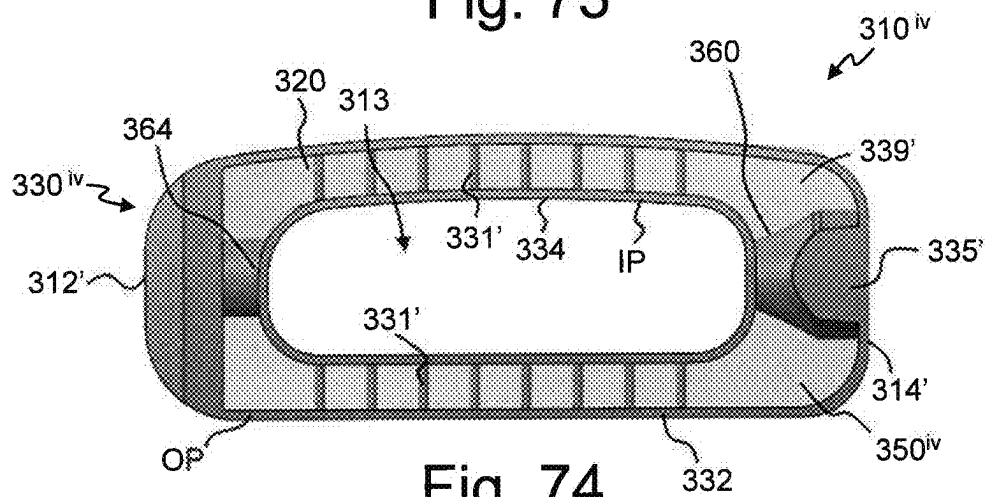


Fig. 74

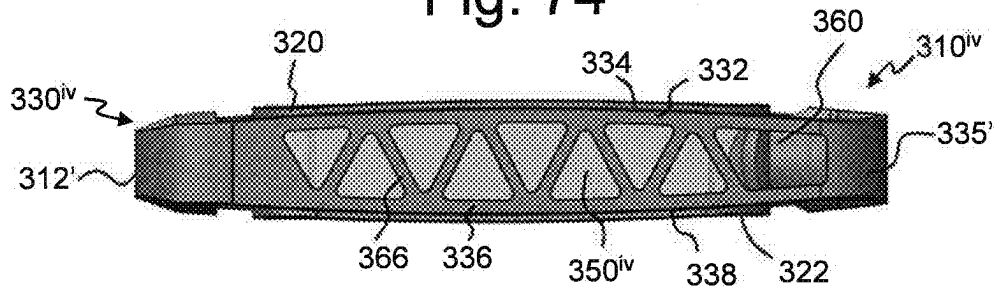


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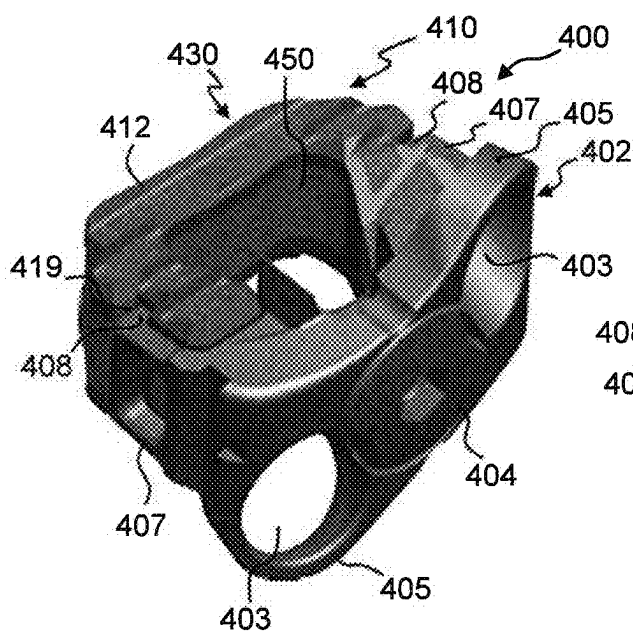


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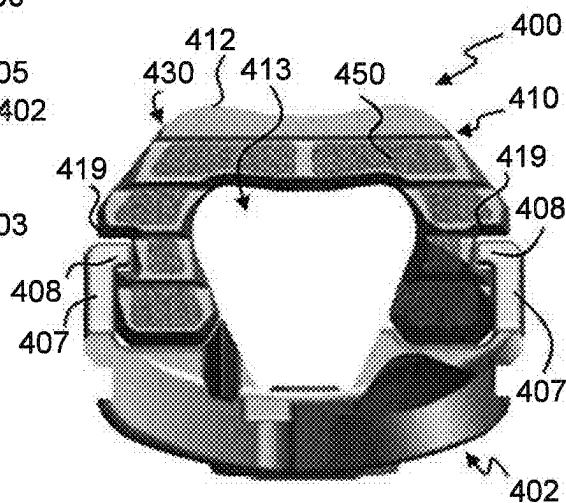


Fig. 77

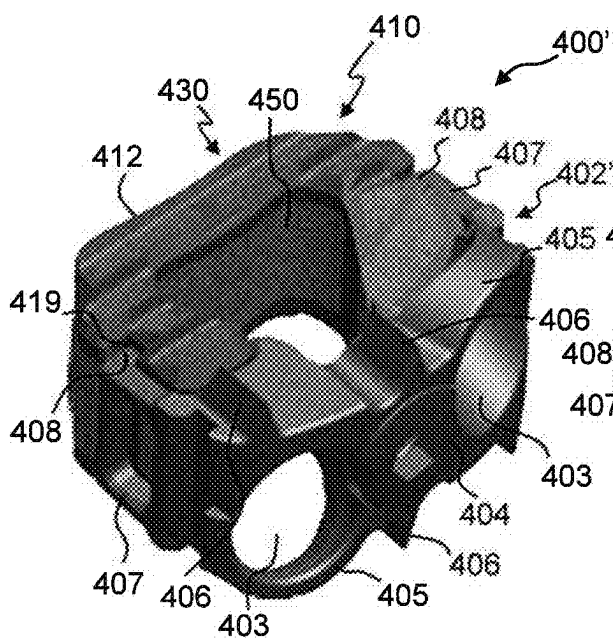


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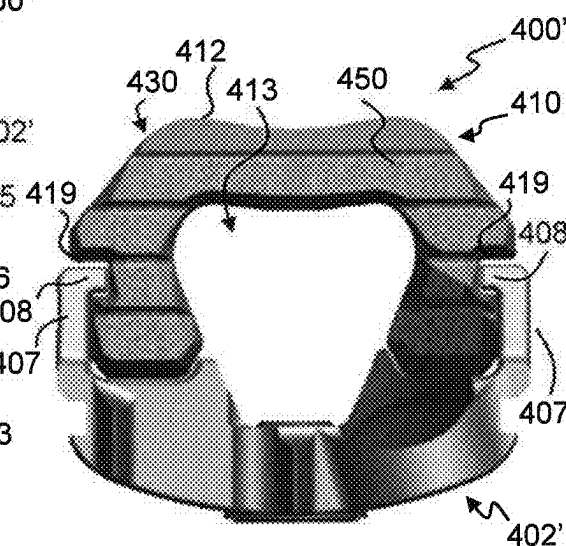


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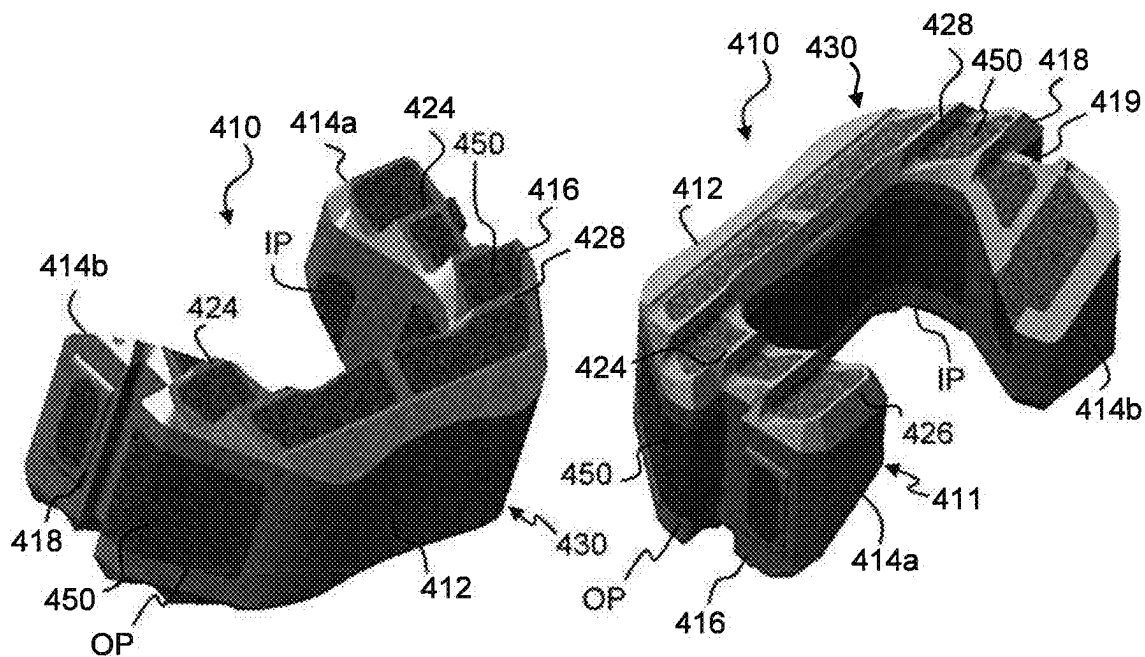


Fig. 80

Fig. 81

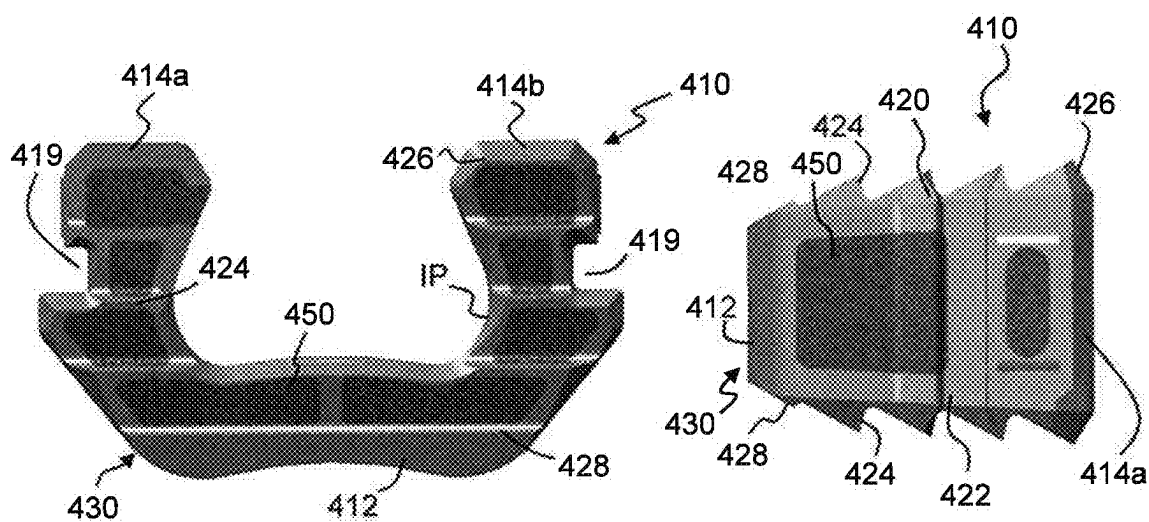


Fig. 82

Fig. 83

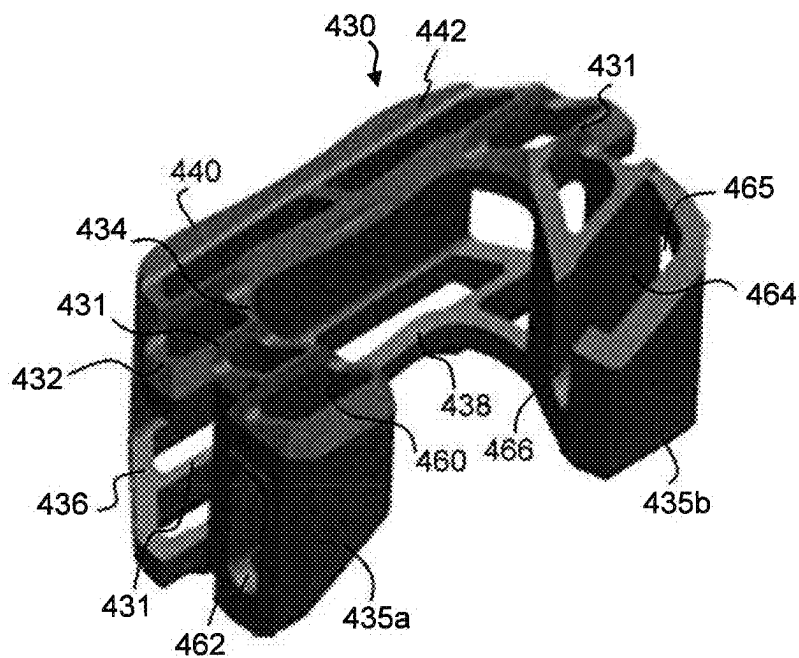


Fig. 84

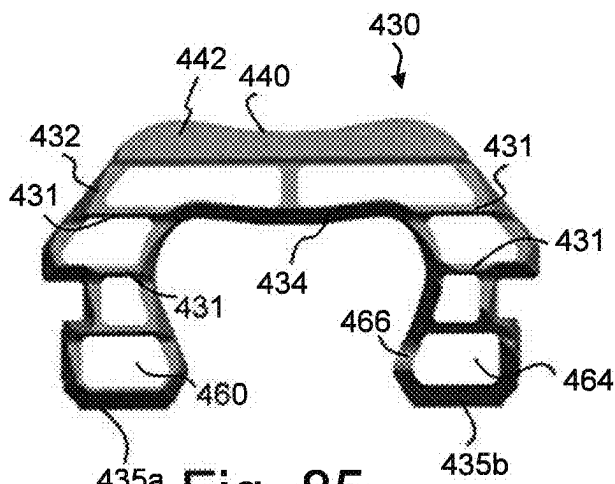


Fig. 85

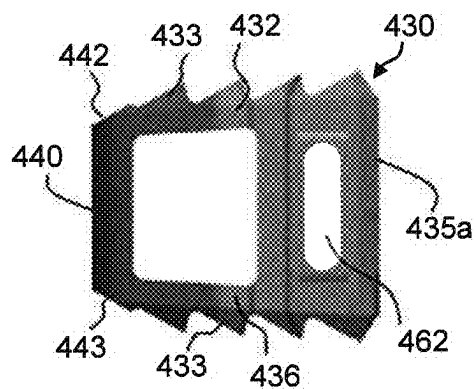


Fig. 86

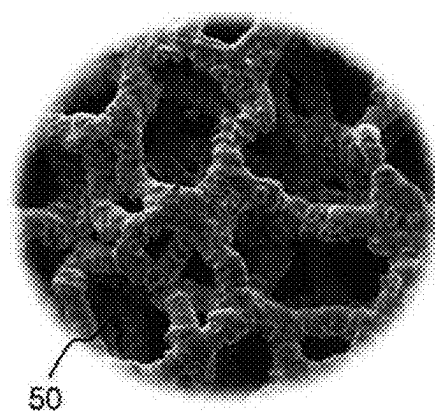


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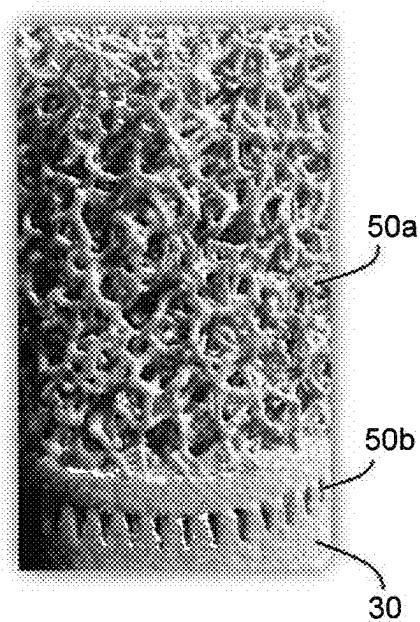


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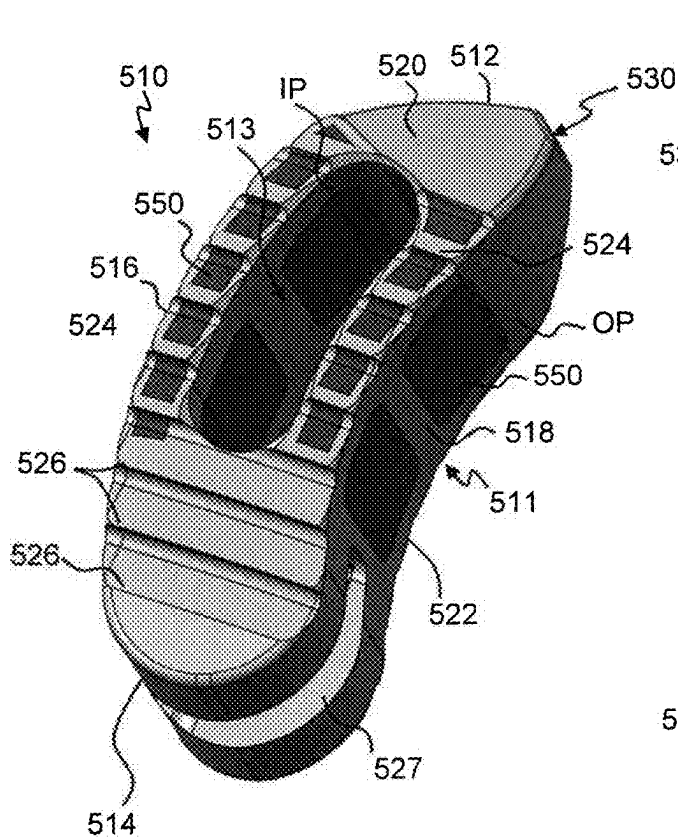


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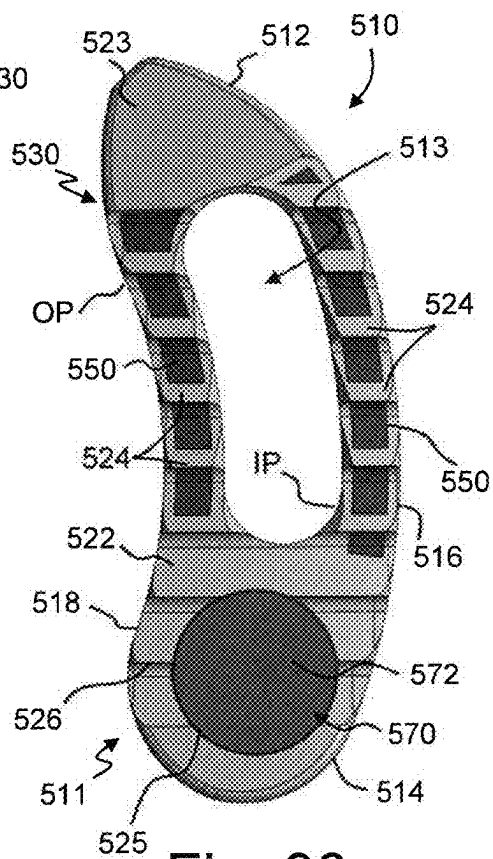


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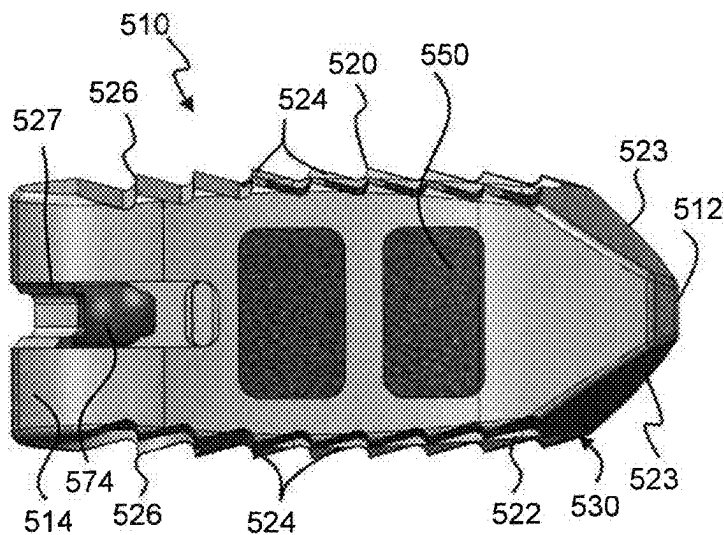


Fig. 91

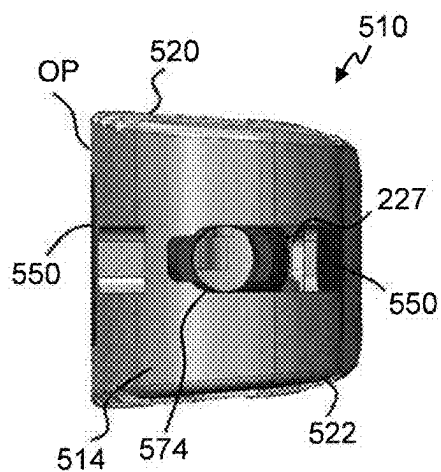


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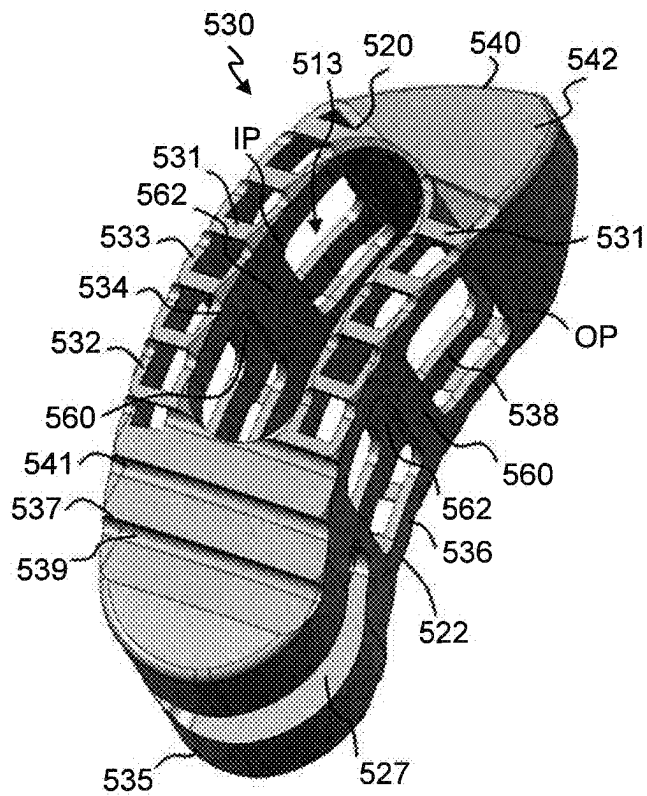


Fig. 93

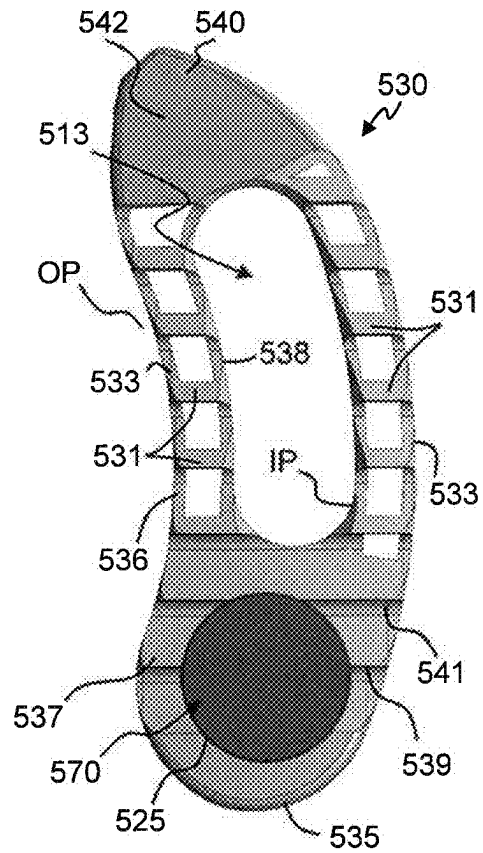


Fig. 94

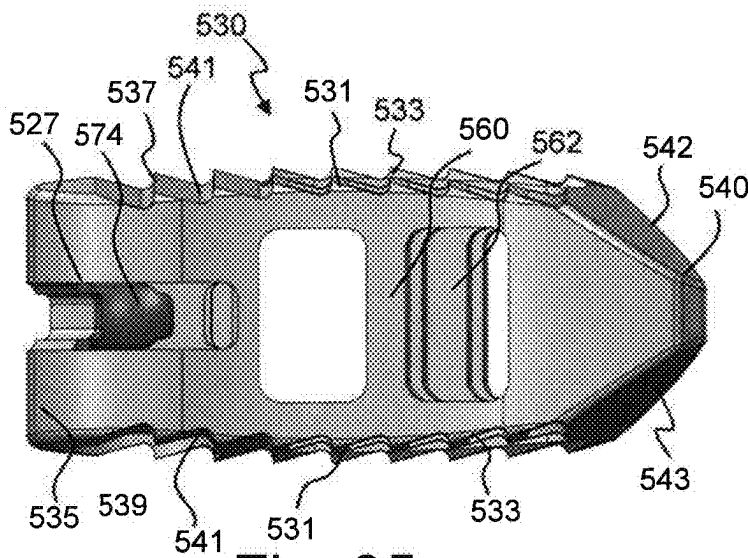


Fig. 95

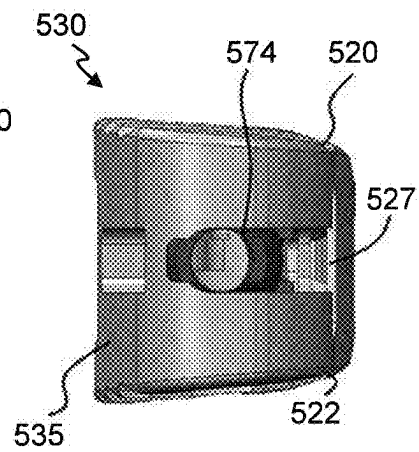


Fig. 96

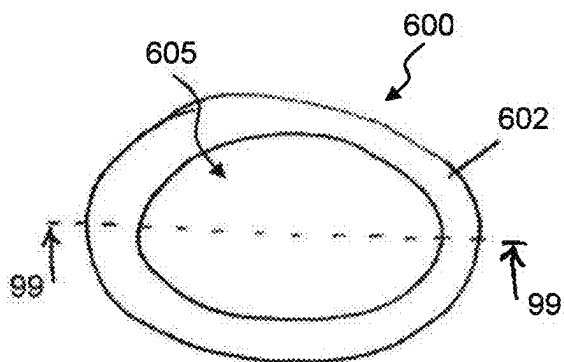


Fig. 97

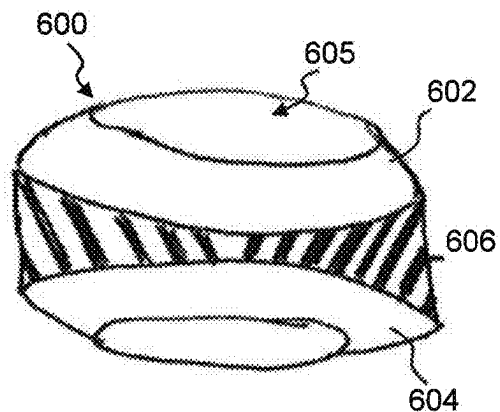


Fig. 98

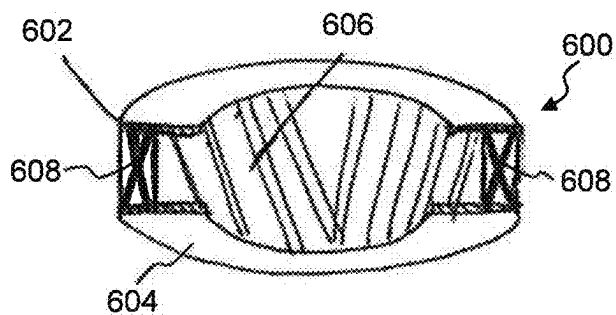


Fig. 99

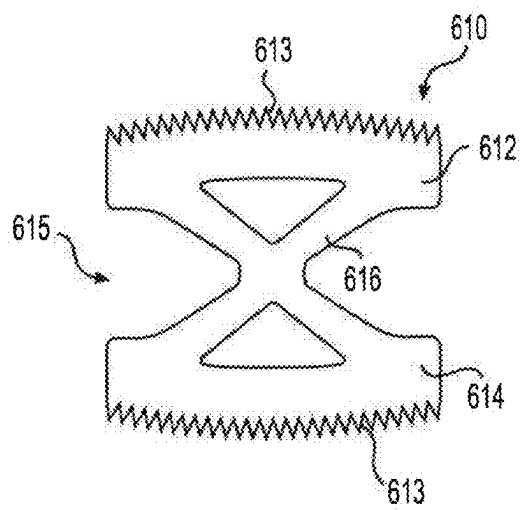


FIG. 100

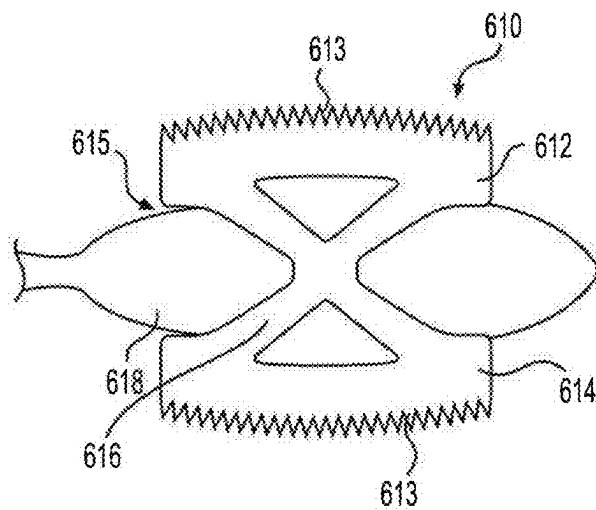


FIG. 101

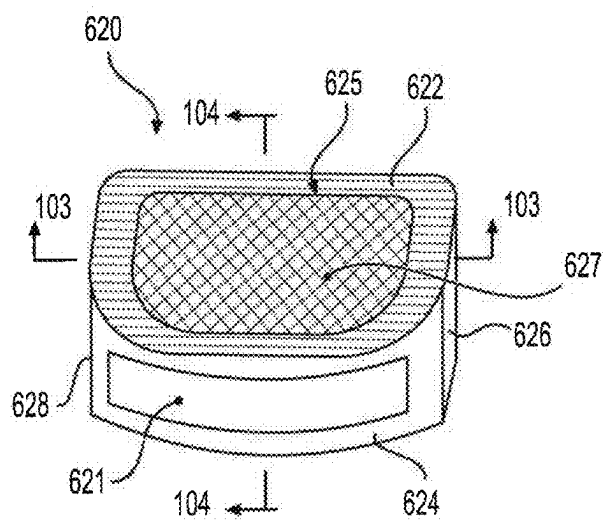


FIG. 102

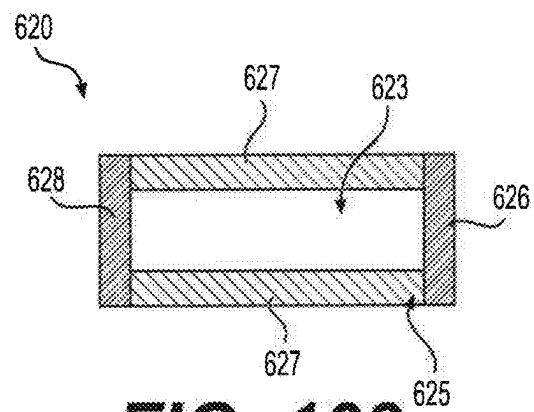


FIG. 103

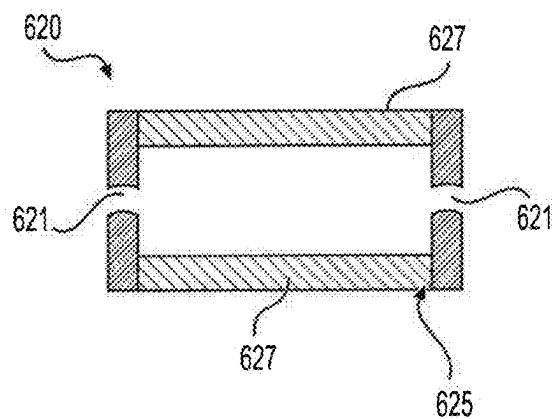


FIG. 104

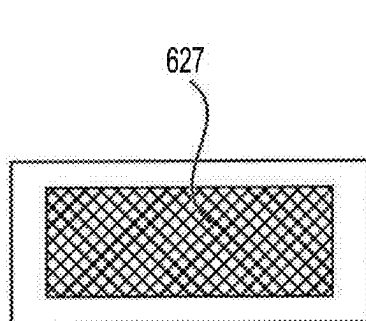


FIG. 105

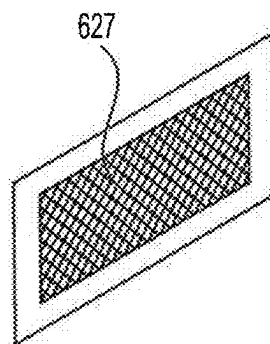


FIG. 106

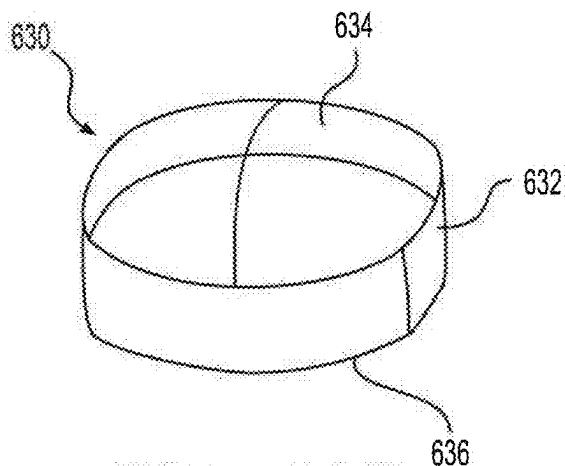


FIG. 107

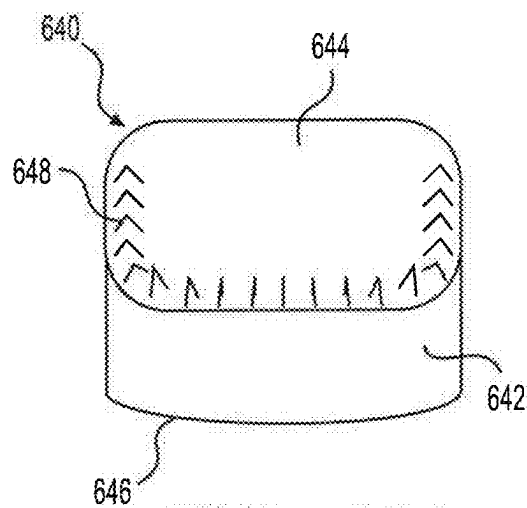


FIG. 108

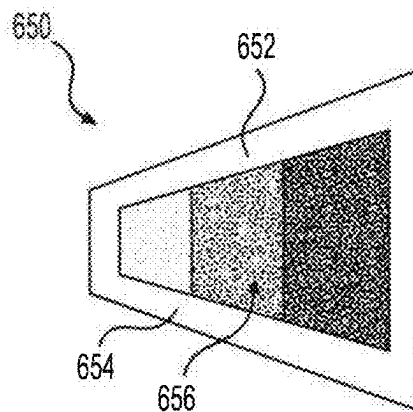


FIG. 109

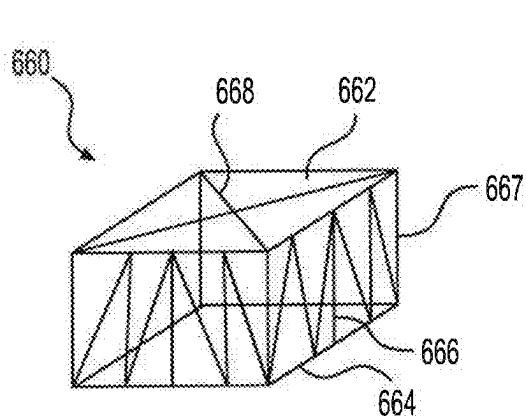


FIG. 110

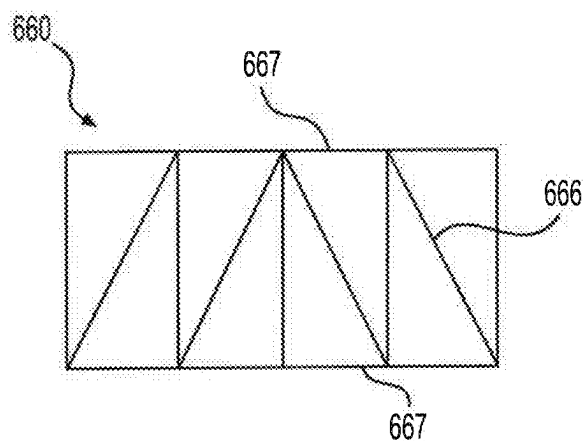


FIG. 111

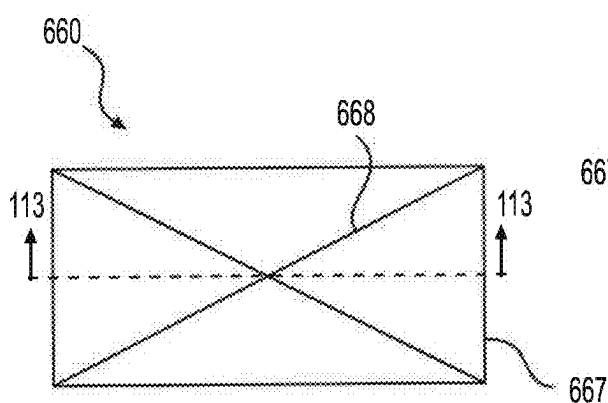


FIG. 112

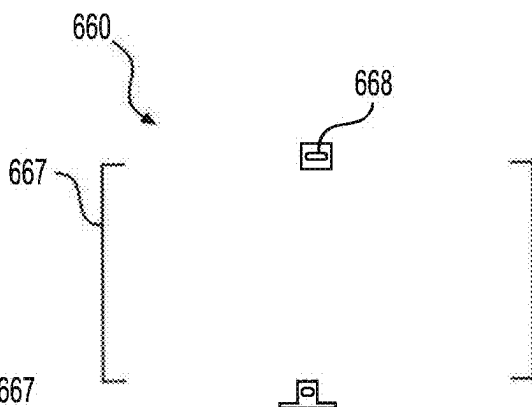


FIG. 113

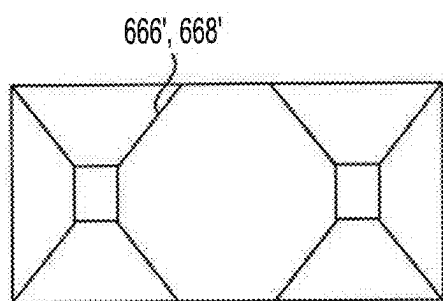


FIG. 114

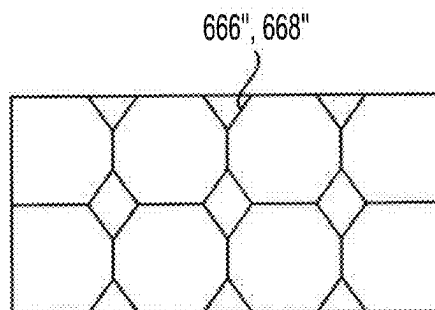


FIG. 115

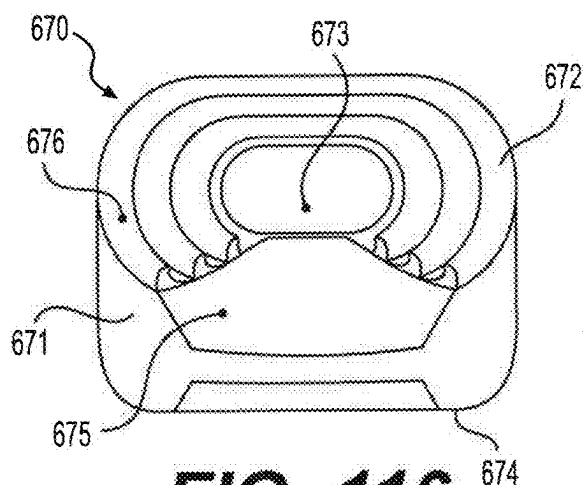


FIG. 116

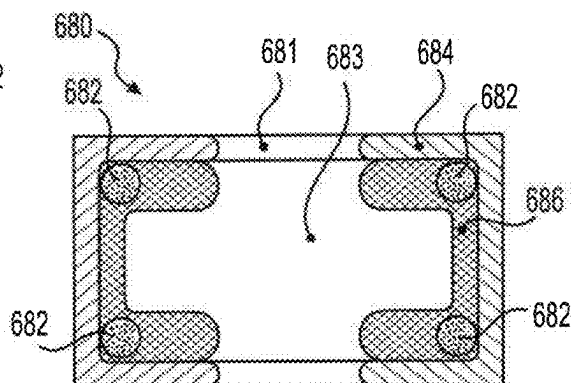


FIG. 117

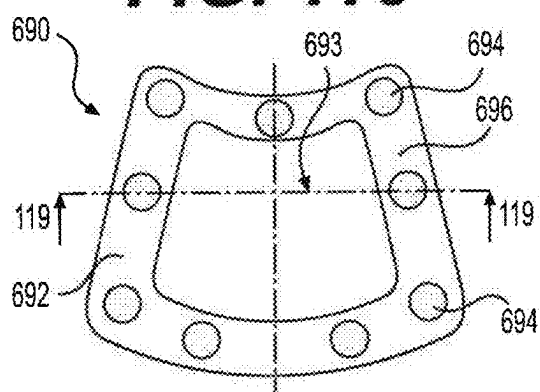


FIG. 118

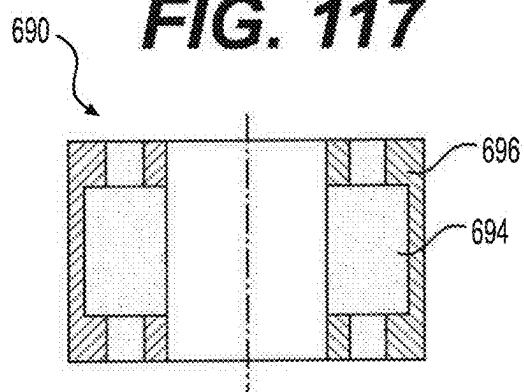


FIG. 119

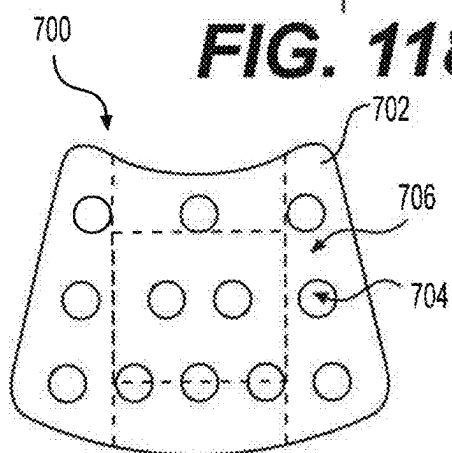


FIG. 120

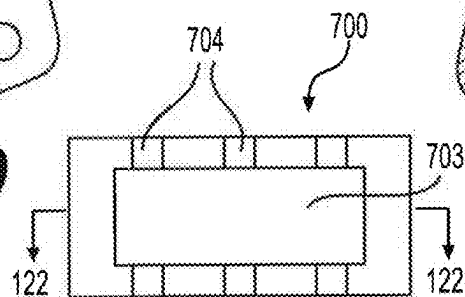


FIG. 121

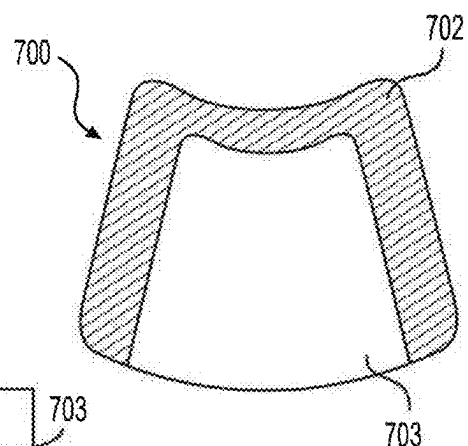


FIG. 122

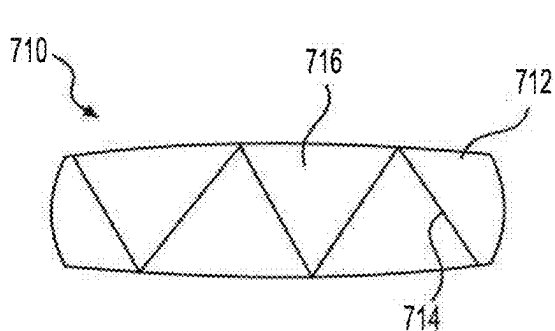


FIG. 123

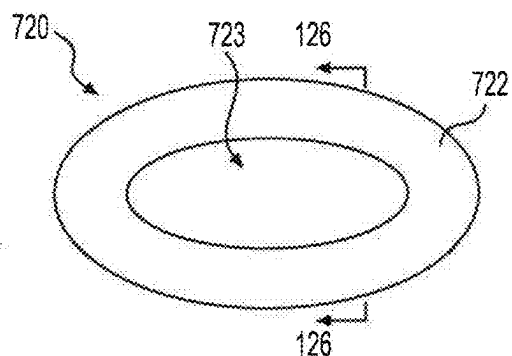


FIG. 124

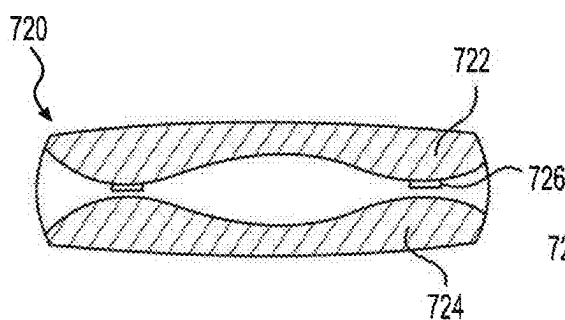


FIG. 125

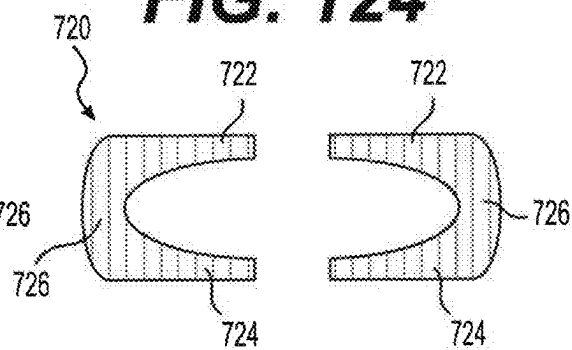


FIG. 126

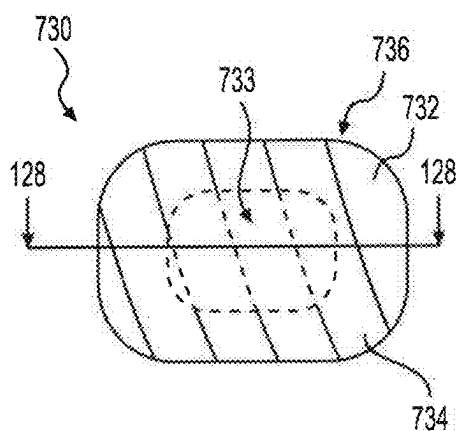


FIG. 127

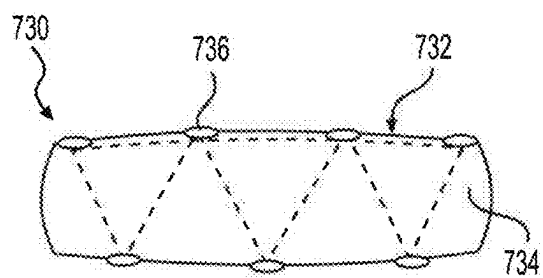


FIG. 128

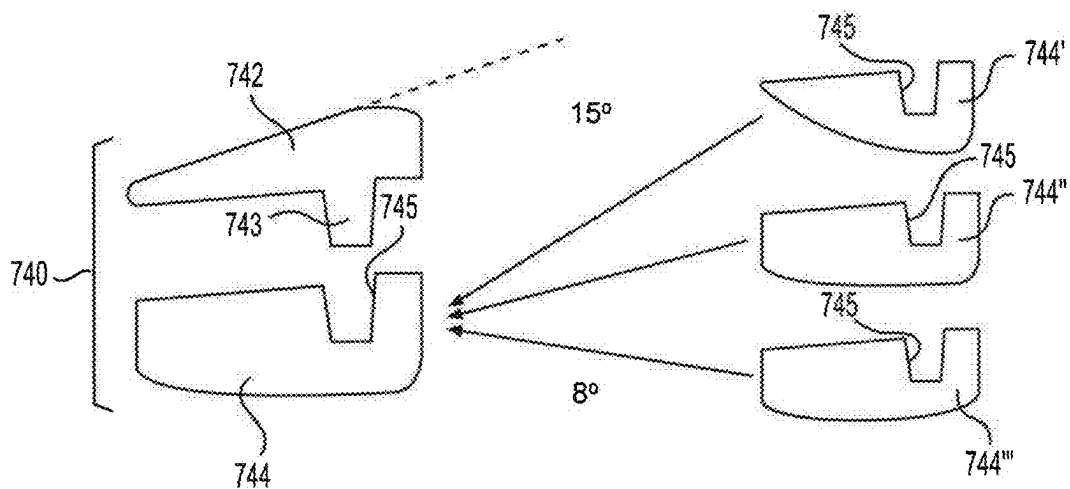


FIG. 129

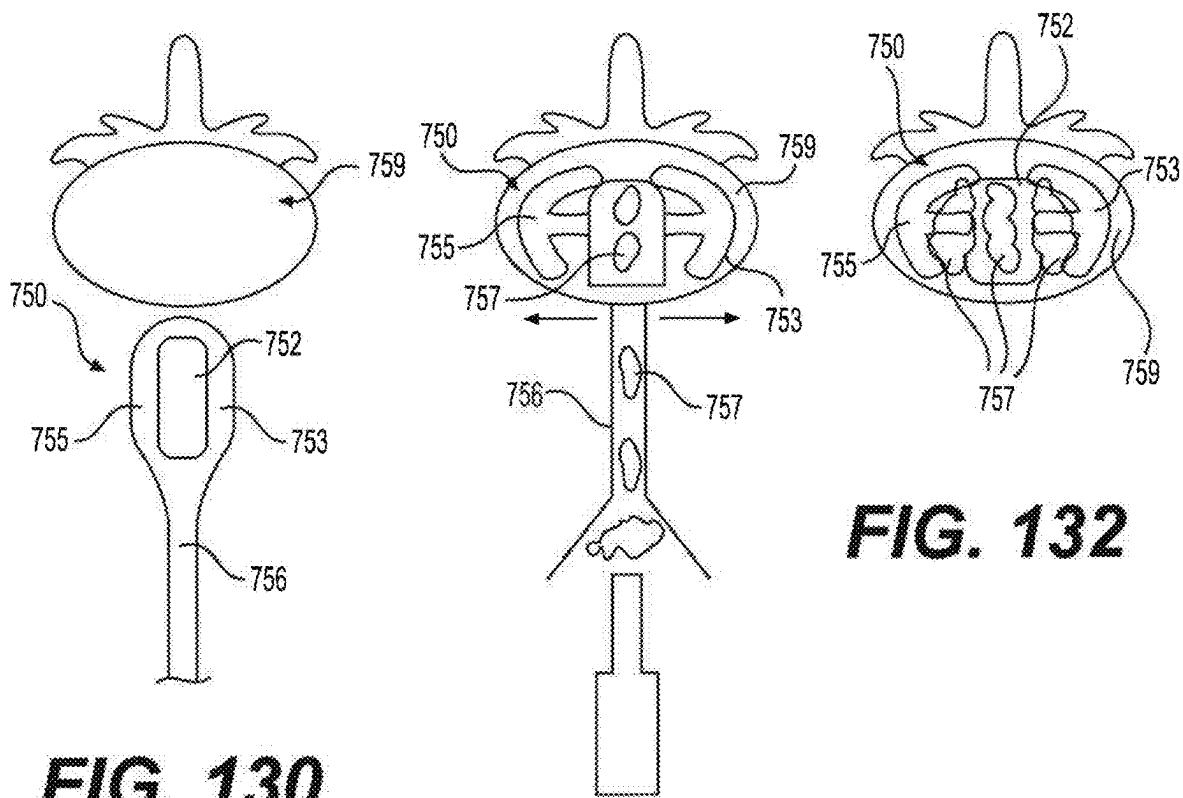


FIG. 130

FIG. 131

FIG. 132

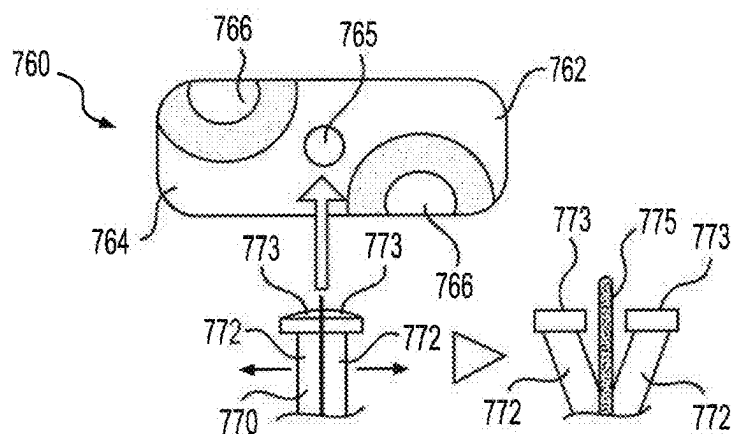


FIG. 133

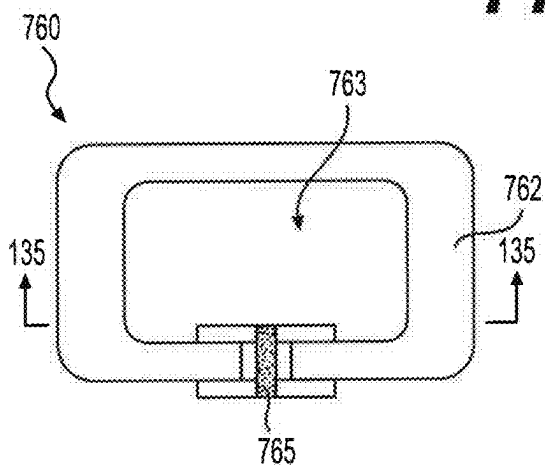


FIG. 134

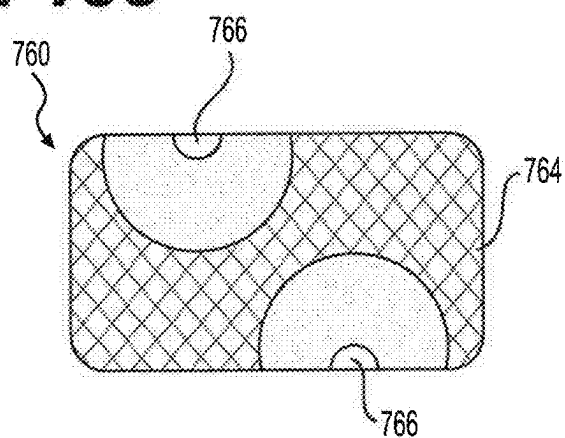


FIG. 135

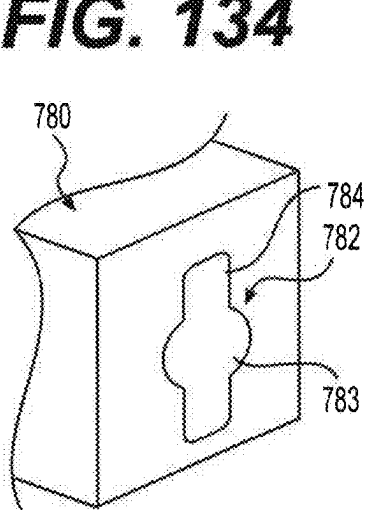


FIG. 136

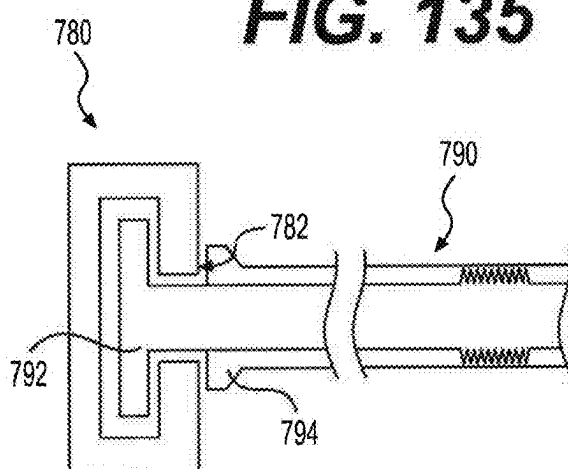


FIG. 137

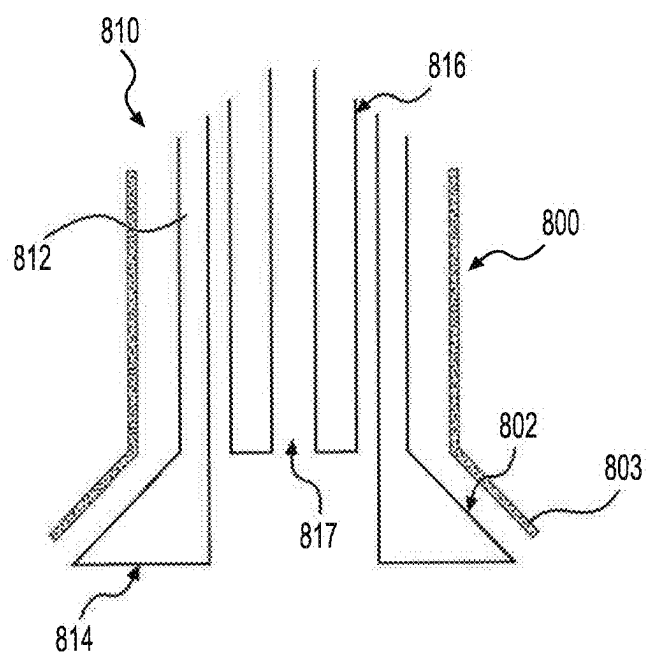
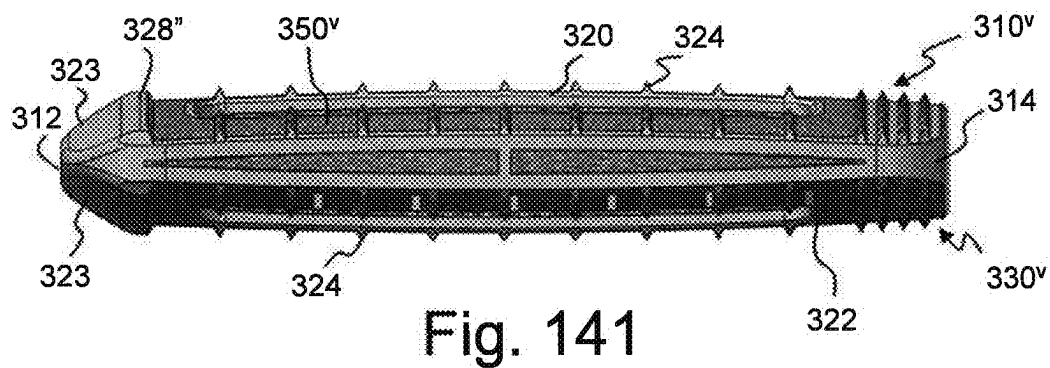
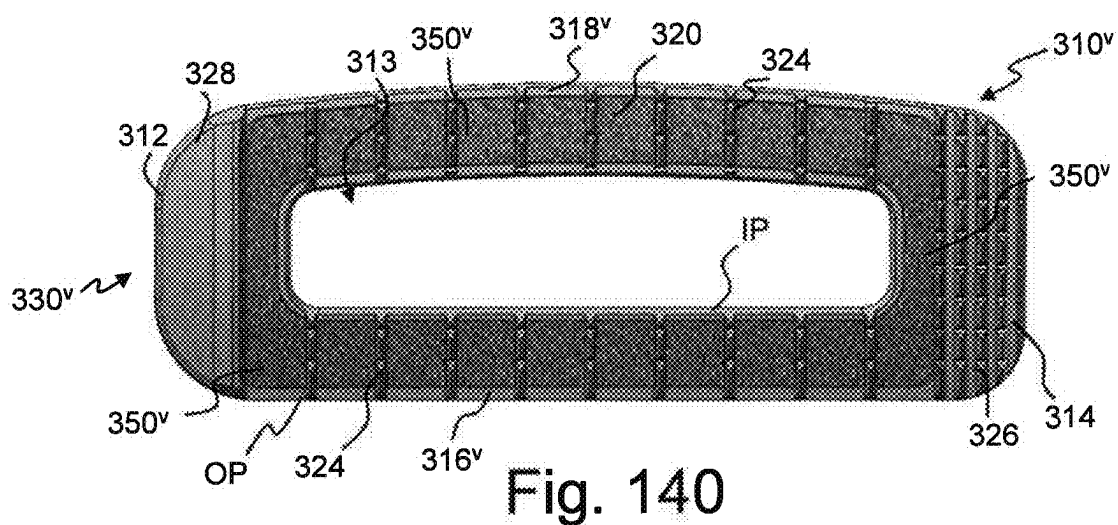
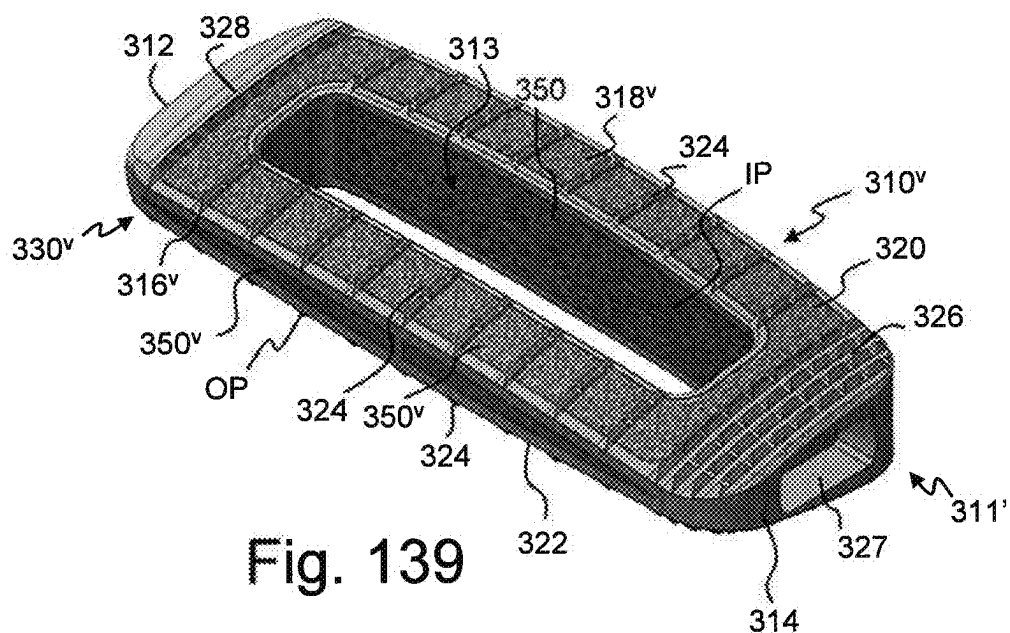


FIG. 138



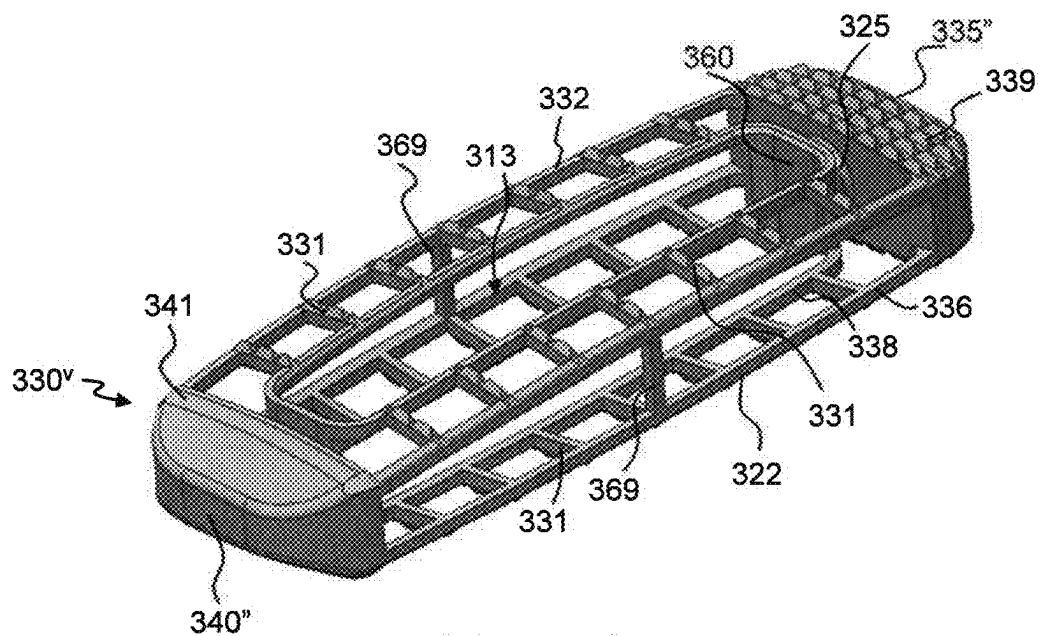


Fig. 142

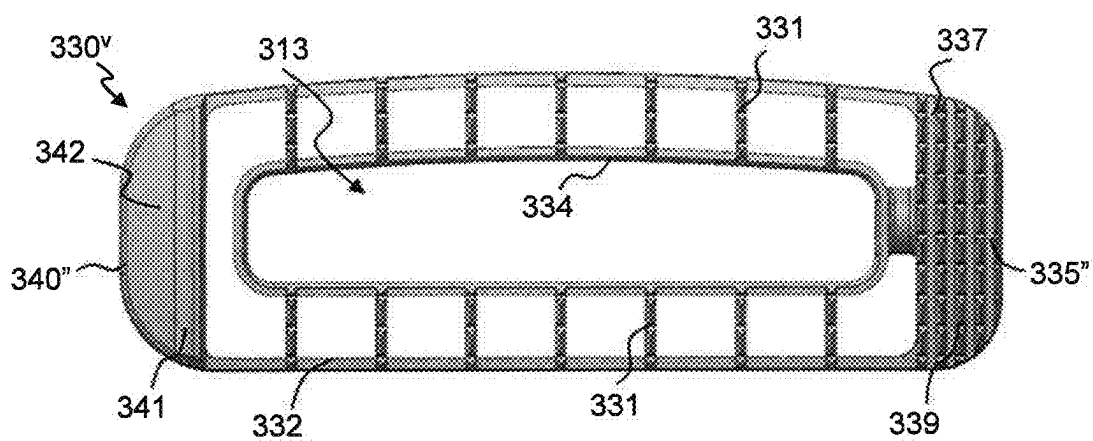


Fig. 143

INTERVERTEBRAL SPINAL IMPLANT

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 17/933,560, filed on Sep. 20, 2022, which is a continuation of U.S. patent application Ser. No. 16/600,632 filed on Oct. 14, 2019 (published as U.S. Pat. Pub. No. 2020-0038198), which is a continuation of U.S. patent application Ser. No. 15/973,756 filed on May 8, 2018, now U.S. Pat. No. 10,555,819, which is a continuation of U.S. patent application Ser. No. 15/973,609 filed May 8, 2018, now U.S. Pat. No. 11,141,286, all of which are incorporated by reference herein in their entireties for all purposes.

FIELD

[0002] The present disclosure generally relates to fixation devices and systems for positioning and immobilizing at least two adjacent vertebrae and methods related to the same. In particular, the present disclosure relates to interbody fusion devices with an integrated solid support structure and porous ingrowth structure.

BACKGROUND

[0003] The spine is the axis of the skeleton on which all of the body parts “hang”. In humans, the normal spine has seven cervical, twelve thoracic and five lumbar segments. The lumbar spine sits upon the sacrum, which then attaches to the pelvis, and in turn is supported by the hip and leg bones. The bony vertebral bodies of the spine are separated by intervertebral discs, which act as joints but allow known degrees of flexion, extension, lateral bending, and axial rotation.

[0004] The typical vertebra has a thick anterior bone mass called the vertebral body, with a neural (vertebral) arch that arises from the posterior surface of the vertebral body. The central of adjacent vertebrae are supported by intervertebral discs. The spinal disc and/or vertebral bodies may be displaced or damaged due to trauma, disease, degenerative defects, or wear over an extended period of time. One result of this displacement or damage to a spinal disc or vertebral body may be chronic back pain. In many cases, to alleviate back pain from degenerated or herniated discs, the disc is removed along with all or part of at least one neighboring vertebrae and is replaced by an implant that promotes fusion of the remaining bony anatomy.

[0005] However, the success or failure of spinal fusion may depend upon several factors. For instance, the spacer or implant or cage used to fill the space left by the removed disc and bony anatomy must be sufficiently strong to support the spine under a wide range of loading conditions. The spacer should also be configured so that it likely to remain in place once it has been positioned in the spine by the surgeon. Additionally, the material used for the spacer should be biocompatible material and should have a configuration that promotes bony ingrowth.

SUMMARY

[0006] To meet this and other needs, intervertebral implants for use with the anterior, antero-lateral, lateral, and/or posterior portions of at least one motion segment unit of the spine, systems, and methods are provided. Tradition-

ally, interbody spacers or implants intended to help facilitate intervertebral fusion may serve as a means to restore intervertebral height and/or lordosis. The implant may feature a central lumen to house bone graft material. It is through this central lumen where most of the fusion may occur. The implants of the disclosure incorporate a volumetric, interconnected porosity throughout the entire spacer. This enables bone to grow into and/or through the spacer, making it part of the fusion mass. The incorporation of a volumetric, interconnected porosity within the implant may encourage faster, stronger intervertebral fusion.

[0007] According to one embodiment, a cervical intervertebral implant for implantation in an intervertebral space between vertebrae is provided. The implant includes a body having a front end, a rear end and opposed side walls extending between the ends. The body has an outer perimeter with a trapezoidal shape and an inner perimeter about an internal chamber. The body includes an upper surface and a lower surface with the upper surface defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface defined by a solid lower outer rim and a spaced apart solid lower inner rim. A solid front wall extends at the front end between at least the solid upper outer rim and the solid lower outer rim. A solid rear wall extends at the rear end between at least the solid upper outer rim and the solid lower outer rim. Each of the side walls includes at least one solid cross strut extending between the solid upper rims and at least one solid cross strut extending between the solid lower rims. Each of the side walls includes at least one solid support structure extending between the upper surface and the lower surface with the solid support structure occupying a minimal space within each side wall. A porous structure is integrally formed with the solid upper rims, the solid lower rims, each of the solid cross struts, and each of the solid structures in each of the side walls and extends from the body outer perimeter to the body inner perimeter.

[0008] According to another embodiment, an intervertebral implant for implantation in an intervertebral space between vertebrae is provided. The implant includes a body having a front end, a rear end and opposed side walls extending between the ends. The body has an outer perimeter and an inner perimeter about an internal chamber. The body includes an upper surface and a lower surface. The upper surface is defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface is defined by a solid lower outer rim and a spaced apart solid lower inner rim. A solid front wall extends at the front end between at least the solid upper outer rim and the solid lower outer rim. A solid rear wall extends at the rear end between at least the solid upper outer rim and the solid lower outer rim. Each of the side walls includes at least one solid support structure extending between the upper and lower surfaces. A porous structure is integrally formed with the solid upper rims, the solid lower rims and the at least one solid support structure in each of the side walls. The porous structure extends from the body outer perimeter to the body inner perimeter while the solid upper and lower outer rims and the solid front and rear walls extend along the outer perimeter such that the porous structure is encased within solid structure.

[0009] According to yet another embodiment, a cervical intervertebral implant for implantation in an intervertebral space between vertebrae is provided. The implant includes a body extending from an upper surface to a lower surface.

The body has a front end, a rear end and a pair of spaced apart first and second side walls extending between the front and rear walls such that an interior chamber is defined within the front and rear ends and the first and second walls. The body defines an outer perimeter with a trapezoidal shape and an inner perimeter extending about the internal chamber. At least one of the walls is defined by a solid support structure and an integral porous structure. The porous structure extends from the outer perimeter to the inner perimeter and from the upper surface to the lower surface and encapsulates at least a portion of the solid support structure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] A more complete understanding of the present disclosure, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:

[0011] FIGS. 1-4 are perspective, side, top and rear views, respectively, of an intervertebral implant according to one embodiment of the disclosure with the porous portions shown textured;

[0012] FIGS. 5-7 are perspective, top and side views, respectively, of the intervertebral implant of FIGS. 1-4 with the porous portions removed to show the support structure;

[0013] FIGS. 8-10 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0014] FIGS. 11-13 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0015] FIGS. 14-16 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0016] FIGS. 17-19 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0017] FIGS. 20-22 and 24 are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured, and FIG. 23 is a cross-sectional view along the lines 23-23 in FIG. 21;

[0018] FIGS. 25-27 are perspective, top and side views, respectively, of the intervertebral implant of FIGS. 20-24 with the porous portions removed to show the support structure;

[0019] FIGS. 28-31 are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0020] FIGS. 32-34 are perspective, top and side views, respectively, of the intervertebral implant of FIGS. 28-31 with the porous portions removed to show the support structure;

[0021] FIGS. 35 and 36 are perspective and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0022] FIGS. 37 and 38 are rear and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0023] FIG. 39 is a perspective view of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0024] FIGS. 40-43 are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0025] FIGS. 44-47 are perspective, top, side and rear views, respectively, of the intervertebral implant of FIGS. 40-43 with the porous portions removed to show the support structure;

[0026] FIGS. 48-51 are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0027] FIGS. 52 and 53 are perspective and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0028] FIGS. 54-57 are perspective, rear, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0029] FIGS. 58-61 are perspective, top, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0030] FIGS. 62-64 are perspective, top and side views, respectively, of the intervertebral implant of FIGS. 58-61 with the porous portions removed to show the support structure;

[0031] FIGS. 65-67 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0032] FIGS. 68 and 69 are perspective and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0033] FIGS. 70-72 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0034] FIGS. 73-75 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown translucently;

[0035] FIGS. 76 and 77 are perspective and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0036] FIGS. 78 and 79 are perspective and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0037] FIGS. 80-83 are front perspective, rear perspective, top and side views, respectively, of the spacer portion of the implants of FIGS. 76-79 with the porous portions shown textured;

[0038] FIGS. 84-86 are perspective, top and side views, respectively, of the spacer portion of FIGS. 80-83 with the porous portions removed to show the support structure; and [0039] FIGS. 87 and 88 are illustrative photos of various porous structures in accordance with embodiments of the disclosure;

[0040] FIGS. 89-92 are perspective, bottom, side and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured;

[0041] FIGS. 93-96 are perspective, bottom, side and rear views, respectively, of the intervertebral implant of FIGS. 89-92 with the porous portions removed to show the support structure;

[0042] FIGS. 97 and 98 are top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 99 is a cross-sectional view along the line 99-99 in FIG. 97;

[0043] FIGS. 100 and 101 are side views of an intervertebral implant according to another embodiment of the disclosure, with FIG. 101 illustrating an insertion tool extending through the implant;

[0044] FIG. 102 is a perspective view of an intervertebral implant according to another embodiment of the disclosure, FIG. 103 is a cross-sectional view along the line 103-103 in FIG. 102 and FIG. 104 is a cross-sectional view along the line 104-104 in FIG. 102;

[0045] FIGS. 105 and 106 are front and perspective views, respectively, illustrating a grid porous configuration;

[0046] FIG. 107 is a perspective view of an intervertebral implant according to another embodiment of the disclosure;

[0047] FIG. 108 is a perspective view of an intervertebral implant according to another embodiment of the disclosure;

[0048] FIG. 109 is a side view of an intervertebral implant according to another embodiment of the disclosure;

[0049] FIGS. 110-112 are perspective, side and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 113 is a cross-sectional view along the line 113-113 in FIG. 112;

[0050] FIGS. 114 and 115 illustrate alternative strut patterns of an illustrative support structure;

[0051] FIG. 116 is a perspective view of an intervertebral implant according to another embodiment of the disclosure;

[0052] FIG. 117 is a cross-sectional view of an intervertebral implant according to another embodiment of the disclosure;

[0053] FIG. 118 is a top view of an intervertebral implant according to another embodiment of the disclosure and FIG. 119 is a cross-sectional view along the line 119-119 in FIG. 118;

[0054] FIGS. 120 and 121 are top and rear views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 122 is a cross-sectional view along the line 122-122 in FIG. 121;

[0055] FIG. 123 is a side view of an intervertebral implant according to another embodiment of the disclosure;

[0056] FIGS. 124 and 125 are top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 126 is a cross-sectional view along the line 126-126 in FIG. 124;

[0057] FIG. 127 is a side view of an intervertebral implant according to another embodiment of the disclosure and FIG. 128 is a cross-sectional view along the line 128-128 in FIG. 127;

[0058] FIG. 129 is an exploded side view of an intervertebral implant according to another embodiment of the disclosure;

[0059] FIGS. 130-132 are top views showing sequentially implantation of an expandable intervertebral implant according to another embodiment of the disclosure;

[0060] FIGS. 133 and 134 are front and top views, respectively, of an intervertebral implant according to another embodiment of the disclosure and FIG. 135 is a cross-sectional view along the line 135-135 in FIG. 134;

[0061] FIG. 136 is a perspective view illustrating an example tool hole and FIG. 137 is a cross-sectional view illustrating a tool engaged in such a hole;

[0062] FIG. 138 is a schematic view of an example delivery tool in accordance with an embodiment of the disclosure;

[0063] FIGS. 139-141 are perspective, top and side views, respectively, of an intervertebral implant according to another embodiment of the disclosure with the porous portions shown textured; and

[0064] FIGS. 142 and 143 are perspective and top views, respectively, of the intervertebral implant of FIGS. 139-141 with the porous portions removed to show the support structure.

DETAILED DESCRIPTION

[0065] Embodiments of the disclosure are generally directed to intervertebral implants, systems, and method of use thereof. The implant may be suitable for use with the anterior, antero-lateral, lateral, and/or posterior portions of at least one motion segment unit of the spine. Traditionally, interbody spacers or implants intended to help facilitate intervertebral fusion may serve as a means to restore intervertebral height and/or lordosis. The implants may feature a central lumen to house bone graft material, for example. It is through this central lumen where most of the fusion may occur. The implants of the disclosure may incorporate a volumetric, interconnected porosity throughout the entire spacer or a portion thereof. This enables bone to grow into and/or through the spacer or a portion thereof, making it part of the fusion mass. The incorporation of a volumetric, interconnected porosity may encourage faster, stronger intervertebral fusion, thereby providing for better patient outcomes.

[0066] Various forms of additive manufacturing, or 3D printing, have been developed which allow structures to be formed layer by layer. One illustrative 3D printing technology is Direct Metal Laser Sintering (DMLS) wherein parts are built using a laser to selectively sinter (heat and fuse) a powdered metal material into layers. The process begins once a 3D CAD file is mathematically sliced into multiple 2D cross sections and uploaded into the system. After the first layer is produced, the build platform is lowered, another powder layer is spread across the plate, and the laser sinters the second layer. This process is repeated until the part is complete. Layer-by-layer manufacturing allows for the direct fabrication of complex parts that would be cost-prohibitive, and often impossible, to produce through traditional manufacturing processes. The powder layer thickness used during the fabrication of the spacers may be as thin as 30 μm . The resolution of the laser may be as fine as 70 μm . Although it is envisioned that any suitable thickness or laser resolution may be used or selected.

[0067] The disclosure is not limited to DMLS, but various 3D printing methods may be utilized. For example, VAT Photopolymerization utilizes a vat of liquid photopolymer resin which is cured through selective exposure to light (via a laser or projector) which then initiates polymerization and converts the exposed areas to a solid part. As another example, Powder Bed Fusion, of which DMLS is a subcategory, utilizes powdered materials which are selectively consolidated by melting it together using a heat source such as a laser or electron beam. The powder surrounding the consolidated part acts as support material for overhanging features.

[0068] As yet another example, in Binder Jetting Liquid bonding agents are selectively applied onto thin layers of powdered material to build up parts layer by layer. The binders include organic and inorganic materials. Metal or ceramic powdered parts are typically fired in a furnace after they are printed. Material Jetting is another example of a 3D printing process which may be utilized wherein droplets of material are deposited layer by layer to make parts. Common varieties include jetting a photocurable resin and curing it with UV light, as well as jetting thermally molten materials that then solidify in ambient temperatures. As another example, in Sheet Lamination sheets of material are stacked and laminated together to form an object. The lamination method can be adhesives or chemical (paper/plastics), ultrasonic welding, or brazing (metals). Unneeded regions are cut out layer by layer and removed after the object is built. Another example of a 3D printing process that may be utilized is Material Extrusion wherein material is extruded through a nozzle or orifice in tracks or beads, which are then combined into multi-layer models. Common varieties include heated thermoplastic extrusion and syringe dispensing. Yet another example is Directed Energy Deposition wherein powder or wire is fed into a melt pool which has been generated on the surface of the part where it adheres to the underlying part or layers by using an energy source such as a laser or electron beam.

[0069] The implants of the disclosure may be manufactured from any of these or other additive manufacturing processes currently known or later developed. The implants may also be manufactured utilizing a combination of additive manufacturing processes and other manufacturing processes, for example, laser etching. Additionally, the implants may be further processed during and/or after manufacture utilizing various techniques, for example, abrasion, machining, polishing, or chemical treatment. The implants may be manufactured from various materials, such as biocompatible materials, including metals, polymers, ceramics or combinations thereof. Exemplary materials include Titanium (and Titanium alloys), Cobalt-Chrome, PEEK, and/or Stainless Steel, for example.

[0070] As will be discussed in more detail hereinafter, the implants of the disclosure generally comprise a solid support structure and a porous structure formed integral therewith. The solid support structure may include solid front and rear walls interconnected by upper and lower implant surfaces. The upper and lower surfaces may include spaced apart rims with cross struts interconnecting the rims. In many embodiments, the solid support structure of the upper and lower surfaces includes a plurality of openings in which the integral porous structure is formed such that the porous structure extends along at least a portion of the upper and lower implant surfaces. The side walls extending between

the front and rear walls generally have a minimal solid structure, for example, a plurality of struts extending between the upper and lower rims, but otherwise have open area therebetween in which the integral porous structure is formed. The configuration of the solid structure is selected to provide the implant sufficient structural integrity and mechanical stability while maximizing the area of porous structure which facilitates better integration/incorporation with the adjacent bone. In several embodiments of the disclosure, the solid structure generally encases the corners of the porous structure or otherwise houses the porous structure therein to maintain the structural integrity of the porous structure.

[0071] Referring now to FIGS. 1-7, one embodiment of a cervical intervertebral implant **10** will be described. As illustrated, the implant **10** has a body **11** with a generally trapezoidal shape. The body **11** is defined by a tapered front end **12**, a rectangular rear end **14** and side walls **16** and **18** extending therebetween. The implant **10** has an outer perimeter **OP** extending about the body **11**. A hollow interior chamber **13** is defined within an inner perimeter **IP** of the body **11**. The hollow interior chamber **13** is configured to receive bone growth promoting materials, for example. The implant **10** has an upper surface **20** and a lower surface **22**, with both surfaces having a tapering portion **23** at the front end **12**. The upper and lower surfaces **20**, **22** may be substantially parallel or otherwise configured to provide the proper intervertebral spacing. The upper and lower surfaces **20**, **22** define a plurality of serrations **24** along the side walls **16**, **18** and a plurality of serrations **26** along the rear end **14**. The serrations **24**, **26** are defined by both the solid support structure **30** and the porous structure **50**. As will be described in detail hereinafter, the solid support structure **30** includes spaced apart rims **32**, **34** and **36**, **38** with cross struts **31** and **37**. The solid support structure **30** defines open spaces or recesses adjacent the cross struts **31**, **37** and the porous structure **50** is formed within such open spaces such that the solid structure **30** and the porous structure **50** together form the serrations **24**, **26**. As illustrated in FIGS. 1-4, the porous structure **50** extends to and forms a portion of the implant upper and lower surfaces **20-22**. The rear end **14** of the implant **10** includes a hole **25** and a pair of blind slots **27** for receiving an instrument that is used for inserting the implant **10**. As seen in FIGS. 1-4, the implant **10** is defined by a solid support structure **30** with an interfiled, integral porous structure **50**.

[0072] The solid support structure **30** will be described in more detail with reference to FIGS. 5-7. An outer rim **32** extends about the outer perimeter **OP** of the upper surface **20** and an inner rim **34** extends about the inner perimeter **IP** of the upper surface **20**, i.e. about the interior chamber **13**. Similarly, an outer rim **36** extends about the outer perimeter **OP** of the lower surface **22** and an inner rim **38** extends about the inner perimeter **IP** of the lower surface **22**. A plurality of cross struts **31** extend between the outer rims **32**, **36** and the respective inner rims **34**, **38** along the side wall areas. As seen in the figures, the cross struts **31** along with contoured portions **33** of the rims **32**, **34**, **36**, **38** define the contour of the serrations **24**. In addition to interconnecting the rims within a given upper or lower surface, struts **44**, **46** and **48** extend within each side wall area to interconnect the upper rims **32**, **34** with the lower rims **36**, **38**. In the illustrated embodiment, a first strut **44** extends from the lower inner rim **38** to the upper outer rim **32** near the rear portion of the

support structure 30, a second strut 46 extends from the lower inner rim 38 to the upper outer rim 32 near the front portion of the support structure 30 and an X-shaped strut 48 extends between both lower rims 36, 38 and both upper rims 32, 34 at a central location of the support structure 30. As can be seen in FIGS. 6 and 7, each of the first struts 44 extends from the lower inner rim 38 proximate the rear wall 35 at an angle to approximately the midpoint of the upper outer rim 32, substantially tangent to the curvature of the inner rims 34, 38. Similarly, each of the second struts 46 extends from the lower inner rim 38 proximate the front wall 40 at an angle to approximately the midpoint of the upper outer rim 32, substantially tangent to the curvature of the inner rims 34, 38. Each of the X-shaped struts 48 extends substantially parallel to the upper and lower rims and positioned at the point where the first and second struts 44, 46 meet with the upper outer rim 36. The struts may have other configurations and more or fewer struts may be utilized.

[0073] The solid rear wall 35 additionally interconnects the outer rims 32, 36 and the respective inner rims 34, 38 along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall 35 defines the hole 25 and slots 27. Recessed areas 39 and 41 on the upper and lower sides of the rear wall 35 define receiving areas for porous structure, as seen in FIGS. 1-4. Cross members 37 in this area along with contours of the outer rims 32, 36 define the serrations 26. The solid front wall 40 has a concave configuration and also interconnects the outer rims 32, 36 and the respective inner rims 34, 38 along the front end area. The front wall 40 includes an upper sloped portion 42 extending between the upper outer rim 32 and inner rim 34 and a lower sloped portion 43 extending between the lower outer rim 36 and inner rim 38. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0074] In the illustrations of the support structure 30 shown in FIGS. 5-7 with the porosity omitted for illustration, it is seen that there is significant open space between the upper rims 32, 34 and the lower rims 36, 38 with only the struts 44, 46, 48 therebetween. The struts 44, 46, 48 occupy only a minimal space between the upper and lower rims 32, 34, 36, 38, for example, less than 50% of the wall space, thereby leaving substantial open space for the porous structure 50. Additionally, there is open space between the inside surface of the front wall 40 and the inner rims 34, 38. Furthermore, there is open space on an inside surface and the recesses 39, 41 of the rear wall 35. As illustrated in FIGS. 1-4, in the implant 10, these open spaces are filled with the porous structure 50 such that the porous structure 50 encapsulates the struts 44, 46, 48 and extends from the upper surface 20 to the lower surface 22 and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 50 substantially defines the inner perimeter IP and defines a substantial portion of the side walls 16, 18 along the outer perimeter OP.

[0075] The configuration of the support structure 30 and the porous structure 50 are selected, for example, to provide the implant with an adequate construct strength while maximizing the potential for bony in-growth and allowing for clear radiographic imaging. Referring to FIGS. 87 and 88, the porous structure 50 may have a randomized pattern of open pores 50a or a repeating pattern of open pores 50b. The

porous structure 50 may have a suitable porosity (open volume). For example, the porous structure 50 may be greater than 50% open, greater than 60% open, greater than 70% open, or approximately 70% open, or approximately 75% open. The porous structure 50 may feature interconnected pores or open pores. The porous structure 50 may have pores, for example, ranging from approximately 100 μm -2 mm, approximately 100 μm -1 mm, approximately 200-900 μm , or approximately 300-800 μm in diameter. The pore size may have an average pore size of about 300-800 μm , about 400-700 μm , or about 500-600 μm . The pore size distribution may be unimodal or bi-modal. Although spherical or partially-spherical pores or nodes are exemplified in forming the porous structure, it is envisioned that other suitable pore shapes and configurations may be used, for example, repeating or random patterns of cylinders, cubes, cones, pyramids, polyhedrons, or the like.

[0076] It is contemplated that different areas of the support structure 30 may have varying stiffness or strength, for example, variable A-P stiffness to achieve optimized load on an anterior graft or to achieve a desired level of flexibility within the implant 10. Furthermore, the porous structure 50 may have different porosities or densities in different areas of the implant 10. For example, the porous structure 50 may have a higher porosity or density along the inner perimeter compared to that at the outer perimeter, for example, with the inner area having a cancellous porosity and the outer area having a cortical porosity. The porous structure 50 may have various configurations, for example, a grid or honeycomb pattern which may promote bony in-growth. Additionally, the porous structure 50 may be configured such that when it is turned past a critical angle it may appear opaque, thereby helping with assessment of the implant orientation or positioning. The surface texture of both the support structure and the porous structure may be controlled to provide both macro and micro texturizing. The features and characteristics described with respect to this embodiment may be incorporated in any of the embodiments described herein. Additionally, features described in any of the embodiments herein may be incorporated into any of the other embodiments.

[0077] Referring now to FIGS. 8-10, a cervical intervertebral implant 10' in accordance with another embodiment of the disclosure will be described. The implant 10' is similar to the previous embodiment except for a slight modification in the structure of the support structure 30' and a corresponding modification in the porous structure 50'. Compared to the previous embodiment, the rear wall 35' has a narrower width with a portion of the rear end 14' having an open support structure into which the porous structure 50' extends. With the narrower width, the recesses portions 39', 41' open directly into the open space of the side walls 16', 18' and rear end 14'. To maintain sufficient implant strength, a pair of X-shaped struts 48', 48'' are positioned in each of the side wall areas 16', 18' proximate the rear end 14' of the implant 10'. While the front end 12 of the implant 10' remains substantially the same as in the previous embodiment, an additional X-shaped strut 48''' is positioned in each of the side wall areas 16', 18' proximate the rear end 14' of the implant 10'. Again, in the implant 10', the open spaces are filled with the porous structure 50' such that the porous structure 50' encapsulates the struts 44, 46, 48, 48', 48'', 48''' and extends from the upper surface 20 to the lower surface 22 and from the outer perimeter OP to the inner perimeter IP.

In the illustrated embodiment, the porous structure 50' substantially defines the inner perimeter IP and defines a substantial portion of the side walls 16', 18' and a portion of the rear end 14' along the outer perimeter OP.

[0078] Referring now to FIGS. 11-13, a cervical intervertebral implant 10" in accordance with another embodiment of the disclosure will be described. The implant 10" is similar to the previous embodiment except for slight modification in the structure of the support structure 30" and a corresponding modification in the porous structure 50". In the present embodiment, the struts within the side walls are replaced with external X-shaped struts 60, 62. Outer X-shaped struts 60 extend along each of the side walls 16", 18" along the outer perimeter OP. The outer X-shaped struts 60 extend between the upper and lower outer rims 32 and 36. Inner X-shaped struts 62 extend along each of the side walls 16", 18" along the inner perimeter IP. The inner X-shaped struts 62 extend between the upper and lower inner rims 34 and 38. A generally hollow wall space is defined between the outer and inner X-shaped struts 60, 62 on the sides and the cross struts 31, 37 on the top and bottom. These hollow wall spaces extend from the front wall 40 to the rear wall 35" and are filled with the integral porous structure 50". Again, in the implant 10", the open spaces are filled with the porous structure 50" such that it extends from the upper surface 20 to the lower surface 22 and from the outer perimeter OP to the inner perimeter IP. In the present embodiment, the struts 60, 62 are not encapsulated in the porous structure 50", but instead the struts 60 are coplanar with the porous structure 50" along the outer perimeter OP and the struts 62 are coplanar with the porous structure 50" along the inner perimeter IP. Again, the porous structure 50" substantially defines the inner perimeter IP and defines a substantial portion of the side walls 16", 18" and a portion of the rear end 14' along the outer perimeter OP.

[0079] Referring now to FIGS. 14-16, a cervical intervertebral implant 10"" in accordance with another embodiment of the disclosure will be described. The implant 10"" is similar to the previous embodiment except for slight modification in the structure of the support structure 30" and a corresponding modification in the porous structure 50"". In the present embodiment, the external X-shaped struts 60', 62' have a narrower configuration and have curved portions compared to those of the previous embodiment. Again, outer X-shaped struts 60' extend along each of the side walls 16", 18" along the outer perimeter OP as they extend between the upper and lower outer rims 32 and 36. Inner X-shaped struts 62' extend along each of the side walls 16", 18" along the inner perimeter IP as they extend between the upper and lower inner rims 34 and 38. In the present embodiment, in the rear area 14" of the implant 10"", the rear wall 35" is not connected to the upper or lower rim 32, 36 and instead open spaces 61 extend therebetween. As in the previous embodiment, a generally hollow wall space is defined between the outer and inner X-shaped struts 60', 62' on the sides and the cross struts 31, 37 on the top and bottom. These hollow wall spaces extend from the front wall 40 to the rear wall 35" and are filled with the integral porous structure 50"". As in the previous embodiments, all of the open spaces of the implant 10"" are filled with the porous structure 50"" such that it extends from the upper surface 20 to the lower surface 22 and from the outer perimeter OP to the inner perimeter IP. As in the previous embodiment, the struts 60', 62' are not encapsulated in the porous structure 50"", but instead the

struts 60' are coplanar with the porous structure 50"" along the outer perimeter OP and the struts 62' are coplanar with the porous structure 50"" along the inner perimeter IP. Again, the porous structure 50"" substantially defines the inner perimeter IP and defines a substantial portion of the side walls 16", 18" and a portion of the rear end 14" along the outer perimeter OP.

[0080] Referring now to FIGS. 17-19, a cervical intervertebral implant 10^{iv} in accordance with another embodiment of the disclosure will be described. The implant 10^{iv} is similar to the previous embodiment except for slight modification in the structure of the support structure 30^{iv} and a corresponding modification in the porous structure 50^{iv}. In the present embodiment, the struts are replaced with an internal corrugated wall 64 within each of the side walls 16", 18". Each corrugated wall 64 extends between the upper support structure and the lower support structure. In the illustrated embodiment, each corrugated wall 64 extends from the front wall 12, interconnects with the cross struts 31, 37 and interconnects with the rear wall 35". In the present embodiment, in the rear area 14" of the implant 10^{iv}, open spaces 61 extend between the rear wall 35" and the upper or lower rims 32, 36 as in the previous embodiment, however, additional supports 63 extend between the upper rims 32, 34 and the rear wall 35" and between the lower rims 36, 38 and the rear wall 35". As in the previous embodiments, all of the open spaces of the implant 10^{iv} are filled with the porous structure 50^{iv} such that it extends from the upper surface 20 to the lower surface 22. The porous structure 50^{iv} of the present embodiment encapsulates each corrugated wall 64 and while the porous structure 50^{iv} is not continuous from the outer perimeter OP to the inner perimeter IP, the porous structure 50^{iv} still substantially defines the inner perimeter IP and defines a substantial portion of the side walls 16", 18" and a portion of the rear end 14" along the outer perimeter OP.

[0081] Referring now to FIGS. 20-27, one embodiment of an anterior lumbar interbody fusion (ALIF) implant 110 will be described. As illustrated, the implant 110 has a body 111 with a generally D-shaped configuration. The body 111 is defined by a tapered front end 112, a rectangular rear end 114 and side walls 116 and 118 extending therebetween. The implant 110 has an outer perimeter OP extending about the body 111. A hollow interior chamber 113 is defined within an inner perimeter IP of the body 111. The hollow interior chamber 113 is configured to receive bone growth promoting materials. The implant 110 has an upper surface 120 and a substantially parallel lower surface 122, with both surfaces having a tapering portion 123 at the front end 112. The upper and lower surfaces 120, 122 define a plurality of serrations 124 along the side walls 116, 118 and a plurality of serrations 126 along the rear end 114. The rear end 114 of the implant 110 includes a plurality of screw holes 125 through which screws (not shown) extend to anchor the implant onto the vertebral body. Secondary holes 127 are provided to receive respective blocking set screws (not shown). A threaded hole 128 and a blind slot 129 are provided for receiving an instrument that is used for inserting the implant 110. As seen in FIGS. 20-24, the implant 110 is defined by a solid support structure 130 with an interfiled, integral porous structure 150.

[0082] The solid support structure 130 will be described in more detail with reference to FIGS. 25-25. An outer rim 132 extends about the outer perimeter OP of the upper surface

120 and an inner rim 134 extends about the inner perimeter IP of the upper surface 120, i.e. about the interior chamber 113. Similarly, an outer rim 136 extends about the outer perimeter OP of the lower surface 122 and an inner rim 138 extends about the inner perimeter IP of the lower surface 122. A plurality of cross struts 131 extend between the outer rims 132, 136 and the respective inner rims 134, 138 along the side wall areas. As seen in the figures, the cross struts 131 along with contoured portions 133 of the rims 132, 134, 136, 138 define the contour of the serrations 124. In addition to interconnecting the rims within a given upper or lower surface, struts 144, 146 extend within each side wall area to interconnect the upper rims 132, 134 with the lower rims 136, 138. In the illustrated embodiment, a first multi-leg strut 144 extends from the lower inner rim 138 to the upper outer rim 132 near the rear portion of the support structure 130 and a second multi-leg strut 146 extends from the lower inner rim 138 to the upper outer rim 132 near the front portion of the support structure 130.

[0083] A solid rear wall 135 additionally interconnects the outer rims 132, 136 and the respective inner rims 134, 138 along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall 135 defines the holes 125, 127, 128 and the slot 129. Recessed areas 139 on the upper and lower sides of the rear wall 135 define receiving areas for porous structure, as seen in FIGS. 20-24. Cross members 137 in this area along with contours of the outer rims 132, 136 define the serrations 126. A solid front wall 140 with a concave configuration also interconnects the outer rims 132, 136 and the respective inner rims 134, 138 along the front end area. The front wall 140 includes an upper sloped portion 142 extending between the upper outer rim 132 and inner rim 134 and a lower sloped portion 143 extending between the lower outer rim 136 and inner rim 138. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0084] In the illustrations of the support structure 130 in FIGS. 25-27, it is seen that there is significant open space between the upper rims 132, 134 and the lower rims 136, 138 with only the struts 144, 146 therebetween. Additionally, there is open space between the inside surface of the front wall 140 and the inner rims 134, 138. Furthermore, there is open space on an inside surface and the recesses 139, 141 of the rear wall 135. As illustrated in FIGS. 20-24, in the implant 110, these open spaces are filled with the porous structure 150 such that the porous structure 150 encapsulates the struts 144, 146 and extends from the upper surface 120 to the lower surface 122 and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 150 substantially defines the inner perimeter IP and defines a substantial portion of the side walls 116, 118 along the outer perimeter OP.

[0085] Referring now to FIGS. 28-34, an ALIF implant 110' in accordance with another embodiment of the disclosure will be described. The implant 110' is similar to the previous embodiment except for slight modification in the structure of the support structure 130' and a corresponding modification in the porous structure 150'. Compared to the previous embodiment, the upper and lower surfaces 120', 122' of the present implant 110' are angled relative to one another. Additionally, the rear wall 135' has a narrower

width with a portion of the rear end 114' having an open support structure into which the porous structure 150' extends. With the narrower width, the recess portions 139' open directly into the open space of the side walls 116', 118' and rear end 114'. The rear wall 135' defines a single opening 128' for receipt of an insertion tool. A cylinder 145 is positioned between the upper rims 132, 136 and the lower rims 134, 138 along the rear end 114'. The cylinder 145 defines a through bore 147 configured to also receive an insertion tool. To maintain sufficient implant strength in the rear end 114', a first X-shaped struts 148 extends between the cylinder 145 and the end wall 135' and a second X-shaped strut 148' is positioned on the opposite side of the rear wall 135. The front end 112 of the implant 110' includes a recessed area 141 which defines a forward serration 149. Again, in the implant 110', the open spaces are filled with the porous structure 150' such that the porous structure 150' encapsulates the struts 144, 146, 114, 148' and extends from the upper surface 120' to the lower surface 122' and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 150' substantially defines the inner perimeter IP and defines a substantial portion of the side walls 116', 118' and a portion of the rear end 114' along the outer perimeter OP.

[0086] Referring now to FIGS. 35 and 36, an ALIF implant 110" in accordance with another embodiment of the disclosure will be described. The implant 110" is similar to the previous embodiment except for slight modification in the structure of the support structure 130" and a corresponding modification in the porous structure 150". Compared to the previous embodiment, the rear wall 135" of the rear end 114" includes a plurality of slots 129' positioned about the hole 128'. Additionally, the struts of the previous embodiment are replaced with a plurality of X-shaped struts 152 which are interconnected to one another by a circumferential intermediate rim 154. Each of the struts 152 also interconnects with the upper rims 132, 136 and the lower rims 134, 137. Again, in the implant 110", the open spaces are filled with the porous structure 150" such that the porous structure 150" encapsulates the struts 152 and the intermediate rim 154 and extends from the upper surface 120" to the lower surface 122' and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 150" substantially defines the inner perimeter IP and defines a substantial portion of the side walls 116', 118' and a portion of the rear end 114" along the outer perimeter OP.

[0087] Referring now to FIGS. 37 and 38, an ALIF implant 110''' in accordance with another embodiment of the disclosure will be described. The implant 110''' is similar to the previous embodiment except for slight modification in the structure of the support structure 130''' and a corresponding modification in the porous structure 150'''. Compared to the previous embodiment, the X-shaped struts are replaced by coil struts 156a and 156b. Coil strut 156a of side wall area 118' extends from the rear wall 135" to the front wall 140" and extends about the circumferential intermediate rim 154. Similarly, coil strut 156b of side wall area 116' extends from the rear wall 135" to the front wall 140" and extends about the circumferential intermediate rim 154, however, the cylinder 145 extends through and interconnects with the coil strut 154b. Each of the struts 156a, 156b also interconnects with the upper rims 132, 136 and the lower rims 134, 137. Again, in the implant 110''', the open spaces are filled with

the porous structure **150'''** such that the porous structure **150'''** encapsulates the intermediate rim **154** and struts **156a**, **156b** and extends from the upper surface **120'** to the lower surface **122'** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150'''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116'**, **118'** and a portion of the rear end **114''** along the outer perimeter OP.

[0088] Referring now to FIG. 39, an ALIF implant **110^{iv}** in accordance with another embodiment of the disclosure will be described. The implant **110^{iv}** has a body **111'** with a generally oval configuration. The body **111'** is defined by a tapered front end **112''**, a rectangular rear end **114'''** and side walls **116''** and **118''** extending therebetween. The implant **110^{iv}** has an outer perimeter OP extending about the body **111'**. A hollow interior chamber **113** is defined within an inner perimeter IP of the body **111'**. The hollow interior chamber **113** is configured to receive bone growth promoting materials. The implant **110^{iv}** has an upper surface **120'** and a substantially parallel lower surface **122'**, with both surfaces having a tapering portion **123** at the front end **112**. In the present embodiment, each of the surfaces **120'**, **122'** includes a central surface portion **175**. The upper and lower surfaces **120'**, **122'** define a plurality of serrations **124'** along the side walls **116**, **118** and the central portions **175** and a plurality of serrations **126'** along the rear end **114'''**. The rear end **114'''** of the implant **110** includes a plurality of holes **128''**, **145** configured for receiving an instrument that is used for inserting the implant **110**. As in the previous embodiments, the implant **110^{iv}** is defined by a solid support structure **130^{iv}** with an interfiled, integral porous structure **150^{iv}**.

[0089] The solid support structure **130^{iv}** includes an upper plate **160** extending from the front end **112''** to the rear end **114'''** and defining side wall portions **162**, **164** and central portion **166**. A plurality of recesses **173** in the upper and lower plates **160**, **170** are filled with the porous structure **150^{iv}** to define the serrations **124'**, **126'**. Similarly, a lower plate **170** extends from the front end **112''** to the rear end **114'''** and defines side wall portions **172**, **174** and central portion **176**. The upper and lower plates **160**, **170** are interconnected by a front wall **140''** and a rear wall **135''**. It is noted that in the present embodiment, the side walls **116'**, **118'** are generally open without any support structure and completely filled with the porous structure **150^{iv}**. The rear wall **135''** defines the holes **128''**, **145**. The rear wall **135''** includes a plurality of recesses **171** configured to receive the porous structure **150^{iv}**. As in the previous embodiments, the porous structure **150^{iv}** generally extends from the upper surface **120'** to the lower surface **122'** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **150^{iv}** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **116'**, **118'** along the outer perimeter OP.

[0090] Referring now to FIGS. 40-47, one embodiment of a transforaminal lumbar interbody fusion (TLIF) implant **210** will be described. As illustrated, the implant **210** has a body **211** with a generally rectangular shape. The body **211** is defined by a tapered front end **212**, a rectangular rear end **214** and side walls **216** and **218** extending therebetween. The implant **210** has an outer perimeter OP extending about the body **211**. A hollow interior chamber **213** is defined within an inner perimeter IP of the body **211**. The hollow interior

chamber **213** is configured to receive bone growth promoting materials. The implant **210** has an upper surface **220** and a substantially parallel lower surface **222**, with both surfaces having a tapering portion **223** at the front end **212**. The upper and lower surfaces **220**, **222** define a plurality of serrations **224** along the side walls **216**, **218** and a plurality of serrations **226** along the rear end **214**. The rear end **214** of the implant **210** includes a hole **225** and a pair of slots **227** for receiving an instrument that is used for inserting the implant **210**. The implant **210** is defined by a solid support structure **230** with an interfiled, integral porous structure **250**.

[0091] The solid support structure **230** includes an outer rim **232** extending about the outer perimeter OP of the upper surface **220** and an inner rim **234** extending about the inner perimeter IP of the upper surface **220**, i.e. about the interior chamber **213**. Similarly, an outer rim **236** extends about the outer perimeter OP of the lower surface **222** and an inner rim **238** extends about the inner perimeter IP of the lower surface **222**. A plurality of cross struts **231** extend between the outer rims **232**, **236** and the respective inner rims **234**, **238** along the side wall areas. As seen in the figures, the cross struts **231** along with contoured portions **233** of the rims **232**, **234**, **236**, **238** define the contour of the serrations **224**. In addition to interconnecting the rims within a given upper or lower surface, external radial struts **260**, **262** additionally interconnect the rims **232**, **234**, **236**, **238**. Outer radial struts **260** extend along each of the side walls **216**, **218** along the outer perimeter OP. The outer radial struts **260** have a central portion **261** and legs **263** which extend between the upper and lower outer rims **232** and **236**. Inner radial struts **262** extend along each of the side walls **216**, **218** along the inner perimeter IP. The inner radial struts **262** have a central portion **265** and legs **267** which extend between the upper and lower inner rims **234** and **238**.

[0092] A solid rear wall **235** additionally interconnects the outer rims **232**, **236** and the respective inner rims **234**, **238** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **235** defines the hole **225** and slots **227**. Recessed areas **239** and **241** on the upper and lower sides of the rear wall **235** define receiving areas for porous structure, as seen in FIGS. 40-43. The contours of the outer rims **232**, **236** define the serrations **226**. A solid front wall **240** with a concave configuration also interconnects the outer rims **232**, **236** and the respective inner rims **234**, **238** along the front end area. The front wall **240** includes an upper sloped portion **242** extending between the upper outer rim **232** and inner rim **234** and a lower sloped portion **243** extending between the lower outer rim **236** and inner rim **238**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0093] In the illustrations of the support structure **230** in FIGS. 44-47, it is seen that there is significant open space between the upper rims **232**, **234** and the lower rims **236**, **238**, between the inside surface of the front wall **240** and the inner rims **234**, **238**, and on an inside surface and the recesses **239**, **241** of the rear wall **235**. As illustrated in FIGS. 40-43, in the implant **210**, the open spaces are filled with the porous structure **250** such that it extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the present

embodiment, the struts **260**, **262** are not encapsulated in the porous structure **250**, but instead the struts **260** are coplanar with the porous structure **250** along the outer perimeter OP and the struts **262** are coplanar with the porous structure **250** along the inner perimeter IP. Again, the porous structure **250** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216**, **218** along the outer perimeter OP.

[0094] Referring now to FIGS. **48-51**, a TLIF implant **210'** in accordance with another embodiment of the disclosure will be described. The implant **210'** is similar to the previous embodiment except for slight modification in the structure of the support structure **230'** and a corresponding modification in the porous structure **250'**. In the present embodiment, a connecting ring **268** interconnects the central portion **261** with the central portion **265** of the struts **260**, **262** in each side wall **216**, **218**. Additionally, on the upper and lower surfaces **220**, **222**, additional serrations **229** are provided adjacent the rear end **214'**. Again, in the implant **210'**, the open spaces are filled with the porous structure **250'** such that it extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment the struts **260** are coplanar with the porous structure **250'** along the outer perimeter OP and the struts **262** are coplanar with the porous structure **250'** along the inner perimeter IP. The connecting rings **268** are encapsulated within the porous structure **250'**. Again, the porous structure **250'** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216**, **218** along the outer perimeter OP.

[0095] Referring now to FIGS. **52** and **53**, a TLIF implant **210"** in accordance with another embodiment of the disclosure will be described. The implant **210"** is similar to the embodiment illustrated in FIGS. **40-47** except for slight modification in the structure of the support structure **230"** and a corresponding modification in the porous structure **250"**. In the present embodiment, in addition to the struts **260**, **262**, a plurality of X-shaped struts **244** extend between the upper rims **232**, **234** and the lower rims **236**, **238** in each side wall **216**, **218**. Again, in the implant **210"**, the open spaces are filled with the porous structure **250"** such that it extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment the struts **260** are coplanar with the porous structure **250'** along the outer perimeter OP and the struts **262** are coplanar with the porous structure **250'** along the inner perimeter IP. The X-shaped struts **244** are encapsulated within the porous structure **250"**. Again, the porous structure **250"** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216**, **218** along the outer perimeter OP.

[0096] Referring now to FIGS. **52** and **53**, a TLIF implant **210'''** in accordance with another embodiment of the disclosure will be described. The implant **210'''** is similar to the embodiment illustrated in FIGS. **40-47** except for slight modification in the structure of the support structure **230'''** and a corresponding modification in the porous structure **250'''**. In the present embodiment, the side walls **216'**, **218'** do not include external struts, but instead include a plurality of X-shaped struts **246** extending between the upper rims **232**, **234** and the lower rims **236**, **238** in each side wall **216'**, **218'**. Additionally, an intermediate plate **248** interconnects the X-shaped struts **246** within each side wall area. Again, in the implant **210'''**, the open spaces are filled with the porous

structure **250'''** such that the porous structure **250'''** encapsulates the struts **246** and the intermediate plate **248** and extends from the upper surface **220** to the lower surface **222** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **250'''** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **216'**, **218'** along the outer perimeter OP.

[0097] Referring now to FIGS. **58-64**, one embodiment of a lateral lumbar interbody fusion (LLIF) implant **310** will be described. As illustrated, the implant **310** has a body **311** with a generally rectangular shape. The body **311** is defined by a tapered front end **312**, a rectangular rear end **314** and side walls **316** and **318** extending therebetween. The implant **310** has an outer perimeter OP extending about the body **311**. A hollow interior chamber **313** is defined within an inner perimeter IP of the body **311**. The hollow interior chamber **313** is configured to receive bone growth promoting materials. The implant **310** has an upper surface **320** and a substantially parallel lower surface **322**, with both surfaces having a tapering portion **323** at the front end **312**. The upper and lower surfaces **320**, **322** define a plurality of serrations **324** along the side walls **316**, **318**, a serration **328** along the front end **312** and a plurality of serrations **326** along the rear end **314**. The illustrated serrations **324**, **326**, **328** have micro serrations defined thereon. The rear end **314** of the implant **310** includes a hole **325** surrounded by a slot **327** for receiving an instrument that is used for inserting the implant **310**. The implant **310** is defined by a solid support structure **330** with an interfiled, integral porous structure **350**.

[0098] Referring to FIGS. **62-64**, the solid support structure **330** includes an outer rim **332** extending about the outer perimeter OP of the upper surface **320** and an inner rim **334** extending about the inner perimeter IP of the upper surface **320**, i.e. about the interior chamber **313**. Similarly, an outer rim **336** extends about the outer perimeter OP of the lower surface **322** and an inner rim **338** extends about the inner perimeter IP of the lower surface **322**. A plurality of cross struts **331** extend between the outer rims **332**, **336** and the respective inner rims **334**, **338** along the side wall areas. As seen in the figures, the cross struts **331** along with contoured portions **333** of the rims **332**, **334**, **336**, **338** define the contour of the serrations **324**. In addition to interconnecting the rims within a given upper or lower surface, a plurality of X-shaped struts **344** extend within each side wall area to interconnect the upper rims **332**, **334** with the lower rims **336**, **338**.

[0099] A solid rear wall **335** additionally interconnects the outer rims **332**, **336** and the respective inner rims **334**, **338** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **335** defines the hole **325** and slot **327**. A portion of the rear wall **335** defining the hole **325** extends to a secondary rear wall **360** which extends between the upper and lower inner rims **334**, **338**. A cross strut **339** on the upper and lower sides of the rear wall **335** and a portion of the rear wall **335** define the serrations **326**. A solid front wall **340** also interconnects the outer rims **332**, **336** and a secondary front wall **362** interconnects the inner rims **334**, **338** along the front end area. The front wall **340** includes an upper sloped portion **342** extending between the upper outer rim **332** and inner rim **334** and a lower sloped portion **343** extending between the lower outer rim **336** and inner rim **338**. While the rims and walls are described as specific elements for clarity, it is

understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements. A portion of the front wall 340 defines the serration 328.

[0100] In the illustrations of the support structure 330 in FIGS. 62-64, it is seen that there is significant open space between the upper rims 332, 334 and the lower rims 336, 338, between the inside surface of the front wall 340 and the secondary wall 362 and on an inside surface of the rear wall 335. As illustrated in FIGS. 58-61, in the implant 310, the open spaces are filled with the porous structure 350 such that it encapsulates the struts 344 and extends from the upper surface 320 to the lower surface 322 and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 350 substantially defines the inner perimeter IP and defines a substantial portion of the side walls 316, 318 along the outer perimeter OP.

[0101] Referring now to FIGS. 65-67, an LLIF implant 310' in accordance with another embodiment of the disclosure will be described. The implant 310' is similar to the previous embodiment except for slight modification in the structure of the support structure 330' and a corresponding modification in the porous structure 350'. Compared to the previous embodiment, the cross struts 331' and 339' do not include mini-serrations. Additionally, the rear wall 335' has a narrower width with a portion of the rear end 314' having an open support structure into which the porous structure 350' extends. The front end 312 of the implant 310' includes a cylindrical portion 364 extending between the front wall 340 and the secondary front wall 362. Again, in the implant 310', the open spaces are filled with the porous structure 350' such that the porous structure 350' encapsulates the struts 344 and extends from the upper surface 320 to the lower surface 322 and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 350' substantially defines the inner perimeter IP and defines a substantial portion of the side walls 316', 318' along the outer perimeter OP.

[0102] Referring now to FIGS. 68 and 69, an LLIF implant 310" in accordance with another embodiment of the disclosure will be described. The implant 310" is similar to the previous embodiment except for slight modification in the structure of the support structure 330" and a corresponding modification in the porous structure 350". Compared to the previous embodiment, the upper rims 332', 334' and lower rims 336', 338' each have a wider configuration and do not have cross struts extending therebetween. The serrations 324', 326', 328' are formed directly on the rims 332', 334', 336', 338', on the rear wall 335' and the front wall 340'. Additionally, the struts 344 are interconnected to one another by a circumferential intermediate rim 365. Again, in the implant 310", the open spaces are filled with the porous structure 350" such that the porous structure 350" encapsulates the struts 344 and intermediate rim 365 and extends from the upper surface 320 to the lower surface 322 and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure 350" substantially defines the inner perimeter IP and defines a substantial portion of the side walls 316", 318" along the outer perimeter OP.

[0103] Referring now to FIGS. 70-72, an LLIF implant 310"" in accordance with another embodiment of the disclosure will be described. The implant 310"" is similar to the embodiment illustrated in FIGS. 65-67 except for slight

modification in the structure of the support structure 330"" and a corresponding modification in the porous structure 350"". Compared to the embodiment illustrated in FIGS. 65-67, the rims 332, 334, 336, 338 do not have cross struts extending therebetween. Additionally, external struts 366 are provided along the outside perimeter OP along each side wall 316"", 318"" and external struts 368 are provided along the inside perimeter IP along each side wall 316"", 318"". Again, in the implant 310"", the open spaces are filled with the porous structure 350"" such that the porous structure 350" encapsulates the struts 344 and extends from the upper surface 320 to the lower surface 322 and from the outer perimeter OP to the inner perimeter IP. The struts 366 are coplanar with the porous structure 350"" along the outer perimeter OP and the struts 368 are coplanar with the porous structure 350"" along the inner perimeter IP. In the illustrated embodiment, the porous structure 350"" substantially defines the inner perimeter IP and defines a substantial portion of the side walls 316"", 318"" along the outer perimeter OP.

[0104] Referring now to FIGS. 73-75, an LLIF implant 310^{iv} in accordance with another embodiment of the disclosure will be described. The implant 310^{iv} is similar to the previous embodiment except for slight modification in the structure of the support structure 330^{iv} and a corresponding modification in the porous structure 350^{iv}. Compared to the previous embodiment, the rims 332, 334, 336, 338 have cross struts 331' extending therebetween but do not include X-shaped struts within the side walls 316^{iv}, 318^{iv}. Again, in the implant 310^{iv}, the open spaces are filled with the porous structure 350^{iv} such that the porous structure 350^{iv} extends from the upper surface 320 to the lower surface 322 and from the outer perimeter OP to the inner perimeter IP. The struts 366 are coplanar with the porous structure 350^{iv} along the outer perimeter OP and the struts 368 are coplanar with the porous structure 350^{iv} along the inner perimeter IP. In the illustrated embodiment, the porous structure 350^{iv} substantially defines the inner perimeter IP and defines a substantial portion of the side walls 316^{iv}, 318^{iv} along the outer perimeter OP.

[0105] Referring now to FIGS. 139-143, an LLIF implant 310^v in accordance with another embodiment of the disclosure will be described. The implant 310^v is similar to the embodiment illustrated in FIGS. 58-64 except for slight modification in the structure of the support structure 330^v and a corresponding modification in the porous structure 350^v. In the present embodiment, the implant body 311' has a wedge configuration, tapering from a thickest height along the outer edge of the side wall 318^v to a thinnest height along the outer edge of the side wall 316^v. The front wall 340" and rear wall 335" are correspondingly tapered. Additionally, the serration 328" along the front wall 340" does not include micro serrations. As in the embodiment illustrated in FIGS. 58-64, the rims 332, 334, 336, 338 have cross struts 331 extending therebetween, but do not include X-shaped struts within the side walls 316^v, 318^v. Instead, each side wall 316^v, 318^v has a single linear strut 369 extending perpendicularly between the upper outer rim 332 and the lower outer rim 336 at an approximate mid-point of the side wall 316^v, 318^v. Again, in the implant 310^v, the open spaces are filled with the porous structure 350^v such that the porous structure 350^v extends from the upper surface 320 to the lower surface 322 and from the outer perimeter OP to the inner perimeter IP. The struts 369 are coplanar with the porous structure 350^v along the outer perimeter OP. In the

illustrated embodiment, the porous structure **350'** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **316'**, **318'** along the outer perimeter OP.

[0106] Referring now to FIGS. 76-86, embodiments of two-piece cervical implants **400**, **400'** will be described. Each of the implants **400**, **400'** includes a plate **402**, **402'** and connectable spacer **410**. Each of the plates **402**, **402'** defines bone screw holes **403** and one or more blocking set screws **404**. The plates **402**, **402'** may have varying configurations with tabs **405** and projections **406**. The plate configurations are not limited to those shown. Each of the plates **402**, **402'** includes arms **407** with inward projections **408** which engage in slots **419** in the spacer **410** to connect the spacer **410** with the respective plate **402**, **402'**. The illustrated plates **402**, **402'** are solid structures and do not include porous structure, however, the plates may be made with portions having a porous structure.

[0107] Referring to FIGS. 80-86, the spacer **410** has body **411** with a generally U-shaped configuration. The body **411** is defined by a tapered front end **412** with side walls **416** and **418** extending to free ends **414a**, **414b**. The spacer **410** has an outer perimeter OP extending about the body **411** and an inner perimeter IP within the U-shape of the body **411**. When the spacer **410** is connected with a plate **402**, **402'**, a hollow interior chamber **413** is defined which is configured to receive bone growth promoting materials. The spacer **410** has an upper surface **420** and a substantially parallel lower surface **422**, with both surfaces having a tapering portion **423** at the front end **412**. The upper and lower surfaces **420**, **422** define a plurality of serrations **424** along the side walls **416**, **418**, a serration **428** along the front end **412** and serrations **426** at the free ends **414a**, **414b**. Each side wall **416**, **418** defines a connection slot **419** forward of the respective free end **414a**, **414b**. The slots **419** are engaged by the projections **408** on the plate arms **407** to attach the spacer **410** to the respective plate **402**, **402'**. The spacer **410** is defined by a solid support structure **430** with an interfiled, integral porous structure **450**.

[0108] Referring to FIGS. 84-86, the solid support structure **430** includes an outer rim **432** extending about the outer perimeter OP of the upper surface **420** and an inner rim **434** extending about the inner perimeter IP of the upper surface **420**. Similarly, an outer rim **436** extends about the outer perimeter OP of the lower surface **422** and an inner rim **438** extends about the inner perimeter IP of the lower surface **422**. A plurality of cross struts **431** extend between the outer rims **432**, **436** and the respective inner rims **434**, **438** along the side wall areas. As seen in the figures, the cross struts **431** along with contoured portions **433** of the rims **432**, **434**, **436**, **438** define the contour of the serrations **424**.

[0109] A solid front wall **440** interconnects the outer rims **432**, **436** along the front end area. The front wall **440** includes an upper sloped portion **442** and a lower sloped portion **443**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements. A portion of the front wall **440** defines the serration **428**. A rear wall structure **435a**, **435b** at each free end **414a**, **414b** additionally interconnects the outer rims **432**, **436** and the respective inner rims **434**, **438** along the rear end area as well as further connecting the upper and lower structures together. The rear wall structure **435a** extends

about an open area **460** with a side opening **462** and rear wall structure **435b** extends about an open area **464** with opposed side openings **465**, **466**.

[0110] In the illustrations of the support structure **430** in FIGS. 84-86, it is seen that there is significant open space between the upper rims **432**, **434** and the lower rims **436**, **438**, between the inside surface of the front wall **440** and the inner rims **434**, **438** and within the open areas **460**, **462**. As illustrated in FIGS. 80-83, in the spacer **410**, the open spaces are filled with the porous structure **450** such that it extends from the upper surface **420** to the lower surface **422** and from the outer perimeter OP to the inner perimeter IP. In the illustrated embodiment, the porous structure **450** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **416**, **418** along the outer perimeter OP.

[0111] Referring now to FIGS. 89-96, an embodiment of an articulating TLIF implant **510** will be described. As illustrated, the implant **510** has a body **511** with a generally arcuate shape. The body **511** is defined by a tapered front end **512**, a rectangular rear end **514** and arcuate side walls **516** and **518** extending therebetween. The implant **510** has an outer perimeter OP extending about the body **511**. A hollow interior chamber **513** is defined within an inner perimeter IP of the body **511**. The hollow interior chamber **513** is configured to receive bone growth promoting materials. The implant **510** has an upper surface **520** and a substantially parallel lower surface **522**, with both surfaces having a tapering portion **523** at the front end **512**. The upper and lower surfaces **520**, **522** define a plurality of serrations **524** along the side walls **516**, **518** and a plurality of serrations **526** along the rear end **514**. The rear end **514** of the implant **510** includes a hole **525** through one of the surfaces **522** and a slot **527** in communication therewith. An articulating member **570** is positioned within the hole **525** and slot **527** and is pivotal relative to the body **511**. The articulating member **570** includes a body **572** received in and rotatable within the hole **525**. A threaded tool receiving opening **574** extends from the body **572** and is aligned within the slot **527**. A threaded implant tool is received in the threaded tool receiving opening **574** and may be utilized for implanting and articulating the position of the implant **510** in a known manner. The articulating member **570** may be manufactured during the 3D manufacturing process of the body **511** or may be manufactured separately and thereafter positioned within the body **511**. The implant **510** is defined by a solid support structure **530** with an interfiled, integral porous structure **550**.

[0112] The solid support structure **530** includes an outer rim **532** extending about the outer perimeter OP of the upper surface **520** and an inner rim **534** extending about the inner perimeter IP of the upper surface **520**, i.e. about the interior chamber **513**. Similarly, an outer rim **536** extends about the outer perimeter OP of the lower surface **522** and an inner rim **538** extends about the inner perimeter IP of the lower surface **522**. A plurality of cross struts **531** extend between the outer rims **532**, **536** and the respective inner rims **534**, **538** along the side wall areas. As seen in the figures, the cross struts **531** along with contoured portions **533** of the rims **532**, **534**, **536**, **538** define the contour of the serrations **524**. In addition to interconnecting the rims within a given upper or lower surface, external linear struts **560**, **562** additionally interconnect the rims **532**, **534**, **536**, **538**. An outer linear strut **560** extends along each of the side walls **516**, **518** along the

outer perimeter OP. The outer linear struts **560** extend substantially perpendicular to the upper and lower rims at an approximate midpoint of the respective wall. An inner linear strut **562** extends along each of the side walls **516**, **518** along the inner perimeter IP. The inner linear struts **562** extend substantially perpendicular to the upper and lower rims at an approximate midpoint of the respective wall.

[0113] The solid rear wall **535** additionally interconnects the outer rims **532**, **536** and the respective inner rims **534**, **538** along the rear end area as well as further connecting the upper and lower structures together. The solid rear wall **535** defines the hole **525** and slots **527**. Recessed areas **539** and **541** on the upper and lower sides of the rear wall **535** extend between the cross struts **537** and define the serrations **526**, however, in the present embodiment, these recesses generally do not receive porous structure, as seen in FIGS. **89-92**. It is understood that these recesses could receive porous structure as in previous embodiments. The solid front wall **540** also interconnects the outer rims **532**, **536** and the respective inner rims **534**, **538** along the front end area. The front wall **540** includes an upper sloped portion **542** extending between the upper outer rim **532** and inner rim **534** and a lower sloped portion **543** extending between the lower outer rim **536** and inner rim **538**. While the rims and walls are described as specific elements for clarity, it is understood that the elements are formed as a unitary structure and may be formed as a smooth structure without any distinction between the elements.

[0114] In the illustrations of the support structure **530** in FIGS. **93-96**, it is seen that there is significant open space between the upper rims **532**, **534** and the lower rims **536**, **538** and between the inside surface of the front wall **540** and the inner rims **534**, **538**. As illustrated in FIGS. **89-92**, in the implant **510**, the open spaces are filled with the porous structure **550** such that it extends from the upper surface **520** to the lower surface **522** and from the outer perimeter OP to the inner perimeter IP. In the present embodiment, the struts **560**, **562** are not encapsulated in the porous structure **550**, but instead the struts **560** are coplanar with the porous structure **550** along the outer perimeter OP and the struts **562** are coplanar with the porous structure **550** along the inner perimeter IP. Again, the porous structure **550** substantially defines the inner perimeter IP and defines a substantial portion of the side walls **516**, **518** along the outer perimeter OP.

[0115] Referring to FIGS. **97-99**, another embodiment of an intervertebral implant **600** will be described. The implant **600** includes an upper plate **602** and a lower plate **604** supported by a plurality of solid struts **606**, **608** extending therebetween. An open graft window **605** is defined within the implant **600**. The struts **606**, **608** are configured to provide desired load bearing properties of the implant **600**. The end plates **602**, **604** may be formed as a porous structure, for example, having a trabecular porosity. In addition to the porosity, the surfaces of the end plates **602**, **604** may have a nanoscale roughness formed thereon. As seen in FIG. **98**, in the illustrated embodiment, the end plates **602**, **604** and the struts **606**, **608** are configured such that the implant **600** has a biconvex superior and inferior geometry.

[0116] Referring to FIGS. **100** and **101**, another embodiment of an intervertebral implant **610** will be described. The implant **610** includes an upper plate **612** and a lower plate **614** supported by X-shaped struts **616** extending between the plates **612**, **614** on each lateral side of the implant **610**.

An open graft window **615** is defined within the implant **610** between the plates **612**, **614**. As shown in FIG. **101**, the window **615** may be configured to receive an insertion tool **618**. The end plates **602**, **604** and struts **616** may be formed as a porous structure. Alternatively, the struts **616** may be formed as a solid structure. In addition to the porosity, the surfaces of the end plates **612**, **614** may have a roughness **613** formed thereon.

[0117] Referring to FIGS. **102-106**, another embodiment of an intervertebral implant **620** will be described. The implant **620** includes an upper plate **622** and a lower plate **624** supported by opposed solid walls **616**, **618** extending therebetween. Anterior and/or lateral windows **621** open into an interior chamber **623** configured to receive graft material. Each of the upper and lower plates **622**, **624** has a central open area **625** wherein a porous webbing surface **627** is formed. FIGS. **105** and **106** illustrate examples of a porous webbing surface **627**. The webbing may be formed with a grid or honeycomb pattern which is porous to bone ingrowth. The webbing may also be configured such that it may appear opaque when turned past a critical angle. Such a feature may be used for assessing the implants orientation or position.

[0118] Referring to FIGS. **107** and **108**, intervertebral implants **630**, **640** illustrating various features will be described. The implant **630** of FIG. **107** includes a body **632** with an upper surface **634** and a lower surface **636**. One or both of the surfaces **634**, **636** may be formed with a complex configuration to optimize end plate contact. For example, the upper surface **634** has a biconcave configuration. Similarly, the implant **640** of FIG. **108** includes a body **642** with an upper surface **644** and a lower surface **646**. One or both of the surfaces **644**, **646** may be formed with various teeth or surface roughness features to minimize subsidence or migration. For example, the upper surface **646** has a plurality of teeth **648** defined thereon. For each of the implants **630**, **640**, the body **632**, **642** may include both solid structure and porous structure as described herein.

[0119] Referring to FIG. **109**, another embodiment of an intervertebral implant **650** will be described. The implant **650** includes an upper plate **652** and a lower plate **654** supported by a plurality of solid struts **656** extending therebetween. The density of the struts **656** increases moving posteriorly to provide a varying A-P stiffness to the implant, for example, to optimize the load of an anterior graft.

[0120] Referring to FIGS. **110-113**, another embodiment of an intervertebral implant **660** will be described. The implant **660** includes an upper surface **662** and a lower surface **664** supported by a plurality of solid struts **666** extending therebetween. The upper and lower surfaces **662**, **664** are defined by perimeter struts **667** and diagonal struts **668**. As illustrated, the struts **666**, **667**, **668** that make up the support structure extend generally along the supporting edges while the remainder of the implant structure remains open for either porous structure or ingrowth chambers. FIGS. **114** and **115** illustrate alternative strut configurations for either the side walls or top or bottom surfaces of the implant **650**. In the embodiment illustrated in FIG. **114**, the struts **666'**, **668'** have a rectangular central region with radial supports extending from the corners thereof. In the embodiment illustrated in FIG. **115**, the struts **666''**, **668''** have a diamond shaped central region with octagon patterned struts thereabout.

[0121] Referring to FIG. 116, another embodiment of an intervertebral implant 670 will be described. The implant 670 includes a body 671 with an upper surface 672 and a lower surface 672 extending about an opening 613 into an ingrowth cavity. The front end of the implant 670 includes sloped surfaces 675 to define a smooth leading edge. One or both of the surfaces 672, 672 may include a radial sawtooth configuration 676. As in previous embodiments, the body 671 may include both solid structure and porous structure as described herein.

[0122] Referring to FIG. 117, another embodiment of an intervertebral implant 680 will be described. The implant 680 includes a solid support frame 682 encased within porous structure 684, 686. The implant 680 defines at least one opening 681 into an ingrowth chamber 683. In the illustrated embodiment, the porous structure includes areas of different densities. In the illustrated embodiment, the outer porous structure 684 has a lower density than the inner porous structure 686. As described herein, the various implants may be formed with porous structures having varying densities, not just with respect to outer and inner layers, but also within a given layer at different areas of the implant, e.g. anterior versus posterior.

[0123] Referring to FIGS. 118 and 119, another embodiment of an intervertebral implant 690 will be described. The implant 690 includes a body 692 extending about an ingrowth chamber 693. The body 692 is defined by a porous portion 696 surrounding a plurality of support struts 694. The support struts 694 are not interconnected to one another. The support struts 694 may be a solid structure or may be a porous structure with a greater density than the porous portion 696. For example, the struts 694 may be a porous structure having a cancellous density while the porous structure 696 has a cortical density. Since the body 692 includes a significant porous structure, the ingrowth chamber 693 may be smaller than compared to an implant having a non-porous body.

[0124] Referring to FIGS. 120-122, another embodiment of an intervertebral implant 700 will be described. The implant 700 includes a body 702 with an ingrowth pocket 703 extending into the posterior side thereof. The body 702 is defined by a porous portion 706 surrounding a plurality of support struts 704. The support struts 704 are not interconnected to one another. The support struts 704 may be a solid structure or may be a porous structure with a greater density than the porous portion 706. For example, the struts 704 may be a porous structure having a cancellous density while the porous structure 706 has a cortical density.

[0125] Referring to FIG. 123, another embodiment of an intervertebral implant 710 will be described. The implant 710 includes a body 712. The body 712 is defined by an external area of higher density porous structure 714 surrounding a less dense porous structure 716. The higher density porous structure 714 acts as struts to provide the implant 710 strength. The area of higher density porous structure 714 is shown in a zig zag pattern, but may have other patterns and configurations.

[0126] Referring to FIGS. 124-126, another embodiment of an intervertebral implant 720 will be described. The implant 720 includes an upper plate 722 and a lower plate 724. Each of the plates 722 and 724 has a wave configuration with minimum struts 726 interconnecting the plates 722, 724. With such a configuration, the implant 720 has spring between the plates. An open graft window 723 is defined

within the implant 720. The plates 722, 724 and the struts 726 may include both solid structure and porous structure, as described herein.

[0127] Referring to FIGS. 127 and 128, another embodiment of an intervertebral implant 730 will be described. The implant 730 includes a body 732 surrounding an ingrowth chamber 733. The body 732 is defined by a porous portion 734 which is surrounded a continuously wrapped support rib 736. The support rib 736 may be a solid structure or may be a porous structure with a greater density than the porous portion 734.

[0128] Referring to FIG. 129, another embodiment of an intervertebral implant 740 will be described. The implant 740 includes interconnectable body portions 742 and 744. In the illustrated embodiment, the upper body portion 742 includes a post 743 configured to friction fit within a hole 745 of the lower body portion 744. Various interconnectable upper and lower body portions 742, 744 may be printed and interconnected to form a customized implant 740. In the illustrated embodiment, three additional lower body portions 744', 744'', 744''' and provided and all may be interconnected with the upper body portion 742 to achieve implants having different configurations.

[0129] Referring to FIGS. 130-132, another embodiment of an intervertebral implant 750 and a method of implantation thereof will be described. The implant 750 includes a central body 752 and a pair of expandable wings 753, 755. The implant 750 is connected to a hollow insertion tool 756 and moved within an intervertebral space 759. Once in position, graft material 757 is fed through the tool 756 and into the implant 750 whereby the wings 753, 755 are caused to expand. Graft material 757 is supplied until the area within the central body 752 and the expanded wings 753, 755 is filled. The packed graft material 757 maintains the expanded configuration of the implant 750.

[0130] Referring to FIGS. 133-135, another embodiment of an intervertebral implant 760 and a method of inserting such implant will be described. The implant 760 includes a body 762 extending about an ingrowth chamber 763. The body 762 is generally defined by a porous structure 764. Solid bore holes 766 are defined within the body 762 and are configured to receive screws or other anchors to facilitate fixation of the implant 760. The body 762 also defines a tool receiving opening 765. In the illustrated embodiment, an expandable tool 770 is inserted through the opening 765. The expandable tool 770 includes a pair of legs 772 with flanges 773 on the end thereof. In a collapsed condition, the flanges 773 pass through the opening 765. To engage the implant 760, a needle 775 or the like is extended between the legs 772, thereby forcing the flanges 773 outward such that the engaged the inside surface of the implant and do not pass through the opening 765 until the needle 775 is removed.

[0131] Referring to FIGS. 136 and 137, another tool and implant opening interconnection assembly will be described. In the present embodiment, the implant 780 has a tool receiving opening 782 with a wider central region 783 and narrower end regions 784. The insertion tool 790 has a head 792 with a complimentary configuration. The head 792 is positioned within the opening 782 with the orientation of the head 792 matching that of the opening 782 such that the head 792 passes through the opening 782. Once inside, the tool head 792 is rotated, for example, a quarter turn, such that the tool head 792 engages the inside surface of the implant 780 and is no longer removable through the opening

782. A securing sleeve **794** is threadably advanced along the tool **790** and engages the outside surface of the implant **780**, securing the implant **780** between the tool head **792** and the sleeve **794**.

[0132] Referring to FIG. **138**, another tool and implant opening interconnection assembly will be described. In the present embodiment, the implant **800** has a tool receiving opening **802** with an outward taper **803** at the inner side of the opening **802**. The insertion tool **810** has an outer hollow tube **812** with a collet **814** on the free end thereof and an inner hollow tube **816** with a through passage **817**. The inner hollow tube **816** is axially moveable relative to the outer hollow tube **812**. Initially, the inner hollow tube **816** is withdrawn into the outer hollow tube **812** such that it is clear of the collet **814** such that the collet **814** may collapse and pass through the implant opening **802**. Once the collet **814** passes through the opening **802**, the inner hollow tube **816** is moved forward such that the collet **814** is expanded outward and maintained in the outward position, engaging the tapered portion **803** of the implant opening **802**. Graft material or the like may be passed into the implant **800** through the through passage **817**.

[0133] As described herein, the implants of the disclosure generally comprise a solid or higher density support structure and a porous structure formed integral therewith. The solid support structure may include solid front and rear walls interconnected by upper and lower implant surfaces. In several of the embodiments, the upper and lower surfaces include spaced apart rims with cross struts interconnecting the rims. In many embodiments, the solid support structure of the upper and lower surfaces includes a plurality of openings in which the integral porous structure is formed such that the porous structure extends along at least a portion of the upper and lower implant surfaces. The side walls extending between the front and rear walls generally have a minimal solid structure, for example, a plurality of struts extending between the upper and lower rims, but otherwise have open area therebetween in which the integral porous structure is formed. The configuration of the solid structure is selected to provide the implant sufficient structural integrity and mechanical stability while maximizing the area of porous structure which facilitates better integration/incorporation with the adjacent bone. In several embodiments of the disclosure, the solid structure generally encases the corners of the porous structure or otherwise houses the porous structure therein to maintain the structural integrity of the porous structure.

[0134] Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that all ranges broadly recited in this document include within their scope all narrower ranges which fall within the broader ranges. It is also intended that the components of the various devices disclosed above may be combined or modified in any suitable configuration.

What is claimed is:

1. A lateral lumbar intervertebral implant for implantation in an intervertebral space between vertebrae, the implant comprising:

a body having a tapered front end, a rear end and opposed side walls extending between the ends, the body having an outer perimeter with a rectangular configuration and an inner perimeter about an internal chamber, the body including:

an upper surface and a lower surface, the upper surface defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface defined by a solid lower outer rim and a spaced apart solid lower inner rim;

a solid front wall extending at the front end between at least the solid upper outer rim and the solid lower outer rim;

a solid rear wall extending at the rear end between the solid upper outer rim and the solid lower outer rim;

each of the side walls including at least one solid support strut extending between the upper outer rim and the lower outer rim;

a plurality of upper solid cross struts connected between the upper outer rim and the upper inner rim and arranged in parallel to each other;

a plurality of lower solid cross struts connected between the lower outer rim and the lower inner rim and arranged in parallel to each other and

a porous structure integrally formed with the solid upper rims, the solid lower rims, each of the solid cross struts, and the solid struts in the side walls, the porous structure extending from the body outer perimeter to the body inner perimeter.

2. The intervertebral implant of claim 1, wherein the body has a wedge configuration tapering from a thicker height along the outer edge of one of the side walls to a thinner height along the outer edge of the other one of the side walls.

3. The intervertebral implant of claim 1, wherein the at least one solid support strut of each of the sidewalls includes a single support strut extending perpendicularly between the upper outer rim and the lower outer rim at an approximate mid-point of the side wall.

4. The intervertebral implant of claim 1, wherein the at least one solid support strut extends perpendicularly relative to the upper outer rim and the lower outer rim.

5. The intervertebral implant of claim 1, wherein the plurality of upper solid cross struts are connected perpendicularly between the upper outer rim and the upper inner rim.

6. The intervertebral implant of claim 1, wherein each of the cross struts defines a portion of a surface serration.

7. The intervertebral implant of claim 6, wherein all of the solid support struts and cross struts are embedded within the porous structure.

8. The intervertebral implant of claim 1, wherein the tapered front end includes a solid structure having an upper sloped portion and a lower sloped portion together defining the taper.

9. The intervertebral implant of claim 1, wherein the rear end is defined substantially by a solid structure.

10. The intervertebral implant of claim 9, wherein the solid structure defined by the rear end includes a recess for receiving an insertion instrument.

11. The intervertebral implant of claim 1, wherein the body further includes a solid secondary rear wall extending between the upper inner rim and the lower inner rim.

12. The intervertebral implant of claim **1**, wherein the upper inner rim and the lower inner rim are both a closed rim.

13. The intervertebral implant of claim **1**, wherein the porous structure has a greater than 60% porosity.

14. The intervertebral implant of claim **1**, wherein the porous structure has an average pore size of about 300-800 μm .

15. The intervertebral implant of claim **1**, wherein the porous structure has a greater than 60% porosity and has an average pore size of about 300-800 μm .

16. A lateral lumbar intervertebral implant for implantation in an intervertebral space between vertebrae, the implant comprising:

a body having a tapered front end, a rear end and opposed side walls extending between the ends, the body having an outer perimeter with a rectangular configuration and an inner perimeter about an internal chamber, the body including:

an upper surface and a lower surface, the upper surface defined by a solid upper outer rim and a spaced apart solid upper inner rim and the lower surface defined by a solid lower outer rim and a spaced apart solid lower inner rim;

a solid front wall extending at the front end between at least the solid upper outer rim and the solid lower outer rim;

a solid rear wall extending at the rear end between the solid upper outer rim and the solid lower outer rim;

each of the side walls including at least one solid support strut extending between the upper outer rim and the lower outer rim;

a plurality of upper solid cross struts connected between the upper outer rim and the upper inner rim and arranged in parallel to each other;

a plurality of lower solid cross struts connected between the lower outer rim and the lower inner rim and arranged in parallel to each other and

a porous structure integrally formed with the solid upper rims, the solid lower rims, each of the solid cross struts, and the solid struts in the side walls, the porous structure extending from the body outer perimeter to the body inner perimeter, the porous structure having a greater than 60% porosity and has an average pore size of about 300-800 μm .

17. The intervertebral implant of claim **16**, wherein the at least one solid support structure of each of the sidewalls includes a single support strut extending perpendicularly between the upper outer rim and the lower outer rim at an approximate mid-point of the side wall.

18. The intervertebral implant of claim **16**, wherein the body further includes a solid secondary rear wall coupled between the upper inner rim and the lower inner rim.

19. The intervertebral implant of claim **16**, wherein the upper inner rim and the lower inner rim are both a closed rim.

20. The intervertebral implant of claim **16**, wherein the at least one solid support strut extends perpendicularly relative to the upper outer rim and the lower outer rim.

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